Technologies to Generate Pseudothermal Light

DISSERTATION

Presented in Partial Fulfillment of the Requirements for the Degree Doctor of Philosophy in the Graduate School of The Ohio State University

By

ANITA BHAGAT, MS

GRADUATE PROGRAM IN PHYSICS

THE OHIO STATE UNIVERSITY

2024

DISSERTATION COMMITTEE:

DANIEL J. GAUTHIER, ADVISOR

ALEXANDRA LANDSMAN

ROLAND KAWAKAMI

LOUIS F. DIMAURO

Abstract

In the twenty-first century, many applications in science and engineering are based upon the coherence theory. Some applications demand a light source with spatial or temporal coherence properties equivalent to a thermal light source (TLS), an incoherent classical light source such as sun or incandescent bulb filament. Although the fundamentals of the emission of TLS have been studied and understood for nearly a century, engineering such light sources is still a difficult task. This is because a true thermal light source has a short coherence time of a femtosecond (10^{-15} s). In recent years, pseudothermal light sources (PTLS) have gained more attention as the coherence properties of the PTLS are analogous to a true thermal source. However, even the most recent PLTS has only achieved a long coherence time of a 10^{-5} to 1 s compared to a 10^{-15} s of TLS.

In this thesis, I demonstrate a novel technology to generate a pseudothermal light source with a short coherence time of 10^{-8} to 10^{-6} s in a single-mode fiber. It is a simple, cost-effective device built using a laser light source, a Mach-Zehnder modulator (MZM) driven by a Gaussian noise generator, and a bias voltage. I characterize the PTLS using a temporal second-order coherence function at a delay time equal to zero, i.e., $g^{(2)}(0)$. I measure the light intensity fluctuation at the output of the MZM and then implement the intensity correlation function to calculate the $g^{(2)}(0)$

for different noise amplitudes. I propose that the light source I built is a super bunching pseudothermal light source as the $g^{(2)}(0)$ is 2.86 ± 0.01, which is higher than a bunching thermal or pseudothermal light, i.e., 2. This value can be further enhanced by using an ideal Gaussian noise generator.

In the latter part of the thesis, I present a spatial second-order coherence $g^{(2)}$ function executed using the intensity correlation function in an optical multi-mode (MM) fiber. I propagate a coherent light into a MM fiber and measure the intensity of speckles created at the exit of the fiber. This method is straightforward and does not require any customization of the input light except that I ensure a maximum number of modes are excited inside the MM fiber. I put forward that the speckle pattern generated at the fiber's output has a super-Gaussian intensity distribution, manifesting as fewer bright spots in the speckle images. The super-Gaussian intensity-intensity correlation function implements a super bunching spatial second-order coherence $g^{(2)}(0) > 2$. The mechanism responsible for this behavior is a mode-dependent loss (MDL) arising from material absorption and coupling of the guided modes to leaky modes in the MM fiber. I obtain the maximum $g^{(2)}(0)$ value to be 2.70 ± 0.05 for the fiber with the high MDL. The spatial second-order correlation function scales as a square root of the length for the jacketed low and high absorption fibers. Whereas it directly relates to the length of the bare low-absorption fibers.

I also demonstrate that these speckle patterns are insensitive to the change in frequency of the incident coherent light. The fibers with high MDL are more insensitive to the input light frequency change compared to the Low MDL fibers. Studying further and finding if the speckle's spatial and spectral are related will be interesting.

Technologies characterized by super bunching second-order coherence are significant in systems that require a high degree of coherence, such as high-order coherence theory. The PTLS in a single-mode fiber can be employed with a rotating ground glass disc to enhance the visibility of images in temporal ghost imaging.

This thesis is dedicated to my parents, Rajendra Bhagat and Ahuliya Devi Bhagat,

In loving memory of my grandmother, Domni Devi Bhagat

Acknowledgments

I especially thank my advisor, Daniel J. Gauthier, for guiding and motivating me. He taught me self-care and to see the big picture before deciding. I thank him for his expert advice, without which this thesis would not have been possible.

I want to thank all my committee members, Roland Kawakami, Louis F. DiMauro, and Alexandra Landsman, for their advice and patience with the time-consuming processes.

I am also thankful to Professor Gregory Lafyatis for guiding me.

I am grateful to Professor Samir Mathur and Kristina Dunlap for always supporting and being kind to me.

I thank Shrivatch Sankar from OSU's electrical and computer department for lending me a camera. It was essential equipment for my experiment, so thank you for being kind enough to lend it to me whenever I requested it.

I want to thank all my colleagues, Abdullah Alhag, Robert Kent, and Roderick D. Cochran, for being there, sharing thoughts, and having long chats. You all made my PhD journey less stressful and filled with happiness. I will always cherish all these good memories.

I thank my friend and brother Daram Ramdim, my cat Vayu, and my family for their unconditional love and support.

Lastly, I want to thank all my teachers and professors from my school and undergraduate, who have always been my well-wishers and inspired me to improve and be a better person.

Vita

July, 2012 .... B.S. Petroleum Engineering
University of Oklahoma

May, 2015 .... B.S. Physics
University of Oklahoma

May, 2017 .... M.S. Physics,
University of Oklahoma

August 2018 - present .... Ph.D. in Physics,
The Ohio State University.

Publications

Research Publications

1. Anita Bhagat and Kieran Mullen. "Pseudo-magnetic quantum Hall effect in oscillating graphene." Solid State Communication 287(2019): 31-34.
2. Bhagat Anita, Carl H. Sondergeld and Chandra S. Rai. "The Petrophysical Study on UAE Carbonates." Technical Program Expanded Abstracts 2012, pp. 1-5. Society of Exploration Geophysicists, 2012.

Fields of Study

Major Field: PHYSICS

Table of Contents

List of Tables

List of Figures

Chapter 1
Introduction

The earliest observation and characterization of coherence in optical physics date back to the mid-nineteenth century. A light source is categorized as coherent if a fixed amplitude and phase relationship exists between the electric field values at different times and spatial locations [1]. Before the 1950s, the concept of coherence was based only on interference phenomena: spatially fixed interference field created by the superposition of two electromagnetic beams. However, in 1957, two astronomers, Robert Hanbury Brown and Richard Q. Twiss, built a new type of interferometer known as an intensity interferometer, which measured the random fluctuations in the intensity (square of the light-wave amplitude) of a light source, i.e., a star. They performed a series of experiments and demonstrated that the coherence properties of the light source are strongly related to the correlation of the intensity fluctuations in the two light beams of the source [2]. Therefore, the outcome of this experiment, along with other successive experiments to the invention of the laser, reformed the concept of coherence and led to the modern definition that the two beams are considered second-order coherent if their intensity fluctuations are correlated [3].

The intensity-coherence concept was developed to describe and understand the electromagnetic field or the emission of incoherent classical sources, i.e., chaotic or thermal light sources (TLS). The natural light generated by sources like the sun, stars, light bulbs, incandescent lamps, etc, is considered thermal light. They are incoherent light sources comprised of many sub-sources, like atoms, where all atoms radiate independently, leading to high fluctuating intensity values. Such sources with random emission were expected not to correlate; however, the Hanbury-Brown-Twiss experiment (HBT) [4] demonstrated that thermal sources show a high-order correlation.

1.1 Hanbury Brown-Twiss effect (HBT)

Before applying the intensity interferometer technique to measure the angular size of the star, Hanbury Brown and Twiss performed a tabletop experiment in the lab using a mercury vapor lamp, rectangular aperture, one half-silvered mirror for splitting the light beam in two, two photomultipliers for detection, one correlator and some optics like lens and filters [4]. The schematic representation of the apparatus is shown in Fig. 1.1. They measured the photon correlations between the photons emitted at different spatial locations. The photon detection at two points was done by varying the spatial separation Δx by sliding one of the detectors transverse normal to the incident light and keeping the other detector fixed. The result was astonishing; no correlations existed between the photons at larger Δx, whereas the correlations increased with decreased Δx.

In conclusion, they observed the unrelated photons emitted arrived at the detector together, exhibiting a bunching effect. The HBT experiment proposed that randomly emitted photons of a thermal source tend to come in bunches at the detection point. Therefore, the bunching effect is a characteristic of a continuous randomly

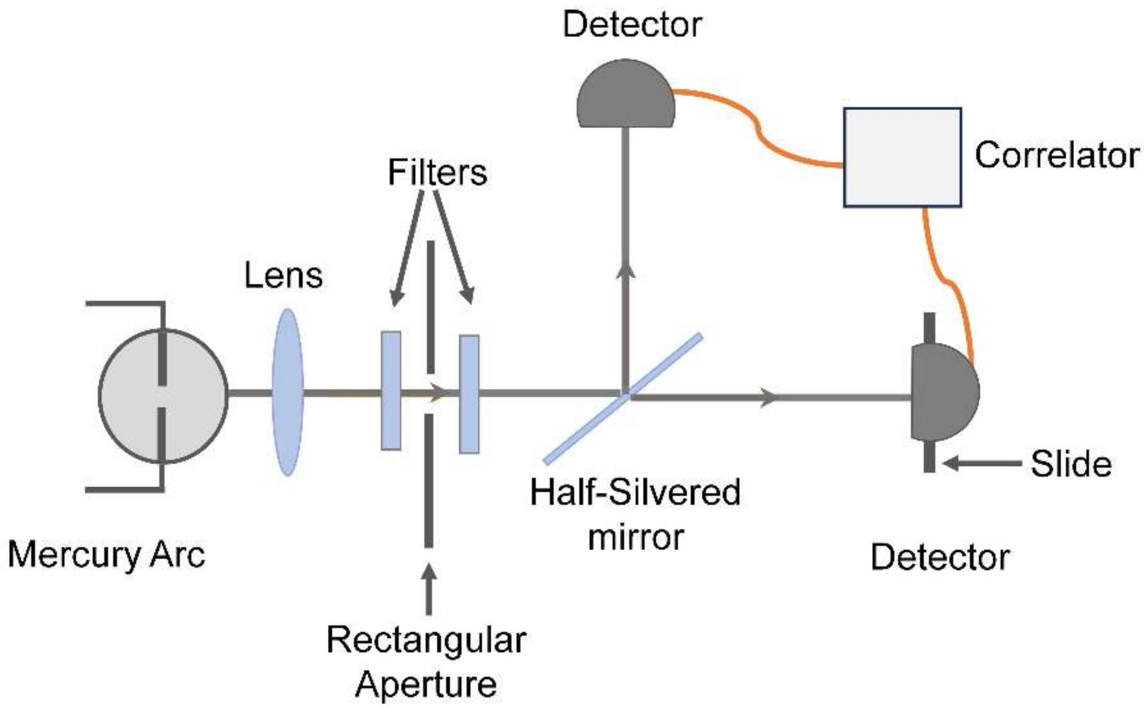


Figure 1.1: The schematic diagram of Hanbury Brown Twiss experiment. The setup measures a photon-photon correlation of a thermal light using a mercury lamp and some optical equipment like lens, filters, and silvered mirror. Two photo multipliers were used to measure the photons, and a correlator was used to calculate the second-order correlation function.

emitting light source like a thermal source. This experiment was significantly crucial for demonstrating the photon bunching effect. The HBT experiment motivated many researchers around the world but also raised fundamental questions regarding the nature of light as the photons in an incoherent, chaotic classical light beam were seemingly uncorrelated and thus could not be detected in nearby pairs. However, the debate ended, and the Hanbury-Brown and Twiss effect prevailed, leading to the photon statistics theory and the light characterization based on the intensity and coherence correlation function. The observation of the bunching effect stimulated the birth of quantum optics [5].

Furthermore, the high order of coherence, i.e., the second order of coherence applied in the HBT effect, can be explained and experimentally measured within the framework of classical electromagnetism using the intensity correlation function. The intensities in these correlation functions can be measured using a classical square-law detector. However, Hanbury Brown and Twiss used detectors that measured a single photon. They performed a photon-photon correlation. The result for the photon-photon correlation for the thermal light obtained was analogous to the intensity-intensity correlation. Later, the photon-photon correlation was measured for a stable wave, and it was found that photons of such light sources show randomness at the detection point. The intensity-intensity correlation or the second-order coherence of such a stable wave equals 1, whereas, for thermal light, it is 2 [6]. So, based on stable light correlation, it was proposed that thermal light photons be bunched as more than one random photon events arrive at the detection point. And this is why the thermal source is called a bunched.

I use intensity correlation, a classical approach, to obtain the results in this thesis. However, a similar result can be obtained if the classical detectors are switched to a single photon detector.

1.2 Second-Order Coherence

Characterizing different light sources based on their emission, i.e., is the light emitted by a coherent or an incoherent classical source or a quantum source, is an important task in light source-driven physics experiments. The intensity correlation between the two fields corresponding to the second order of the coherence plays an important role in distinguishing different light sources. In general, the normalized form of the second-order coherence, $g^{(2)}(\vec{r}_1, t_1; \vec{r}_2, t_2)$ function with intensities $I_1(\vec{r}_1, t_1)$ and $I_2(\vec{r}_2, t_2)$ of the light source measured at points $(\vec{r}_1, t_1)$ and $(\vec{r}_2, t_2)$ in space and time is given by

$$g^{(2)}(\vec{r}_1, t_1; \vec{r}_2, t_2) = \frac{\langle I_1(\vec{r}_1, t_1) I_2(\vec{r}_2, t_2)\rangle}{\langle I_1(\vec{r}_1, t_1)\rangle \langle I_2(\vec{r}_2, t_2)\rangle}. \tag{1.1}$$

The angle bracket notation is for statistical averages.

The second order of coherence can be realized in spatial or temporal form. The $g^{(2)}(\vec{r}_1, t_1; \vec{r}_2, t_2)$ function at zero spatial difference or time delay has different values for the various light sources. However, both the realizations have analogous consequences. Thus, considering the temporal second order of coherence, if a light source has wide-ranging stationary statistics, the intensity correlation will depend only on the delay time τ between the two intensity values such that the statistical properties are invariant in time. Then, the normalized form of the second-order temporal coherence measured at the same location, $g^{(2)}(\tau)$ with intensity $I(t)$ is given by

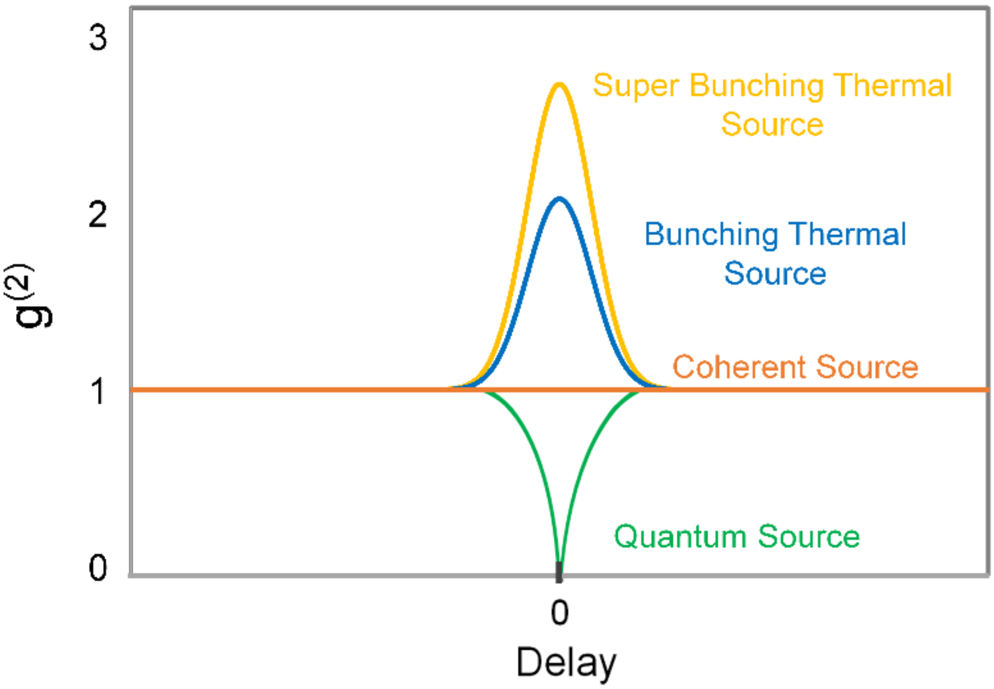


Figure 1.2: The plot shows $g^{(2)}(\tau)$ as a function of delay for different light sources such as thermal light (super bunching and bunching), coherent light, and quantum light.

$$g^{(2)}(\tau) = \frac{\langle I(t)I(t+\tau)\rangle}{\langle I(t)\rangle^2}. \tag{1.2}$$

The time evolution of the second-order coherent function $g^{(2)}(\tau)$ exhibiting distinguishable features of different light sources is shown in Fig. 1.2. The figure presents the $g^{(2)}(\tau)$ for the typical sources used in classical and quantum fields. The $g^2(0)$ values categorizes these sources as:

1. $g^{(2)}(\tau{=}0) > 2$, Super Bunching Thermal Source
2. $g^{(2)}(\tau{=}0) = 2$, Bunching Thermal Source

3. $g^{(2)}(\tau{=}0) = 1$, Coherent Source

4. $g^{(2)}(\tau{=}0) < 1$, Quantum Source

1.3 Thermal and Pseudothermal Lights

Thermal light sources (TLS) have been historically significant in optical physics [7]. Before the development of coherent sources like laser, thermal sources were plausibly the only light sources available for application and experimental purposes. Although lasers are widely used for many applications in modern optics, thermal light sources have applications in science and engineering. They are still important for understanding some fundamental concepts of physics [4, 8]. Therefore, engineering to develop a light source with the coherence properties of true thermal light has been taking place for decades. However, a fundamental challenge exists in using a TLS for application purposes. These light sources have a very short coherence time around femtoseconds (10^{-15} s), which is difficult to detect using the present low-time resolution detectors [7].

Nevertheless, this problem was resolved in 1964 by Martienssen and Spiller when they created a "new light source" and called it a pseudothermal light source (PTLS) [9]. The source was created by focusing a coherent laser beam on a rotating ground glass; the resulting scattered light created a dynamic speckle pattern that imitated the intensity variability of a TLS. The formation of the light field was definite and clearly visible, thus making the rotating ground disc operate as an excellent imitator of the thermal light emitted by a collection of atoms. Many scientific groups [10–15] studied the properties of this innovative light as the PTLS had a similar coherence properties to that of a true thermal source, except that its coherence time was much longer than that of a true TLS. It varied between 10^{-5} to 1 s [7]. Since its invention, the PTLS has been broadly employed for applications where a true TLS is needed, as its usage is much easier and simpler. It is used in experiments to study and develop the higher degree of coherence theory [16–19], for imaging applications like ghost imaging [20–24], for intensity interferometer [25] and many other applications.

Eventually, the high-order coherence experiment with PTLS led to the invention of a new type of PTLS called super bunching pseudothermal light [16]. This new light's second-order coherence $g^{(2)}(\tau = 0)$ was much larger than 2, representing the scattered photons being more bunched than pseudothermal and thermal light [26] as a result acquired its name super bunching pseudothermal light. The super-bunched light was created using coherent laser light and N ($N \gg 2$) number of rotating ground glasses. However, adding more rotating ground glasses increased the uncertainty in obtaining the $g^{(2)}(\tau = 0)$ values, which limited their usage for imaging applications like ghost imaging. Later, to overcome this problem, other techniques were employed, i.e., modulating the intensity of incident laser light using an electronic [18, 19] or an acoustic [27] modulator. These modulators are expensive equipment, and thus, these setups are not cost-effective techniques to create a super bunching pseudothermal light source. Some of the other disadvantages of all these methods in creating the super bunching pseudothermal source are, firstly, they all are complex setups; they either use a large number of rotating ground glasses or a high-voltage modulator setup. Secondly, all these setups are conducted in free space, making them lose huge amounts of light. Finally, the coherence time of the super bunching light using these methods is around 10^{-5} to 1 s, much longer than a true TLS.

1.4 Outline and Summary

In the first part of the thesis, I primarily develop a technology that resolves these issues and generates a super bunching PTLS in a lab. The technology is simple and cost-effective. I discuss the light source's temporal second-order coherence and describe how this technology can be a better alternative compared to other pseudothermal light sources, as it facilitates a shorter tunable coherence time. My experiment is entirely single-mode fiber-based, an essential factor in controlling light loss in the scattering mechanism.

In the later part of the thesis, I present another economic experimental setup using a multi-mode fiber to demonstrate a spatial second-order correlation function with non-Gaussian intensity distribution. Most research groups use a complex method where the input light is tailored to create speckles with non-Gaussian statistics. However, I show that such intensity distributions can manifest without customizing the input beam using only a multi-mode fiber and a camera. The speckles can be created by propagating a coherent light in a multi-mode fiber, and the images of speckle patterns can be recorded using a CCD camera.

Chapter 2: Spatial and Temporal Coherence of a Classical Light

In this chapter, I give an overview of the background of pseudothermal light sources and why developing such sources was an important invention. I describe each method used to generate the PTLS light and its pros and cons. I then argue with supportive reasons why building a new technology to create pseudothermal light is necessary.

Chapter 3: Technology to Generate: Super bunching Pseudothermal Light in a Single-Mode Fiber

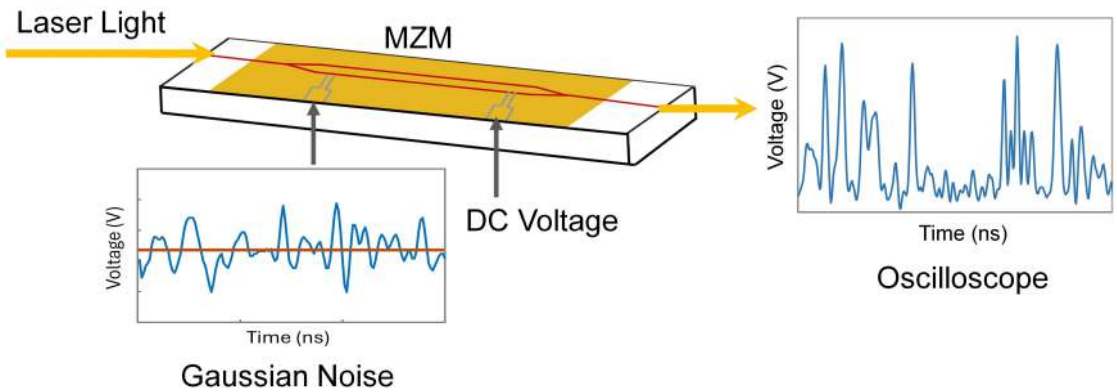


Figure 1.3: MZM: Mach Zehnder modulator. The schematic illustrates the procedure to create an intensity fluctuation at the output of a Mach Zehnder modulator experiment. The modulator is driven by fast (Gaussian noise) and slow (DC) voltages.

In this chapter, I introduce a simple and cost-effective technology for generating a super bunching pseudothermal light source using a laser light and a Mach-Zehnder modulator (MZM) driven by a Gaussian noise generator and a bias Voltage. I discuss a simple classical approach to model the super bunching temporal second-order coherence in devices like MZM. Then, I describe the experimental setup and the procedure for collecting the intensity fluctuation data. In addition, I explain the importance of the bias voltage in achieving the super bunching temporal second-order coherence at a delay time equal to zero. The schematic of the procedure is depicted in Fig. 1.3.

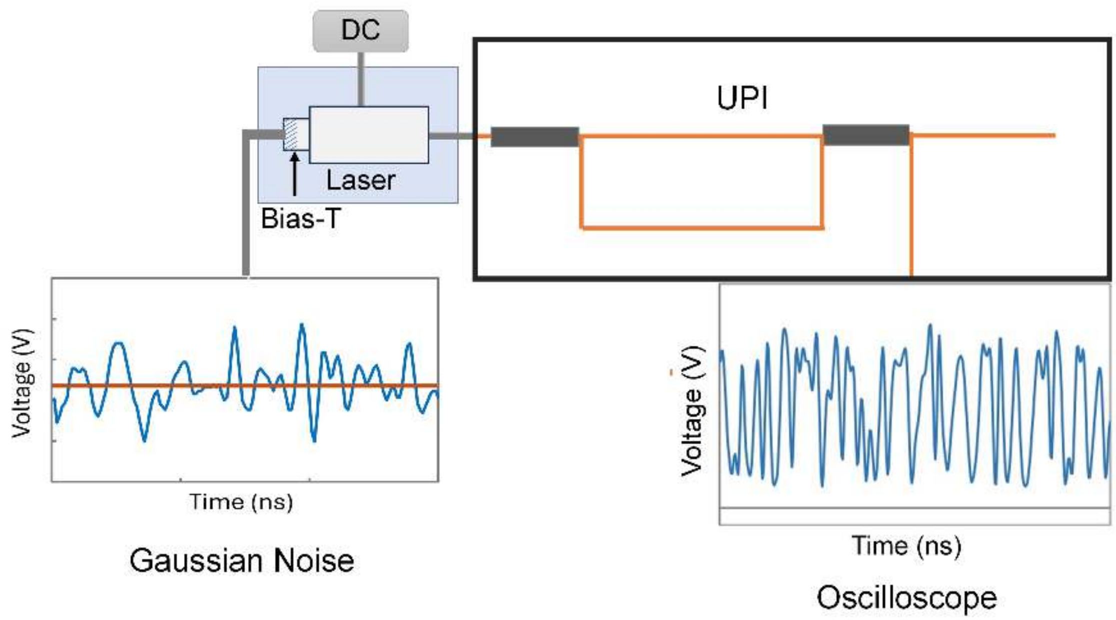


Figure 1.4: DC: Direct current Voltage and UPI: unequal-path interferometer. The schematic diagram shows the procedure to generate intensity fluctuation at the output of UPI when connected to a diode laser modulated by a Gaussian noise generator. The noise generator is connected to the laser using a bias-T. The modulated laser light propagates through the UPI system, and as a result, a fluctuating intensity is observed at the oscilloscope.

Chapter 4: Intensity Correlation Function for Light Passing Through an Unequal-Path Interferometer

In this chapter, I present a technique to modulate a diode laser's frequency by changing its current. I describe how this modulated laser light can be coupled to an unequal-path interferometer (UPI) and obtain intensity fluctuation at the output of the UPI. So, this is a simple device that can convert an optical frequency modulation to an intensity fluctuation. Moreover, the intensity fluctuations measurement can be further implemented to calculate the second-order coherence. The schematic shown in Fig. 1.4 outlines the UPI device mechanism to generate an intensity fluctuation when connected to a modulated diode laser.

Chapter 5: Coherent light propagating in a Multi-Mode Optical Fiber Manifest a Super Bunching Effect with Super Gaussian Intensity Distribution

In Chapter 3, I introduced a new technique for generating a temporal super bunching pseudothermal light in a single-mode fiber. Moreover, in this chapter, I demonstrate a multi-mode fiber-based spatial super bunching second-order correlation function. Optical multi-mode fiber has an intricate composition, yet the cylindrical symmetry, flexibility, and versatility make it an ideal tool for studying optical transmission. I present a simple system where coherent light propagates in a step-index multi-mode fiber, creating speckles with interesting statistics at the exit of the fiber. The input light is not customized as performed by other groups. The ray diagram in Fig. 1.5 displays light propagation in an ideal and a real step-index multi-mode fiber. In this chapter, I explain why the real fibers' behavior differs from the ideal fibers and

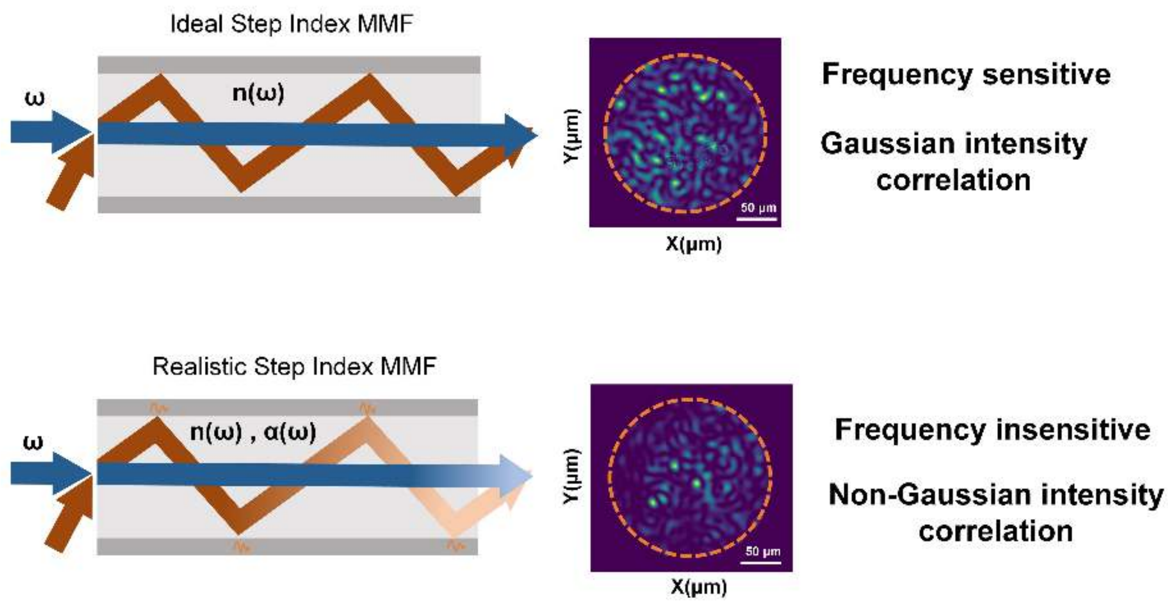


Figure 1.5: The ray diagram illustrates the ideal case and the case when absorption is introduced in a step-index multi-mode fiber. The top figure is the ideal mode representation. When light propagates through a fiber that does not have any material absorption, the speckles generated at the output are more sensitive to the frequency of the laser. There are more bright speckles. Whereas vice versa is observed if the absorption is present in the fiber, less sensitive to incident light frequency, and has fewer bright speckles.

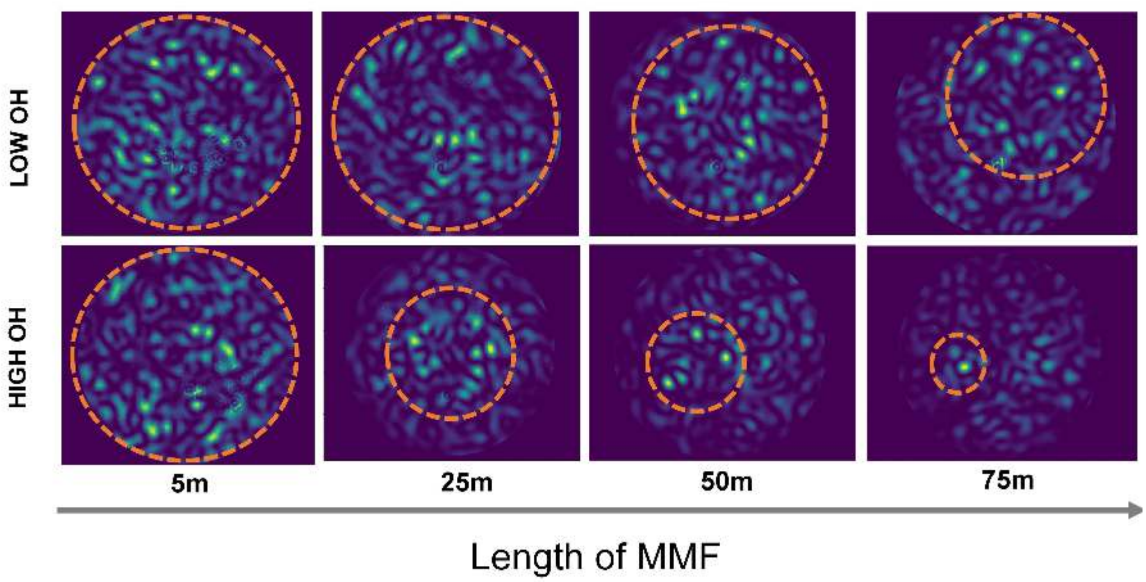


Figure 1.6: OH: Hydroxyl ions. Low OH fibers are fibers with low material absorption, and High OH with high material absorption. The intensity profiles of the speckles recorded at the end of MM fiber are shown for the Low OH (top row) and High OH (bottom row). The speckle images are for the different lengths of fibers: 5m, 25m, 50m, and 75m, respectively. The images display that fibers with longer lengths have fewer bright spots. Also, if material absorption increases, the number of bright spots is further reduced.

how a factor like material absorption introduced as Hydroxyl ions (OH) plays a vital role in obtaining spatially correlated super-Gaussian intensity profile speckle patterns. The images shown in Fig. 1.6 depict the effect of material absorption on the speckle patterns. Finally, in this chapter, I study the intensity distribution and different correlation functions, such as spatial second-order coherence and spectral correlation of the speckle patterns, and deduce their relations with the physical properties of the fiber.

Summary and Future Work

Finally, in this chapter, I summarize all the projects I presented in this thesis and suggest possibilities for future research work.

Chapter 2

Spatial and Temporal Coherence of a Classical Light

In this chapter, I introduce the statistical properties of a thermal light source (TLS) in the context of the second-order coherence-intensity correlation function. This method is further implemented to characterize the pseudothermal light source (PTLS) I have created experimentally in Ch. 3 and Ch. 5. In Ch. 3, I explicitly use temporal second-order coherence to characterize the super bunching PTLS I have realized in a single-mode fiber. Later, in Ch. 5, I execute the spatial second-order coherence of the speckles created at the output of a multi-mode (MM) fiber. It is, therefore, important to understand the coherence properties of a light source, either temporal or spatial. Fundamentally, the first order of coherence can be quantified by calculating the correlations of the electric fields. However, in this thesis, I study the second-order coherence of the light by measuring the intensity correlation functions.

2.1 Coherence

If the attitude (amplitude, phase) of the light has a high degree of coherence, then it is highly correlated at different locations (spatial coherence) or at times (temporal coherence). A high degree of coherence means interference can be easily created and observed. For better understanding, the concept of coherence can be divided into two types: spatial and temporal.

2.1.1 Temporal

Temporal coherence measures the correlation of two points at the same spatial locations but at different times, as shown in Fig. 2.1(b). The temporal coherence describes the time in which the frequency and amplitude of the wave train will remain constant. For an ideal coherent monochromatic light, the coherence time is infinite because the frequency of the light does not change; thus, the phase at any point can be predicted based on the phase of a known arbitrary point. However, in reality, all the light sources have a finite spectral width, which means that the frequency of the wave train is not constant over time. Therefore, the phase can only be predicted for a certain duration for which the frequency of the wave train remains stable. This duration is called coherence time τ_c. Using coherence time, coherence length can also be obtained as the coherence length is $l_c = c\tau_c$. Here, c is the speed of light in the vacuum. This length represents the distance a wave covers while maintaining the same amplitude and frequency. It is the largest optical length difference that is required for the two waves to interfere and form fringes.

The coherence time is also related to the spectral width of the light source. According to the Wiener-Khintchine theorem, the coherence time is given by the inverse of the spectral width as $\Delta\nu = 1/\tau_c$ [6]. Thus, the broader the light source spectrum, the shorter its coherence time.

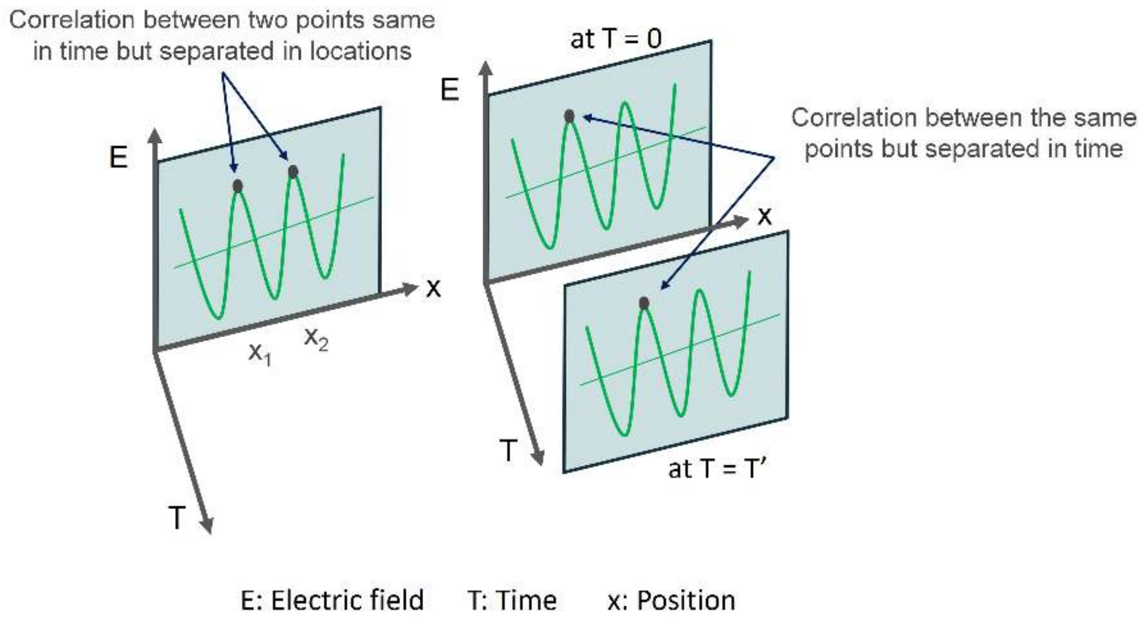


Figure 2.1: The schematic representation of spatial and temporal coherence. The left figure shows the spatial coherence, whereas the right one shows the temporal coherence.

2.1.2 Spatial

Spatial coherence is used to determine the correlation between two phase-related points that are at the same time but are separated in space. Analogous to temporal coherence, spatial coherence is also shown in Fig .2.1(a), which specifies two spatially coherent points located on the same wavefront.

2.2 The degree of second-order coherence

In this thesis, I will only consider intensity correlations and not field correlations. In Ch. 1, I explained how the second order of coherence that plays a significant role in characterizing the light beams can be quantified in terms of intensity correlation functions. In this chapter, I describe the thermal light bunching and super bunching regimes. Later, in this thesis, the temporal second-order coherence is implemented in Ch. 3 to characterize the super bunching pseudothermal source in a single-mode fiber, and the spatial second-order coherence is implemented in Ch. 5 to analyze the super bunched speckle patterns at the output of the multi-mode Fiber.

2.3 Classical Regime: Bunching and Super-Bunching

For a classical light, the $g^{(2)}(\tau)$ is also expressed in terms of intensity variance by defining the intensity as $I(t) = I + \delta\ I(t)$. The mean of $\delta\ I(t)$ is zero. Using this definition of intensity, I classify the thermal light source $g^{(2)}(\tau = 0)$ equal to 2 and $g^{(2)}(\tau = 0) > 2$ regimes. The normalized auto-correlation function of the intensity fluctuation measured as a function of time at $\tau = 0$ as

$$g^2(\tau = 0) = \frac{\langle I(t)^2 \rangle}{\langle I(t) \rangle^2}. \tag{2.1}$$

Now, using the above definition of intensity, I get

$$\begin{aligned} g^2(0) &= \frac{\langle (I + \delta I(t))^2 \rangle}{\langle I + \delta I(t) \rangle^2} = \frac{\langle I \rangle^2 + \langle \delta I(t)^2 \rangle}{\langle I \rangle^2} \\ &= 1 + \frac{\langle \delta I(t)^2 \rangle}{\langle I \rangle^2}. \end{aligned} \tag{2.2}$$

here $\langle \delta I(t)^2 \rangle$ at $\tau = 0$ is the variance of the intensity fluctuation, which always gives a positive value. Therefore, in the normalized form using Eq. 2.1, $g^{(2)}(\tau)$ for classical sources like thermal light is present within the region $1 \leq g^{(2)}(\tau) \leq \infty$ and $g^{(2)}(\tau = 0)$ is expressed as $1 \leq g^{(2)}(\tau = 0)$ and $g^{(2)}(\tau) \leq g^{(2)}(\tau = 0)$.

The bunched thermal or pseudothermal light source has the second-order correlation, $g^{(2)}(\tau = 0)$ equal to 2 [7] and can be explained in terms of intensity fluctuation using Eq. 2.2.

This characteristic is observed when the fluctuations in the intensity of the light reach the size of the average intensity value or when the variance of the intensity, $\langle \delta I(t)^2 \rangle$ is equal to the square of the mean intensity, $\langle I \rangle^2$. If the fluctuation in the intensity becomes larger than the average intensity value, then the $g^{(2)}(\tau = 0) > 2$, and such light sources are called super bunching thermal light.

Coherent and thermal light intensity fluctuations with their second-order correlation functions are shown in Fig. 2.2.

Different Types of Light:

2.4 Laser Light

Laser light is a coherent light source that emits mostly monochromatic light. The light wave has a fixed frequency, amplitude, and a constant relative phase. In an ideal laser, the stable phase gives an infinite coherence length (l_c) and time (τ_c). The intensity has no fluctuations and thus is constant over time. Of course, an ideal laser does not exist. In practice, the laser's emission spectrum is broadened around the center frequency, i.e., the spectrum has a finite bandwidth. This finite bandwidth leads to a finite coherence length and time, which differs for different types of lasers.

2.5 Thermal Light

The fluctuations in a thermal light source can be explained using Loudon's model for chaotic light sources. Loudon, in 1983 [6] proposed a theory in which excited atoms

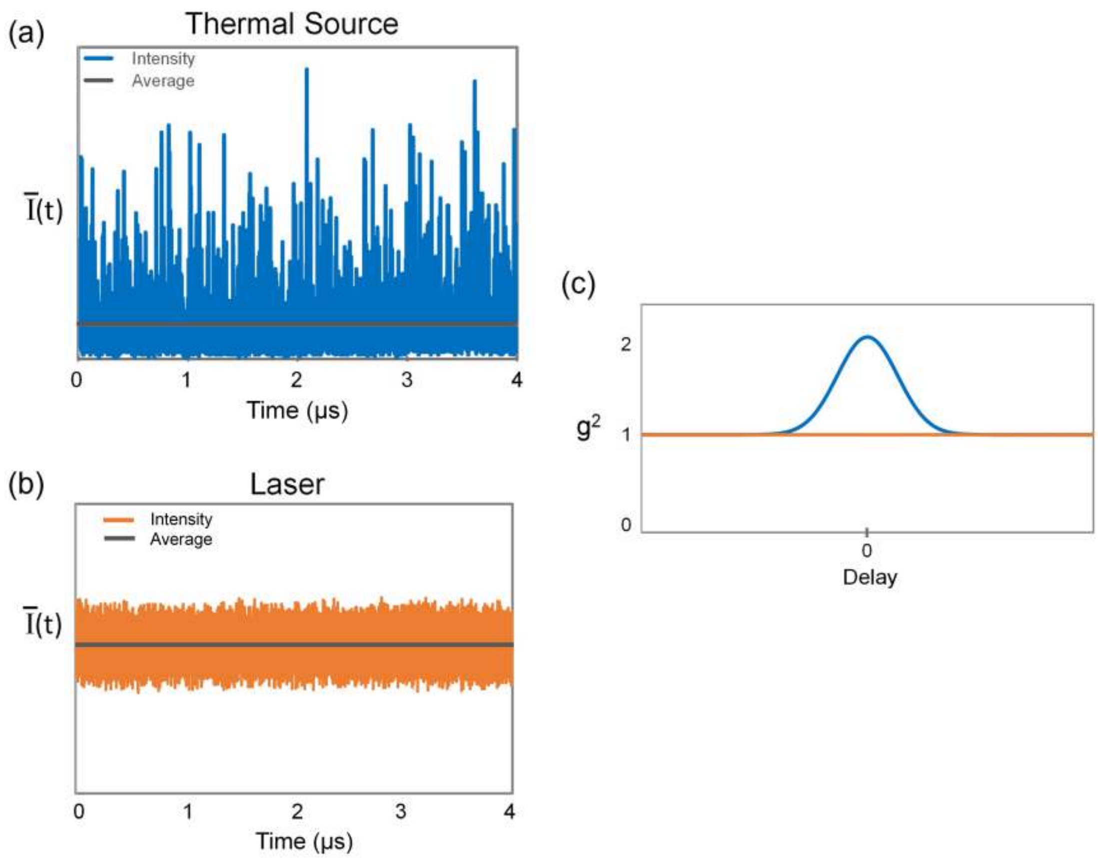


Figure 2.2: (a) The graph shows the intensity time series of an ideal thermal source. (b) The graph shows the intensity time series of a coherent laser light. (c) The $g^{(2)}(\tau)$ plot for the thermal light (blue) with the resulting value of $g^{(2)}(\tau = 0) = 2$ and the laser light (orange) with $g^{(2)}(\tau) = 1$ as such coherent sources have no correlations between the intensity fluctuations.

represented a collision-broadening effect; those atoms radiated at a single frequency ω_o and obtained a random phase shift each time a collision occurred. Hence, the phases of the field before and after collision are no longer associated with each other. However, the phase remains constant between the collisions, and the average time between the collisions gives the temporal coherence τ_c. This model correlates to the collision-broadened thermal light source, thus considers the collision broadening as a predominating and neglects the broadening from the radiation and Doppler effects. The concept behind the chaotic or thermal light can be explained using Loudon's atoms collision model, where an electric field of a single atom written in the complex form as

$$E(t) = E_{\mathrm{o}} e^{(-i\omega_{\mathrm{o}} t + i\varphi(t))}. \tag{2.3}$$

where, amplitude E_{o} and the frequency ω_o of the radiated light is constant for every period. However, the phase φ(t) remains constant during the period of free flight but changes abruptly each time a collision occurs. The total electric-field (E_T) emitted by ν number of radiating atoms is given as

$$\begin{aligned} E_T &= E_1(t) + E_2(t) + \ldots .... + E_\nu(t) \\ &= E_o e^{-i\omega_o t} a(t) e^{i\varphi(t)}. \end{aligned} \tag{2.4}$$

Here, φ(t) and the amplitude $a(t)$ are random variables. I adopt the ergodic convention for such thermal sources. Hence, the time average of the radiated field emitted by an atom is equal to the average of the ν atoms or the average across the ensemble. The electric field from Eq. 2.4 has a zero temporal average over one period. Taking the real part of the electric field in Eq. 2.4, the average of the intensity over one full cycle at frequency ω_o is given by

$$\bar{I}(t) = \frac{1}{2}\epsilon_0 c |E_T(t)|^2 = \frac{1}{2}\epsilon_0 c {E_o}^2 a(t)^2. \tag{2.5}$$

where c is the speed of light in vacuum, and ε_0 is permittivity in air. The over-bar notation is for the average. The intensity average in above Eq. 2.5 depends only on the amplitude modulations, and it remains approximately constant for $t \ll \tau_c$ while has large fluctuations for $t \gg \tau_c$ [6]. If there is a long observation duration, i.e., $T \gg \tau_c$ and the instantaneous intensity measurement assumptions imply, then the time average intensity can be replaced by a statistical average as

$$\begin{aligned} \bar{I}(t) = \langle I(\bar{t}) \rangle &= \frac{1}{2}\epsilon_0 c {E_o}^2 \langle |exp(i\varphi_1(t)) + exp(i\varphi_2(t)) + ... + exp(i\varphi_\nu(t))|^2 \rangle \\ &= \frac{1}{2}\epsilon_0 c {E_o}^2 \nu. \end{aligned} \tag{2.6}$$

The cross terms of the phase factors from different radiating atoms get canceled; thus, those terms have zero contribution. Now, the mean square of the intensity is given as

$$\begin{aligned} \langle I(\bar{t})^2 \rangle &= \frac{1}{4}{\epsilon_0}^2 c^2 {E_o}^4 \langle |exp(i\varphi_1(t)) + exp(i\varphi_2(t)) + ... + exp(i\varphi_\nu(t))|^4 \rangle \\ &= \frac{1}{4}{\epsilon_0}^2 c^2 {E_o}^4 \Bigg(\sum_i \langle |exp(i\varphi_i(t))|^4 \rangle + \sum_{i>j} \langle |2exp[i(\varphi_i(t) + \varphi_j(t))]|^2 \rangle \Bigg) \\ &= \frac{1}{4}{\epsilon_0}^2 c^2 {E_o}^4 (\nu + 2\nu(\nu - 1)). \end{aligned} \tag{2.7}$$

Here, the first term represents the contributions from the individual atoms, whereas the second term is for pairs of atoms. Factor 2 in the second term accounts for the number of occurrences for pairs i and j. Now, using Eq. 2.6 and Eq. 2.7 we get the second-order temporal coherence at $\tau = 0$ to be

$$g^2(\tau = 0) = \frac{\langle I(\bar{t})^2 \rangle}{\langle I(\bar{t}) \rangle^2} = \left(2 - \frac{1}{\nu} \right). \tag{2.8}$$

The number of radiating atoms is large in numbers; hence, ν is very large and therefore, Eq. 2.8 to a good approximation will become

$$g^2(\tau = 0) = 2. \qquad (\nu \gg 1) \tag{2.9}$$

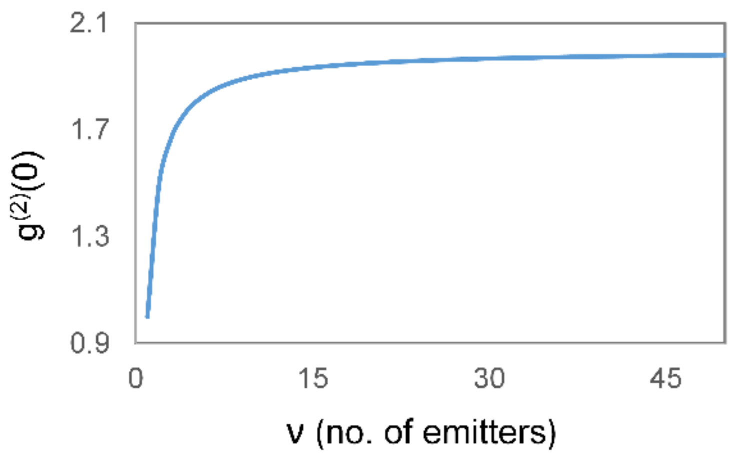


Figure 2.3: The plot showing the number of emitters vs second order coherence $g^{(2)}(0)$. For two emitters, $g^{(2)}(0)$ is equal to 1.5, and for a large number of emitters, it reaches the value 2.

as shown in Fig. 2.3.

Consequently, Loudon's collision broadening model gives the thermal light source's bunching effect. Therefore, Loudon's collision-broadening model is valid for any chaotic light source, whether thermal or pseudothermal. Using this model, the field distribution can also be established. This model states that the colliding atoms radiate the electromagnetic fields with the same frequency but with random phases. The superposition of such radiated fields of many atoms results in a Gaussian distribution, which corresponds to an intensity distribution given as

$$P(I) = \frac{1}{\bar{I}} exp\left[-\frac{I}{\bar{I}}\right]. \tag{2.10}$$

Therefore, the bunching thermal light fits a negative exponential probability distribution.

2.6 Pseudothermal Light with $g^{(2)}(0) = 2$

The true TLS coherence time is associated with their intrinsic photon emission process, making it hard to control and use for applications. As discussed in Sec. 2.1.1, we can use a spectral filter to increase the coherence time. However, it is difficult to reach a regime of τ_c larger than a few picoseconds (10^{-12} s) without a huge loss of intensity. A pseudo-thermal light source (PTLS) is an easier way to proceed. The PTLS has a longer coherence time compared to a true TLS, but it allows for intensity correlation measurements. Similar to TLS, the intensity distribution of PTLS is a negative exponential. The second-order coherence of bunching PTLS is equal to 2. Some prominent examples of the realization of PTLS are using a rotating ground glass disc (RGGD) [9] and an optical multi-mode (MM) fibers [28].

2.6.1 Rotating Ground Glass

As shown in Fig. 2.4, a RGGD experiment is constructed by focusing a single-mode laser light with a stable amplitude on a moving ground glass disc that acts as a scatterer [9]. The grainy surface of the grounded glass provides many scattering centers that, when illuminated by a laser beam, act as independent point-like sub-sources of a TLS. The random phase shift acquired in each scattering process results in intensity distribution caused by the interference. The intensity distribution can be observed as speckle patterns in the far field. The size of the grains of the speckles is determined by the spot size of the laser beam used to illuminate the RGGD. These patterns can be stationary or dynamic depending on the movement of the disc. If the RGGD is static, the speckle patterns remain static; however, if the disc is rotated, different scattering centers are illuminated, resulting in non-stationary or temporarily

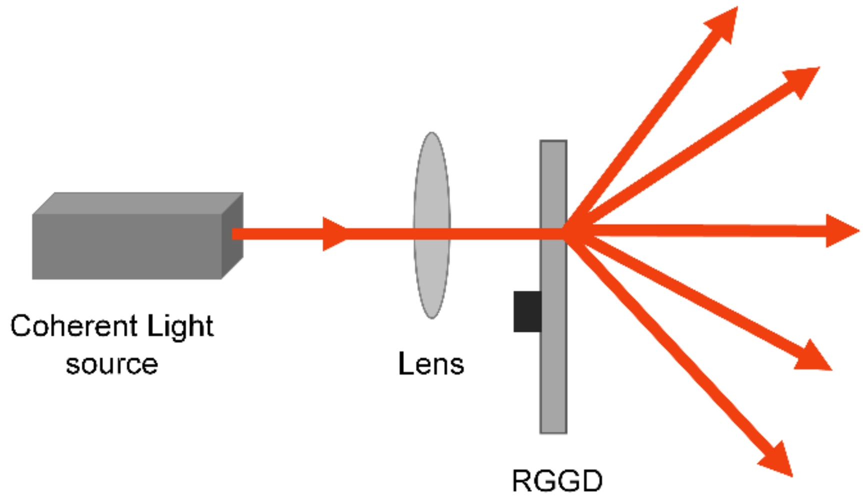


Figure 2.4: RGGD: Rotating Ground Glass Disc. The schematic diagram to generate pseudothermal light using a rotating ground glass disc.

changing speckles. The phases of the respective radiated light yielding to the same speckle are related and thus are coherent.

In contrast, the radiation from different speckles is not phase-related and thus is spatially incoherent. That being so, the temporal coherence of the RGGD is primarily determined by the disc's rotation speed, which is easy to control. The coherence time using RGGD is within the range of microseconds (10^{-6} s) to milliseconds (10^{-3} s) [9]. Also, a spectral filter must be used for the true TLS to achieve a practical coherence time. It is unnecessary for the PTLS as they already emit intensity values within the narrow bandwidth. This is an important advantage of using PTLS over TLS as it uses a laser light for illumination, which can be considered monochromatic.

2.6.2 Optical Multi-Mode Fiber

Another method to realize a PTLS is using an optical multi-mode fiber [28]. A coherent light source is coupled into a MM fiber; the light is guided inside the fiber by a total internal reflection as shown in Fig. 2.5. The actual geometrical parameters of the fiber define the optical path length of the light inside the fiber. When light propagates through the fiber, each mode of the fiber acquires a random phase with respect to the other, imitating the behavior of many independent emitters of a TLS. A randomized grainy speckle pattern is observed in the far field at the output of the MM fiber due to the interference between the different modes. The size of the smallest grains of the speckle at the output of the fiber can be evaluated using the interference patterns of the two counter-propagating rays inside the fiber at a critical angle [29] as shown in Fig. 2.5. Using the grain size and fiber physical parameters, the number of sub-sources and, finally, the number of modes of the MM fiber can be approximated.

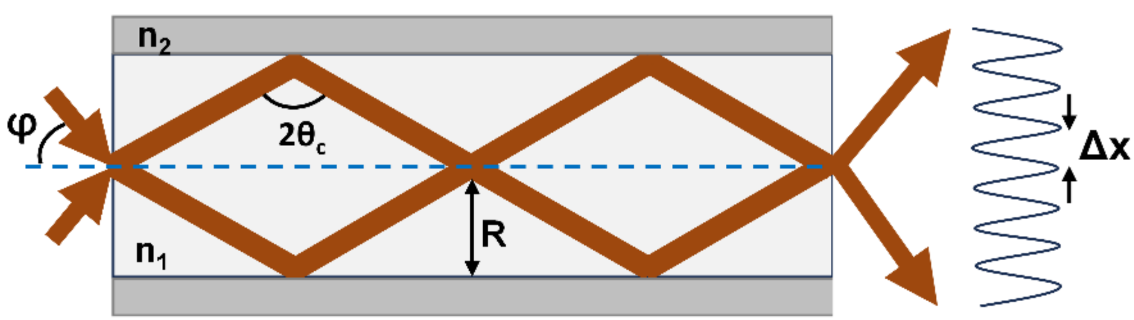


Figure 2.5: Light propagation in MM fiber with fringes at the output of the fiber. R is the radius of the fiber core, n_1 and n_1 are the refractive indices of the core and cladding of the fiber, respectively. The angles φ and θ_c are the incidence and critical angles, respectively.

The fringe spacing at the output, as shown in Fig. 2.5, can be estimated empirically as

$$\Delta x \approx \frac{\lambda \; NA^{-1}}{2}. \tag{2.11}$$

where λ is the wavelength of the laser light used, and NA is the fiber's numerical aperture. Now, using the Δx, the number of sub-sources can be estimated as

$$M_s = \frac{R^2}{\Delta x^2} = \left(\frac{2R \; NA}{\lambda}\right)^2 \tag{2.12}$$

where R is the radius of the fiber core. Also, from [30], we have a total number of modes given as

$$M_m = 2\left(\frac{\pi R \; NA}{\lambda}\right)^2 \tag{2.13}$$

Finally, using Eq. 2.12 and Eq. 2.13 relation between the number of sub-sources and modes of the MM fiber yields

$$M_m = \frac{\pi^2}{2} M_s \tag{2.14}$$

The MM fiber can also have static and varying speckles similar to the rotating glass PTLS. If the MM fiber is fixed, the optical path of each mode will remain fixed, and thus, one can observe a stationary speckle behind the fiber exit. However, if a vibration is applied to the fiber by shaking it, the phases of the modes change due to the geometrical shift, resulting in varying speckle patterns. Changing the frequency of the incoming laser light and keeping the MM fiber fixed is another way to change the speckle patterns. The coherence time can be determined using the characteristic time of the dynamic speckle pattern, i.e., the time scale at which these patterns change. MM fiber PTLS is less preferable for temporal correlation experiments than the RGGD as it is harder to control the coherence time in the MM fiber. The vibration can be automatized mechanically; however, the coherence times obtained are not as stable as the RGGD.

2.7 Pseudothermal Light with $g^{(2)}(0) > 2$

More scientists worldwide have shown interest in super-bunching pseudothermal light sources in the last decade. The scattered photons from rotating ground glass or the MM fiber PTLS are bunched, and the $g^{(2)}(\tau = 0)$ equals 2. However, the Eq. 2.2 in Sec. 2.3 validates that a chaotic light source can have intensity fluctuations much larger than the mean intensity. Zhou et al. in 2017 experimentally generated such light source that gave a $g^{(2)}(\tau = 0) > 2$ [16]. The name given to such a light source is a super-bunching pseudothermal light. In Ch. 3, I demonstrate a technology to create such a light source in the lab with simple and low-cost equipment. Two approaches have been applied to realize such light sources in the past. Method I, where more than one RGGD is used with an invariant laser light to illuminate the disc, whereas Method II, only one RGGD is used, but the laser light is intensity-modulated using an electronic or acoustic modulator before illuminating the RGGD. Both methods are discussed in detail below.

2.7.1 N number of RGGD using invariant incident laser light

As shown in Fig. 2.6, this approach first creates the pseudothermal light using the rotating ground glass described in Sec. 2.6 and using an invariant laser light. Then, the scattered light from the first rotating glass, RGGD_1, is filtered out using a

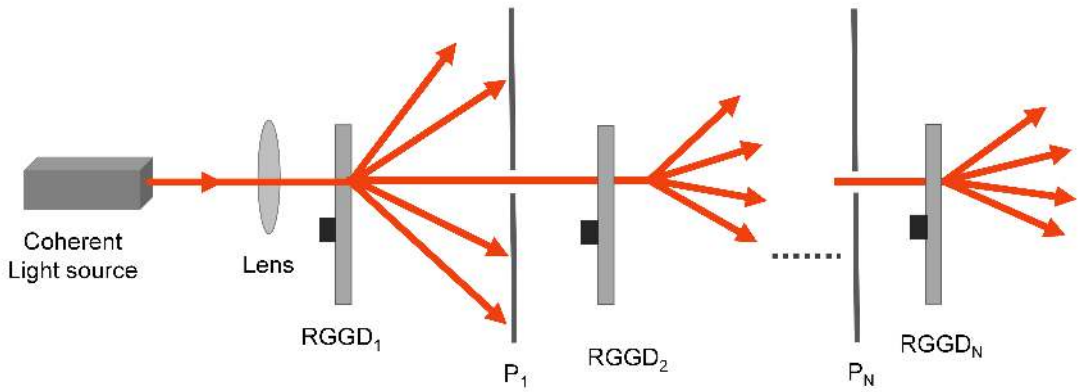


Figure 2.6: RGGD: Rotating Ground Glass Disc, P: Pinhole. 1, 2, and N refer to the numbers of RGGD or pinholes used for the experiment. The schematic diagram to generate pseudothermal light using a rotating ground glass disc.

pinhole with a diameter less than the coherence length l_c. The light passed through the pinhole is then focused and incident onto another rotating glass, $RGGD_2$. As the incident light is coherent, interference will occur after scattering from $RGGD_2$. Likewise, another pinhole and rotating glass can be kept after $RGGD_2$, and the process can be repeated with N number of pinholes and rotating glasses. Implementing such a setup will give a super-bunched pseudothermal light after N ($N \geq 2$) number of RGGDs [16, 17].

2.7.2 One RGGD with varying incident Laser Light

Super bunching pseudothermal light is also created using intensity-modulated laser light and rotating ground glass, as shown in Fig. 2.7. Proper intensity modulation of the incident laser light before the RGGD can change the temporal coherence properties of the scattered light of the RGGD. The laser light can be modulated using an electromagnetic optical modulator (EOM) [18, 19] or an acoustic-optical modulator

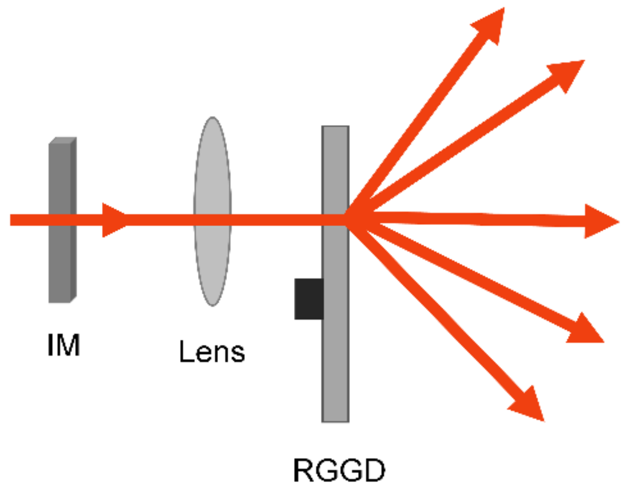


Figure 2.7: RGGD: Rotating Ground Glass Disc, IM: Intensity Modulator. The schematic diagram to generate a super bunching pseudothermal light using a modulated incident laser light, which is here referred to as IM, and rotating ground glass disc.

(AOM) [27]. The intensity of the laser light modulated using a white noise [18] or a binary distribution [31] changes the statistics of the light, and when such light illuminates the RGGD, the scattered light exhibits a super-bunching effect with the $g^{(2)}(\tau = 0)$ values greater than 2.

Spatial coherence can also be changed to achieve $g^{(2)}(\Delta r = 0)$ value higher than 2. Hong [32] proposed a super bunching PTLS by controlling the photon fluctuations of the light source using a spatial modulator. They modulated the laser light at a kiloHertz speed to obtain the desired photon statistics at the output, which gave the $g^{(2)}(\Delta r = 0)$ value higher than 2. Therefore, a spatially super bunched light was created.

2.8 Applications

Soon after the invention, both the bunching and super-bunching PTLS were extensively used for applications in the science and engineering world. These light sources can be used to further study and develop the coherence theory. They are used to measure intensity correlations and understand the physics of the second and higher-order intensity correlation functions [16, 27, 32]. The bunching PTLS is used for ghost imaging [20–24] and super-bunching PTLS is used to increase the visibility of the ghost-imaging [31].

Ghost Imaging

Ghost imaging is an interesting technique for acquiring an image of an object by correlating the intensity measurement of two light beams from two detectors [33–35]. Only one beam views the object and measures the intensity using a single-pixel detector. The other beam never interacts with the object and is detected by a multi-pixel detector. The image of the object is reconstructed by combining the information of both detectors. For a long time, it was considered a quantum correlation phenomenon as neither of the two beams independently carried the information of the object to reproduce the image. Moreover, in 1995, Pittman et al. experimentally demonstrated ghost imaging using entangled photon pairs [33]. However, after a few decades, researchers showed that quantum properties of the light beams were not necessary and ghost imaging can be performed using purely classical light in the spatial domain [35–38] or in time domain [39].

The schematic description in Fig. 2.8 is a standard spatial ghost imaging procedure using a classical incoherent light source, for example, a pseudothermal light [36]. The

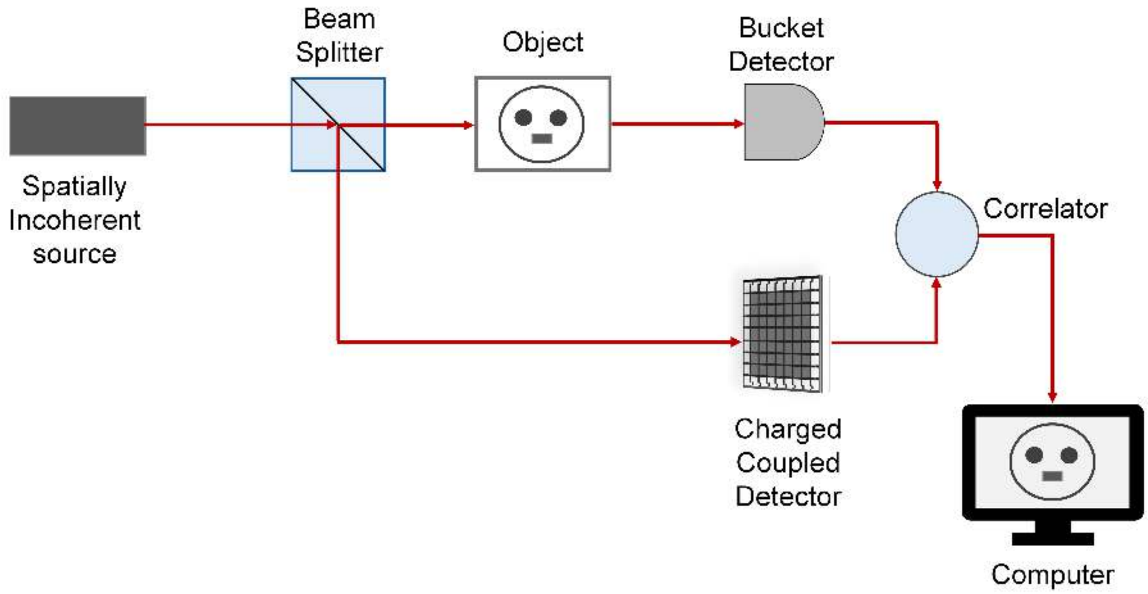


Figure 2.8: The schematic diagram for spatial ghost imaging. A spatially incoherent light source is divided by a beamsplitter. The transmitted light from the beamsplitter is sent toward a spatial object (a mask), where the light is spatially modulated and measured by a bucket detector, which cannot produce the image by itself. The reflected beam from the beamsplitter is sent to a charged coupled detector (CCD), which records a copy of the noisy pattern. The measured signals from the bucket and charged coupled detectors are sent to a computer where the correlation function is calculated. Finally, repeating this process several times with modulated spatial patterns leads to a ghost image.

pseudothermal light generates a random spatial pattern, which gets divided into two copies using a beamsplitter. One copy is used to illuminate the object (a mask), and the total amount of transmitted light from the object is collected using a bucket (single pixel) detector. The other copy is directly sent to a charged coupled (multi-pixel) detector (CCD) without interacting with the object. The CCD records the complete spatial profile of the random spatial pattern. The measurements from both detectors are sent to the computer, where the correlation function is applied. This

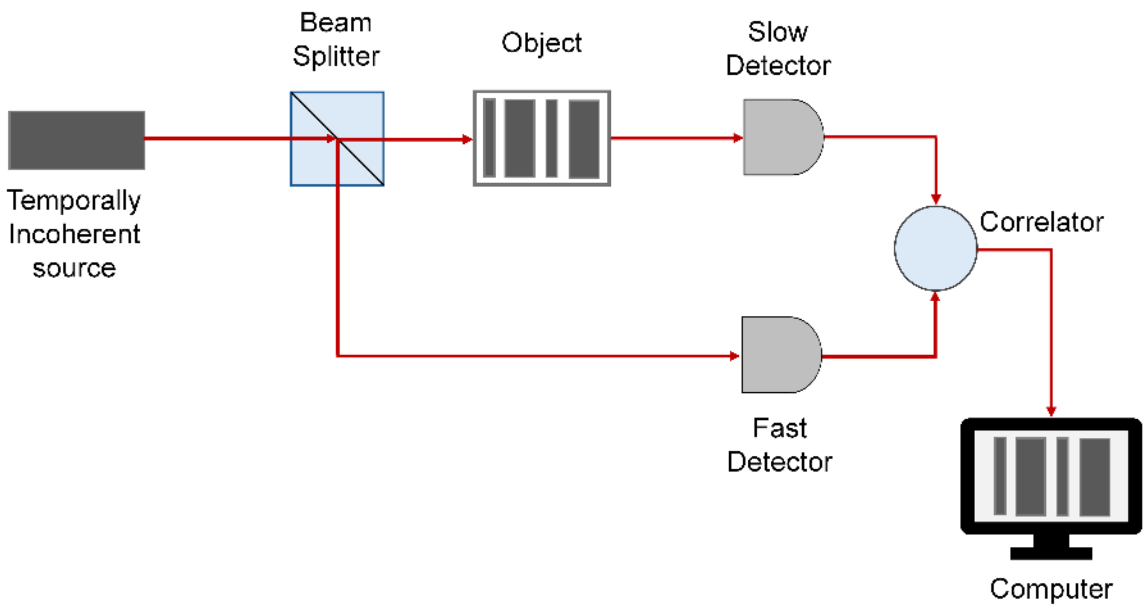


Figure 2.9: The schematic to demonstrate the principle of temporal ghost imaging using a classical incoherent light source. A beamsplitter divides the light from a temporally incoherent source into two parts. The transmitted light beam is sent to an object (some temporal sequence). The light beam is temporally modulated, and the measured signal is detected using a slow detector. The detector cannot resolve the pattern by itself. The reflected light beam from the beamsplitter is sent to a fast detector, which records a copy of the noise signal. Both detectors measured signals and sent them to a computer where the correlation is calculated. Similar to spatial ghost imaging, by repeating this process with several noise ensembles, the modulated temporal image is reconstructed, called a temporal ghost image.

procedure is repeated several times with different random spatial patterns, and finally, each realization is added to construct a final spatial ghost image of the object.

The time domain ghost image was created by Ryczkowski et al. in 2016 [39] using a similar method of spatial ghost imaging as shown in Fig. 2.9. The object, however, was a temporally varying signal generated by an electro-optic modulator. The light source is a temporally incoherent classical source; therefore, spatial random patterns are replaced by temporal random patterns. Like spatial imaging, the temporal random patterns are split into two copies using a beamsplitter. One copy is sent to the object, and the transmitted intensity is measured using a slow detector (photodiode). At the same time, the other copy of the temporal random pattern is sent to a fast detector (photodiode) without any interaction with the object. The measurements from two detectors are sent to a computer, where the correlation function is calculated. Finally, the temporal ghost is created by repeatedly repeating this procedure with varying temporal patterns.

Ghost imaging can help create high-quality 3D object images using a few single-pixel detectors. As ghost imaging uses the intensity measurements from the detectors, the distortion of the signals does not affect the final image until the total intensity measurement data does not vary over the total acquisition time.

2.9 Summary

In this chapter, I reintroduce the coherence theory. Important tools, like the degree of second-order coherence, are discussed in detail. As indicated above, in the next chapters, I will demonstrate the experimental setup and intensity measurements for temporal and spatial second-order coherence with a super-bunching effect in a single-mode and a multi-mode fiber, respectively. The calculation of $g^{(2)}(\tau = 0)$ in Ch. 3 and $g^{(2)}(\Delta r)$ in Ch. 5 will also be shown explicitly.

Chapter 3

Technology to Generate: Super bunching Pseudothermal Light in a Single-Mode Fiber

3.1 Introduction

In this chapter, I present an alternate method to create a pseudothermal light source (PTLS). It is a cost-effective and straightforward technique to generate a super-bunching PTLS in the lab. I implement the temporal second-order coherence to characterize the light source. The chaotic intensity fluctuation is created using a laser and a Mach Zehnder modulator (MZM), i.e., modulated by a Gaussian noise - a fast voltage and a bias - a slow voltage as shown in Fig. 3.1. By varying the noise amplitude and bias voltage, I can change the correlation of the intensity fluctuations at the output of the MZM. As a consequence, this technique allows to tune the second order of coherence $g^{(2)}(\tau = 0)$.

Using this system, I get a minimum value of $g^{(2)}(\tau = 0)$ equal to 1.0 ± 0.01, a coherent source to 2.0 ± 0.01, a bunching thermal source to finally 2.86 ± 0.01, a super bunching thermal source. The minimum value of $g^{(2)}(\tau = 0)$ equal to 1.0 is achieved in the middle of the fringe, whereas the maximum value of 2.86 ± 0.01 is realized at

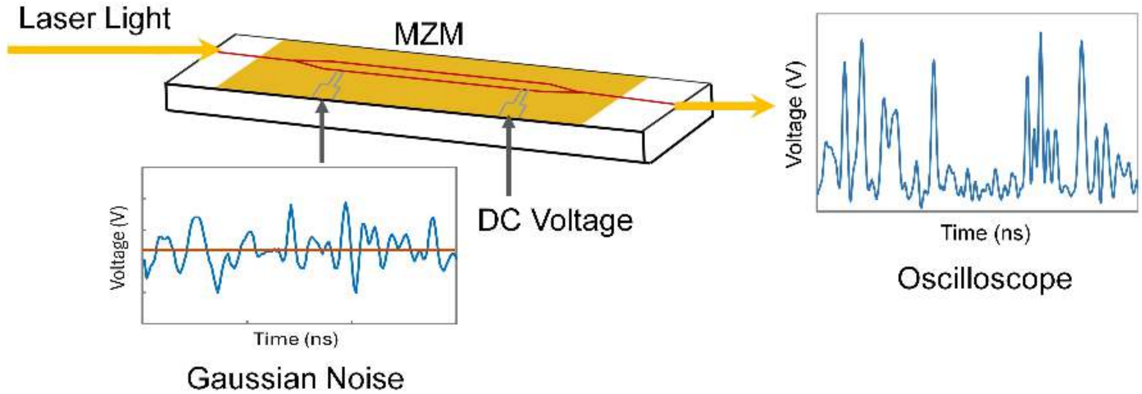


Figure 3.1: MZM: Mach Zehnder Modulator. The schematic of intensity fluctuation at the output of MZM when it is operated by a Gaussian noise and a bias voltage.

the bottom of the fringe. Therefore, I demonstrate that by moving from the middle to the bottom of the fringe using a biased voltage, I am able to attain a coherent to a super-bunch pseudothermal light. Also, the values of $g^{(2)}(\tau = 0)$ have low uncertainty compared to the values reported by different groups [16, 18, 19]. My setup is simpler than the recent ones used to realize the super-bunching pseudothermal light. It is redundant to make the setup complex by adding more rotating ground glasses [16] or modulating the incident laser light with a high voltage [18, 19, 27]. To resolve the uncertainty problem in $g^{(2)}(0)$ by adding rotating ground glasses (RGG) [16], Liu et al. [19] varied the incident laser light before illuminating the light on the RGG. However, this technique is still complex and requires high voltage to modulate laser light using an electronic optical modulator (EOM). Instead, one can use the super-bunching light created using my technique, incident on the RGG, for applications like temporal ghost imaging [40] to increase their quality.

I also obtain the coherence time of the super-bunching PTLS to be 10^{-8} to 10^{-6} s; this is very short compared to 10^{-5} to 1 s achieved by other research groups. Also, all the methods mentioned above are conducted in free space, whereas in my setup, the laser light beam is directed through a single-mode fiber, which comparatively helps to collect more light at the detection point. Finally, I use a simple classical mathematical concept to explain the approach.

The rest of the chapter is categorized as follows: In Section II I discusse the modified second-order correlation function using Gaussian noise and biased voltage that is used to characterize the super bunching pseudothermal source. Then I use Loudon's two-emitter model to explain the theory. In section III I demonstrate the experimental setup with schematic diagrams and in Section IV I cover the experimental results discussion. Finally, in section VI I summarize with the conclusion.

3.2 Mach-Zehnder Modulator

The MZMs are waveguide-based devices meant for very high-speed modulation for telecommunication. The schematic of MZM shown in Fig. 3.2 is a y-coupler modulator. The input electric field E_{in} enters the modulator, gets divided into two paths, and travels down between the two electrodes. These devices are made up of electro-optic materials, implying that the field inside the modulator undergoes a phase shift if a voltage is applied. Like in Fig. 3.2, when a voltage of φ_o is applied across the top electrode, keeping the bottom electrode constant, constructive or destructive interference occurs accordingly at the output. If the two fields passing through the two electrodes create a superposition field, E_{out} as in Eq. 3.2, then constructive interference is observed at the output, whereas destructive interference is observed if the light

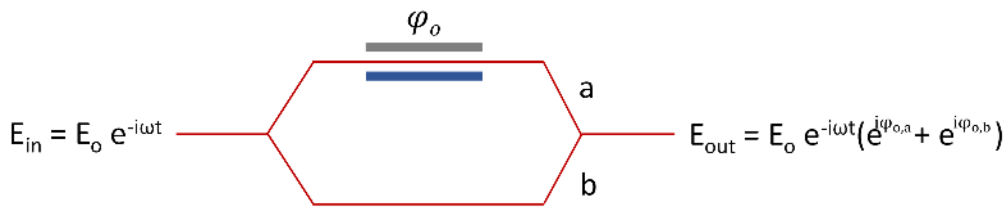


Figure 3.2: Schematic of a Mach-Zehnder Modulator.

gets scattered inside the modulator. The input and output fields of the modulator can be given as

$$E_{in} = E_o e^{-i\omega t}. \tag{3.1}$$

and

$$E_{out} = E_o e^{-i\omega t}(e^{i\varphi_{o,a}} + e^{i\varphi_{o,b}}). \tag{3.2}$$

where, E_{in} is the light sent into the modulator, and E_{out} is the light at the output of the modulator. E_o is the amplitude of the electric field and ω is the frequency of the light source used. $\varphi_{o,a}$ and $\varphi_{o,b}$ are the phases of the field passed through the arms a and b. The intensity of such field is given as

$$I_{out} = I_{in} \cos^2\left(\frac{\varphi_o}{2}\right). \tag{3.3}$$

I_{in} and I_{out} are the input and output intensities of the light of the modulator. The term φ_o gives the phase shift. Using the Eq. 1.2 from Ch. 1 and the MZM intensity, $g^{(2)}(\tau = 0)$ for the MZM can be derived as

$$g^{(2)}(\tau = 0) = \frac{\langle (I_{in} \cos^2(\frac{\varphi_o}{2}))^2 \rangle}{\langle I_{in} \cos^2(\frac{\varphi_o}{2}) \rangle^2}. \tag{3.4}$$

If φ_o uniformly sweeps over 0 to 2π then I can integrate $\cos^2(\varphi_o/2)$ over 0 to 2π which is 1/2. Likewise, $\cos^4(\varphi_o/2)$ can also be integrated over 0 to 2π which will equate to 3/8. Plugging these values in Eq. 3.4, I obtain the $g^{(2)}(\tau = 0)$ for the MZM to be 1.5. Later, in Sec. 3.4, I will show how the MZM $g^{(2)}(\tau = 0)$ is analogous to Loudon's two emitters model.

3.3 Super bunching pseudothermal light using MZM

The MZM I use for my experiment allows me to change the phase of the input field and achieve a superposition of the fields at the device's output. I do this by driving the MZM with a Gaussian noise source connected to one of the MZM electrodes and a DC voltage to another. The noise generator produces random phase shifts in the field, whereas DC bias voltage is used to produce slow changes. Using both of these provisions, i.e., the fast and the slow variations of the voltages, I am able to achieve $g^{(2)}(\tau = 0)$ values equal to 1 for the lower end and 3 for the higher end. As the value of $g^{(2)}(\tau = 0)$ reaches higher than 2, the light created in my experiment is a super bunching pseudothermal light. To demonstrate the super bunching second-order of coherence in my experiment and to find the exact theory, I incorporate the Gaussian noise probability density,

$$P(V) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-V^2}{2\sigma^2}}. \tag{3.5}$$

into the Eq. 3.4 and finally achieve an expression for $g^{(2)}(\tau = 0)$ as

$$g^{(2)}(\tau = 0) = \frac{\frac{3}{2} + 2\cos(\varphi_o)e^{\frac{-\sigma^2\pi^2}{2}} + \frac{1}{2}\cos(2\varphi_o)e^{-2\sigma^2\pi^2}}{\left[1 + \cos(\varphi_o)e^{\frac{-\sigma^2\pi^2}{2}}\right]^2}. \tag{3.6}$$

where σ^2 is the variance, and σ is the standard deviation of the Gaussian noise probability density function. Now, using Eq. 3.6, I estimate the $g^{(2)}(\tau = 0)$ for different values of φ_o and variance.

3.3.1 Top of the fringe: φ_o equals to zero

When the Gaussian noise is applied at the top of the fringe, i.e., φ_o equals zero, the expression for $g^{(2)}(\tau = 0)$ using the Eq. 3.6 is

$$g^{(2)}(\tau = 0) = \frac{\frac{3}{2} + 2e^{\frac{-\sigma^2\pi^2}{2}} + \frac{1}{2}e^{-2\sigma^2\pi^2}}{\left[1 + e^{\frac{-\sigma^2\pi^2}{2}}\right]^2}. \tag{3.7}$$

For two limits of variance, I find the $g^{(2)}(\tau = 0)$ as

Case I: $\varphi_o = 0$ and $\sigma = 0$

$g^{(2)}(\tau = 0) = \frac{\frac{3}{2}+2+\frac{1}{2}}{[2]^2} = 1.$

Case II: $\varphi_o = 0$ and $\sigma = \infty$

$g^{(2)}(\tau = 0) = \frac{\frac{3}{2}}{1} = 1.5.$

These values can be explained in terms of the variance of the intensity fluctuations, as shown in Fig. 3.3. When a Gaussian noise with a low noise amplitude (orange) is applied, the intensity fluctuations produced at the output of the MZM have a variance equal to zero. As implied from Ch. 2, Eq. 2.2 if the variance of the intensity is zero, i.e., no correlation between the intensity fluctuations, then the value of $g^{(2)}(\tau = 0)$ is equal to 1 analogous to coherent light. Similarly, if a large noise amplitude is applied, the intensity fluctuations (blue) have a positive variance. The maximum value it reaches is half of the mean square of intensity fluctuations, resulting in the $g^{(2)}(\tau = 0)$ equal to 1.5.

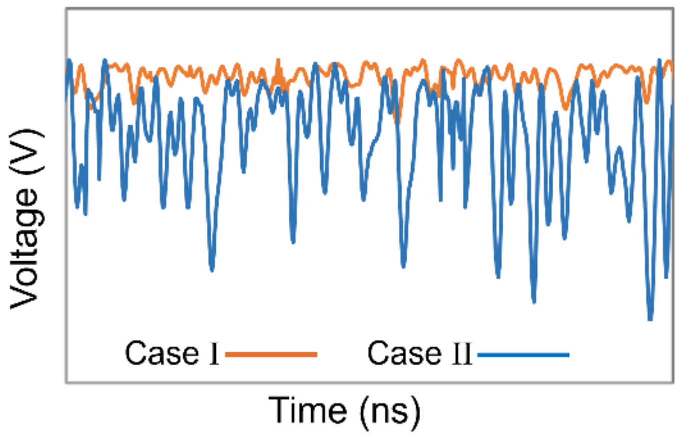


Figure 3.3: The intensity fluctuation at the output of MZM when a Gaussian noise is sent at the top of the MZM fringe.
Case I: $\varphi_o = 0$ and $\sigma = 0$. Case II: $\varphi_o = 0$ and $\sigma = \infty$.

3.3.2 Middle of the fringe: φ_o equals to $\frac{\pi}{2}$

$$g^{(2)}(\tau = 0) = \frac{3}{2} - \frac{1}{2}e^{-2\sigma^2\pi^2}. \tag{3.8}$$

For two limits of variance, I find the $g^{(2)}(\tau = 0)$ as

Case I: $\varphi_o = \frac{\pi}{2}$ and $\sigma = 0$

$g^{(2)}(\tau = 0) = 1.$

Case II: $\varphi_o = \frac{\pi}{2}$ and $\sigma = \infty$

$g^{(2)}(\tau = 0) = 1.5.$

Similar to the top of the fringe, the $g^{(2)}(\tau = 0)$ in the middle of the fringe can also be explained in terms of the variance of the intensity fluctuations, as shown in Fig. 3.4. The low noise amplitude generates an intensity fluctuation (orange) with

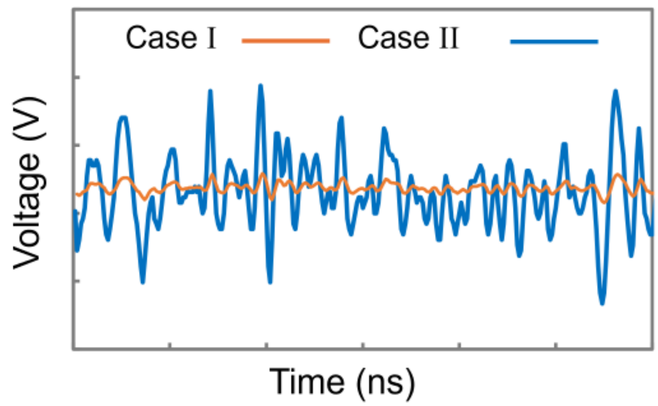


Figure 3.4: The intensity fluctuation at the output of MZM when a Gaussian noise is sent in the middle of the MZM fringe.
Case I: $\varphi_o = \pi/2$ and $\sigma = 0$. Case II: $\varphi_o = \pi/2$ and $\sigma = \infty$.

zero variance resulting $g^{(2)}(\tau = 0)$ equal to 1. Meanwhile, the large noise amplitude creates some correlations in the intensity fluctuations (blue), resulting in a positive variance. Again, the largest value the variance can reach is half of the mean square of the intensity fluctuations, giving the $g^{(2)}(\tau = 0)$ equal to 1.5.

3.3.3 Bottom of the fringe: φ_o equals to π

$$g^{(2)}(\tau = 0) = \frac{\frac{3}{(2)} - 2e^{\frac{-\sigma^2\pi^2}{2}} - \frac{1}{2}e^{-2\sigma^2\pi^2}}{\left[1 - e^{\frac{-\sigma^2\pi^2}{2}}\right]^2}. \tag{3.9}$$

Case I: $\varphi_o = \pi$ and $\sigma = 0$

Expanding exponential term as $e^x = 1 + x$ and using $\lim \sigma \to 0$, I get

$$
g^{(2)}(\tau=0) = \frac{\frac{3}{2} - 2\left(1 - \frac{\sigma^2\pi^2}{2}\right) + \frac{1}{2}\left(1 - 2\sigma^2\pi^2\right)}{\left[1 - \left(1 - \frac{\sigma^2\pi^2}{2}\right)\right]^2}
$$

$$
= \frac{\sigma^2\pi^2 - \sigma^2\pi^2}{\left(\frac{\sigma^2\pi^2}{2}\right)^2} \qquad \longrightarrow 0. \tag{3.10}
$$

The 1^{st} order expansion does not give the correct estimation of $g^{(2)}(\tau = 0)$; therefore, I employ the second order of expansion, i.e., $e^x = 1 + x + \frac{x^2}{2}$

$$
g^{(2)}(\tau=0) = \frac{\frac{3}{2} - 2\left(1 - \frac{\sigma^2\pi^2}{2} + \frac{1}{2}\left(\frac{\sigma^2\pi^2}{2}\right)^2\right) + \frac{1}{2}\left(1 - 2\sigma^2\pi^2 - \frac{1}{2}\left(2\sigma^2\pi^2\right)^2\right)}{\left(\frac{\sigma^2\pi^2}{2}\right)^2}
$$

$$
= \frac{\frac{-\sigma^4\pi^4}{4} + \sigma^4\pi^4}{\frac{\sigma^4\pi 4}{4}} = 3. \tag{3.11}
$$

Case II: $\varphi_o = \pi$ and $\sigma = \infty$

$g^{(2)}(\tau = 0) = 3/2 = 1.5$, using Eq. 3.9.

Again, this result can be explained in terms of the variance of the intensity fluctuations, as shown in Fig. 3.5. When a Gaussian noise with a low noise amplitude (orange) is applied, the intensity fluctuations produced at the output of the MZM have high variance, and Eq. 2.2 from Ch. 2 implies that if the variance of the intensity is higher than the intensity mean square then the $g^{(2)}(\tau = 0) > 2$, i.e., high correlation between the intensity fluctuations. Similarly, if a large noise amplitude is applied, the intensity fluctuations (blue) also have a positive variance. However, the maximum

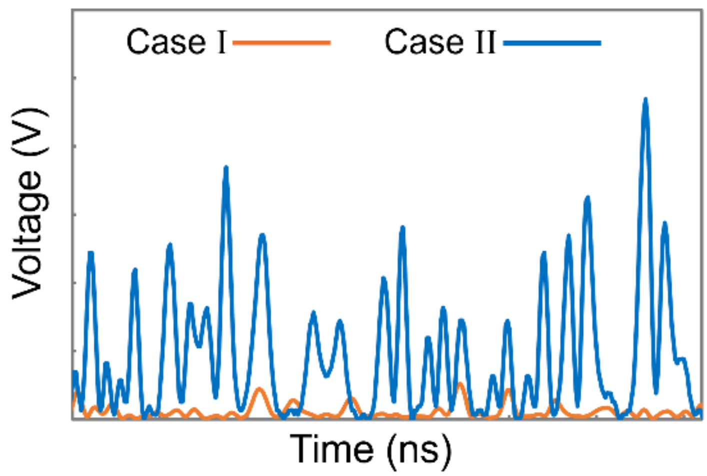


Figure 3.5: The intensity fluctuation at the output of MZM when a Gaussian noise is sent at the bottom of the MZM fringe.
Case I: $\varphi_o = \pi$ and $\sigma = 0$. Case II: $\varphi_o = \pi$ and $\sigma = \infty$.

value of the variance is half of the mean square of the intensity fluctuations, thus resulting in the $g^{(2)}(\tau = 0)$ equal to 1.5.

To sum up, if φ_o is equal to $\pi/2$ or 0 at the middle or top of the fringe (Fig. 3.7) respectively then $1 \leq g^2(\tau = 0) \leq 1.5$ for $0 \leq \sigma^2 \leq \infty$. However, if φ_o is equal to π, i.e., the bottom of the fringe as shown in Fig. 3.7 where the transmission is minimum, $g^{(2)}(\tau = 0)$ is equal to 1.5 for wide variance. Whereas, if the variance gets narrowed down to zero, the value of $g^{(2)}(\tau = 0)$ increases to 3.

3.4 Loudon's Theory for two Emitters

As discussed in Ch. 2, the actual practical form of the $g^{(2)}$ for thermal light sources is based on Loudon's collision-broadening theory [6], the process of how the total electric field is generated. The number of radiating atoms at a single frequency ω

with a random phase imitates the number of emitters in the thermal or pseudothermal source. In this thesis, I use a single MZM; thus, it represents two emitters or radiating atoms in Loudon's Theory. Therefore, the second-order coherence $g^{(2)}(0)$ for two emitters using Ch. 2 Eq. 2.8 when the value of ν is equal to 2 is

$$g^{(2)}(\tau = 0) = 2 - \frac{1}{2} = 1.5. \tag{3.12}$$

From Sec. 3.3, it can be observed that $g^{(2)}(0)$ equals 1.5 when it is at the top and in the middle of the fringe. However, the $g^{(2)}(0)$ is 3 at the bottom of the fringe. This is because Loudon's theory is based on the Gaussian field distribution of the emitted electric fields, whereas, when I am at the bottom of the fringe, the field distribution deviates from Gaussian to super-Gaussian and thus does not fall under the model's prediction.

3.5 Experiment

For the experimental setup shown in Fig. 3.6, I use a continuous wave amplitude-stabilized laser light (Fitel, F0L 15DCWC-A82-19340-B) with an output power of 37.5 mWatt and the centered wavelength to be at 1550 nm. The laser light is coupled into a polarization-maintained single-mode fiber connected to a Mach-Zehnder modulator (EOspace, AX-0K1-12-PFAP-PFA-R3-UL). The modulator is driven by a Gaussian noise generator and a DC power supply. The noise generator is connected to the RF electrode, and the DC power supply is connected to the modulator's DC electrode. I use two types of Gaussian noise generators for my experiment. They are Agilent Signal Generator- E8267D (PSG FG) and Tektronix Function generator-AFG3251 (TEK FG). The PSG FG waveform is set to 1 MHz frequency on ampli-

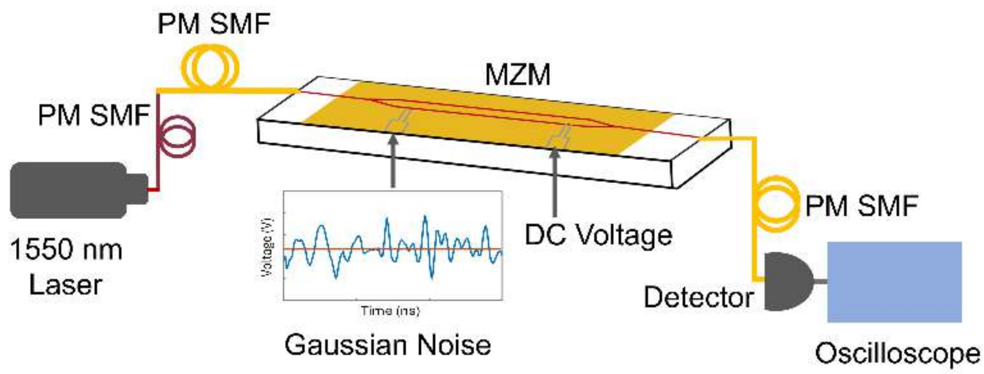


Figure 3.6: PM SMF: polarization-maintained single-mode fibers, MZM: Mach-Zehnder Modulator. The schematic of the experimental setup used to generate a super-bunching pseudothermal source. A continuous single-mode 1550 nm laser is coupled through PM SMF into a MZM. The MZM is driven by a Gaussian noise generator and a DC voltage. The intensity at the output is detected by a photodiode and recorded by a digital oscilloscope.

tude modulation (AM), i.e., it has a period of 1 μs per wave, whereas the TEK FG waveform is much faster, with a period of around 5 ns per wave.

The PSG FG noise generator has a peak-to-peak voltage (V_{pp}) of 7 V, whereas the TEK FG has 5 V. Both of these noise generators do not produce a standard Gaussian noise; their $g^{(2)}(\tau = 0)$ deviates from the ideal Gaussian noise value, i.e., 3. The PSG FG $g^2(\tau = 0)$ is measured to be 2.902, whereas the TEK FG is 2.683. The light at the output of MZM has a coherence time of tens of nanoseconds. Therefore, for detection, I use an InGaAs PIN photodiode (New Focus 1611). The detector's bandwidth is 30 kHz—1 GHz, and its rise time is 400 ps. Using the photodiode, long-time series intensity data are collected at a digital oscilloscope (Agilent Technologies) with a 20 GHz sampling frequency and 4 GHz bandwidth.

The intensity at the output of the modulator driven by an AC noise generator and a DC voltage is given as

$$I_{out} = I_{in} \cos^2 \left({\varphi_o}' + \left(\pi \frac{V_{DC}}{V_{\pi,DC}} \right) - \left(\pi \frac{V_{AC}(t)}{V_{\pi,AC}} \right) \right). \tag{3.13}$$

Here, I_{in} is the intensity of the laser light entering the modulator, and I_{out} is the intensity of the light on the oscilloscope shown in Fig. 3.6. $\varphi_o^{'}$ is the phase present due to optical path imbalance. V_{DC} and V_{AC} are the noise amplitude and bias voltage applied to the modulator. Whereas, $V_{\pi,DC}$ and $V_{\pi,AC}$ are the DC and AC, π voltages respectively.

Theoretically, Mach Zehnder modulators are designed to have equal arms, which means balanced optical paths and, thus, no phase shift. However, this is not true, as there are always some manufacturing tolerances or material inhomogeneities present in these modulators that lead to an imbalance of optical paths. Therefore, the phase ${\varphi_o}'$ is explicitly used in the modulator intensity expression in Eq. 3.13 to specify this optical path imbalance.

The interference or the intensity curve MZM created at the output is a cosine square function, as shown in Fig. 3.7. The graph shows how the intensity of the light at the output of the modulator changes with the phase. This cosine function can be moved back and forth by changing the bias voltage (DC voltage) applied to the modulator. The important features of this graph are V_π and three points A, B, and C. Point "A" is the point where the transmission of the light is minimum, whereas point "C" has the maximum transmission. V_π is the amount of voltage required to get to the point "C" from the point "A". Experimentally measured values of the V_π for my experimental setup are 4.74 V and 5.11 V for the AC (Noise Generator) and

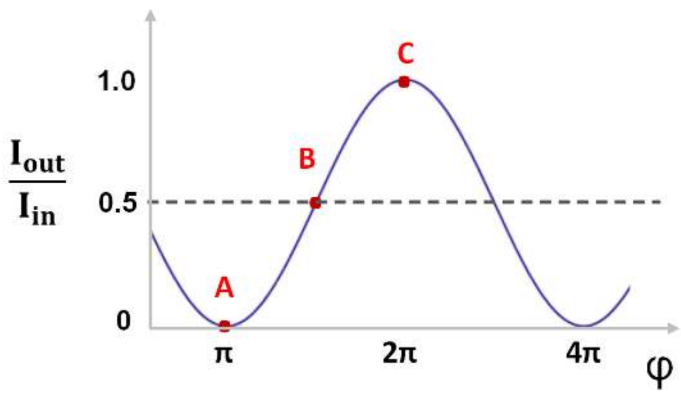


Figure 3.7: Schematic of a fringe of the Mach-Zehnder Modulator (MZM). Intensity of MZM vs the bias phase.

DC (Bias) voltage, respectively. The point “B” is the quadrature point, also referred to as the middle of the fringe in this chapter.

Now, to collect data, firstly, the operating point of the modulator, which implies the point on its fringe or intensity curve (Fig. 3.8b and Fig. 3.9b), is chosen. This is the point where the modulation signal is applied. For my experiment, I chose point “A” (bottom of the fringe) and point “B” (middle of the fringe) as the operating points. Gaussian noise with varying amplitudes is applied to these points, followed by detecting the intensity at the output of the MZM. Fig. 3.8 and Fig. 3.9 show in detail the procedure followed to obtain the intensity time series data that is further used to calculate the second-order temporal coherence values. The features of the output signals from the detector are shown in Fig. 3.8c and 3.9c. Those intensities are obtained when a high noise voltage, V_{pp} of 2 V (blue), and a low noise voltage, V_{pp} of 0.4 V (orange), are sent at the bottom (Fig. 3.8a - 3.8d) and the middle (Fig. 3.9a - 3.9d) of the fringe of the MZM.

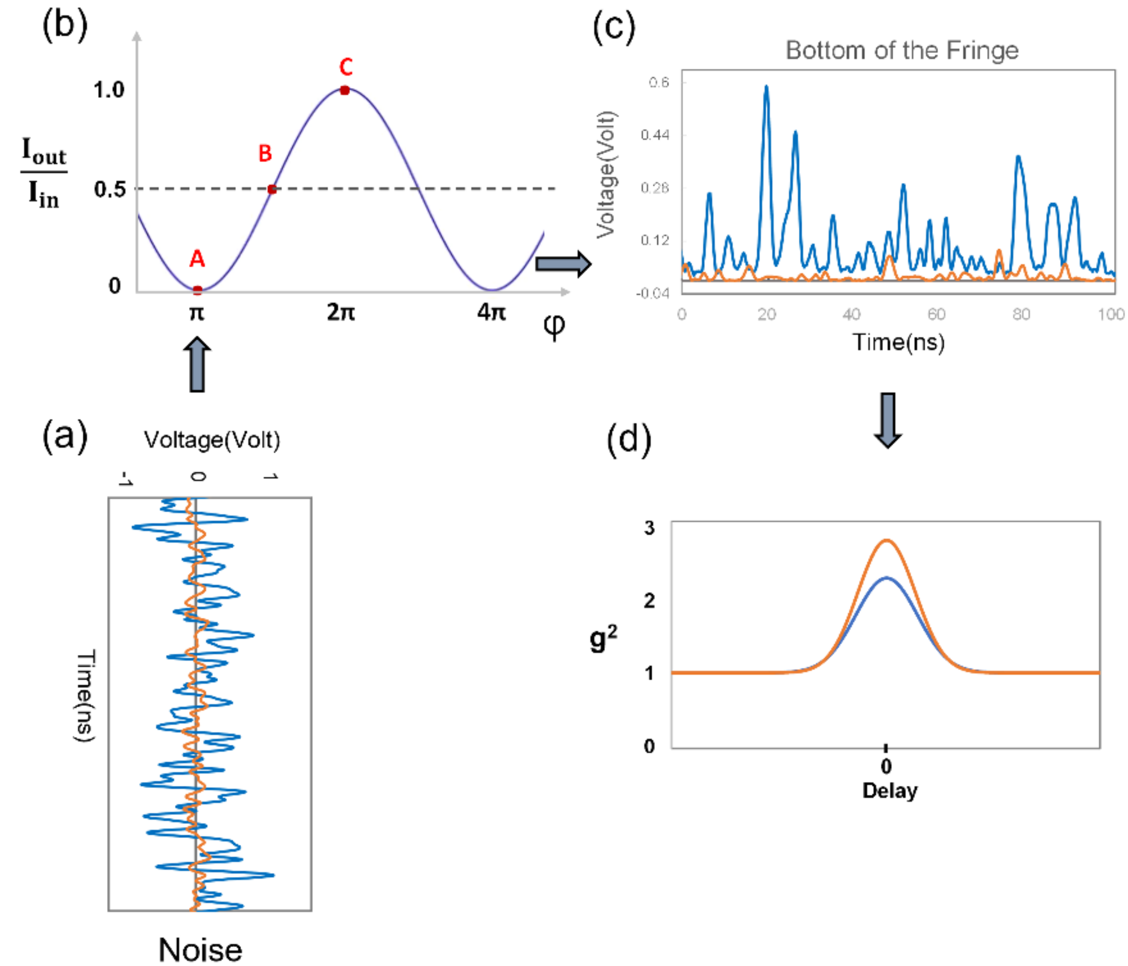


Figure 3.8: The schematic of the procedure followed to obtain the second-order temporal coherence function at the bottom of the fringe of the MZM connected to a Gaussian Noise generator and a bias voltage. (a) Shows the Gaussian noise generated using the TEK FG noise generator. (b) Shows the intensity curve of the MZM obtained using Eq. 3.3. (c) The measured intensity signals are detected by the photodiode. (d) Shows the second-order correlation function plot generated using the output signals shown in (c). Primarily, Gaussian noise in (a) with $V_{pp} = 2$ V (blue) and 0.4 V (orange) is sent at the bottom of the fringe at point "A" on the MZM intensity curve in (b), then the output intensity, voltage as a function of time is collected on the oscilloscope as shown in (c). Finally, in (d), the $g^{(2)}$ correlation functions are obtained using the intensity data of (c) and Eq. 1.2 from Ch. 1.

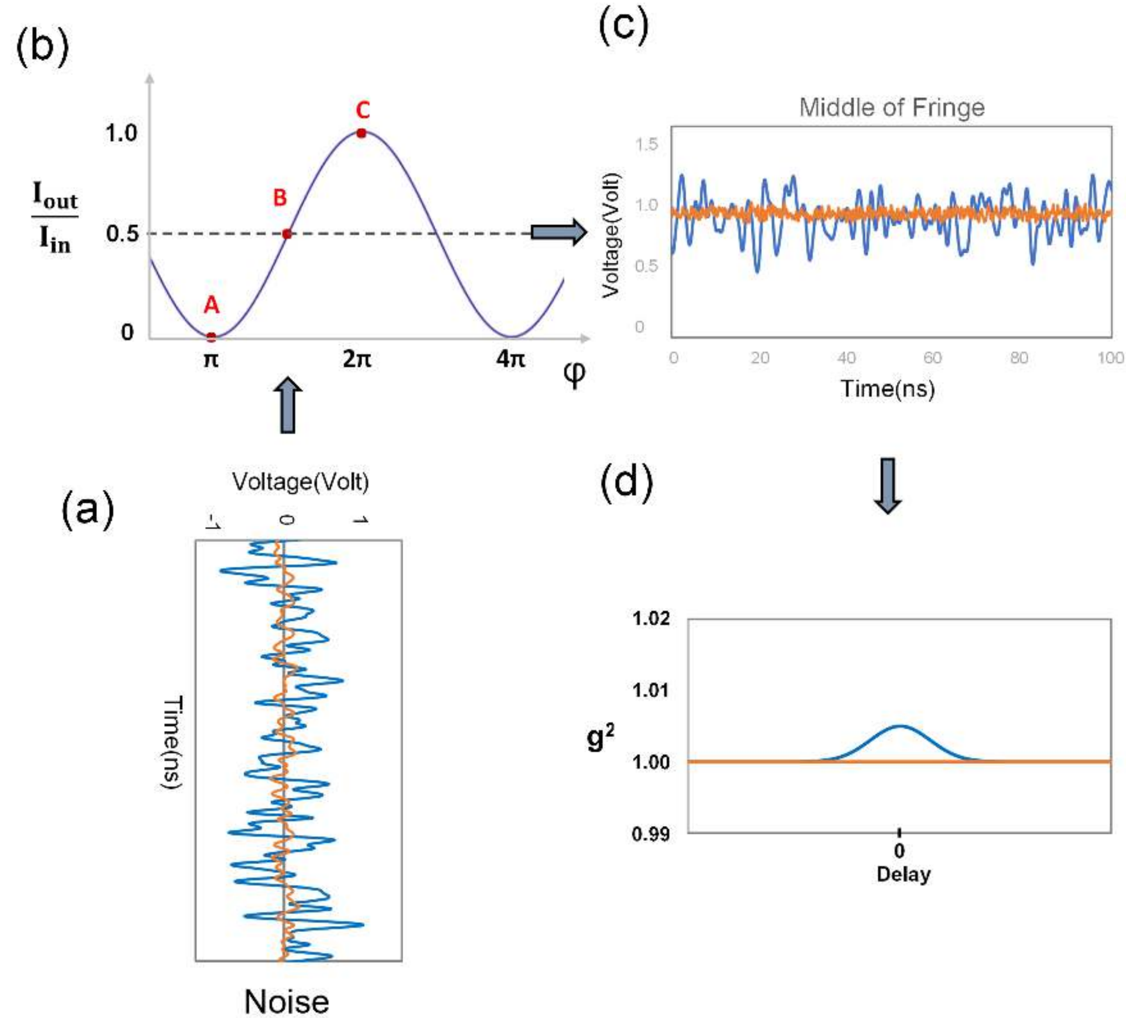
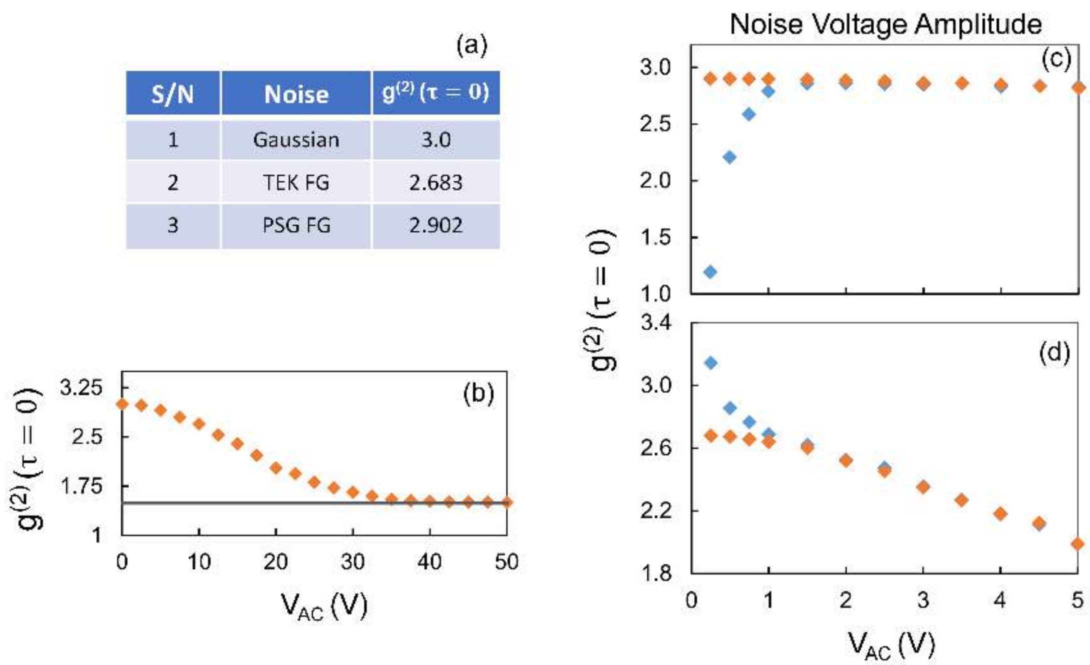


Figure 3.9: The schematic of the procedure followed to obtain the second-order temporal coherence function at the middle of the fringe of the MZM connected to a Gaussian Noise generator and a bias voltage. (a) Shows the Gaussian noise generated using the TEK FG noise generator. (b) Shows the intensity curve of the MZM obtained using Eq. 3.3. (c) The measured intensity signals are detected by the photodiode. (d) Shows the second-order correlation function plot generated using the output signals shown in (c). The Gaussian noise with $V_{pp} = 2$ V (blue) and 0.4 V (orange) is sent at the middle of the fringe at point “B” on the intensity curve shown in (b). The intensity at the output of the photodiode is collected as a voltage-time series for both the noise amplitudes and is shown in (c). Finally, in (d), the $g^{(2)}$ functions for both signals are obtained using the intensity data and using Eq. 1.2 from Ch. 1.

It can be observed that the 2 V noise, when sent to the bottom of the fringe, the intensity distribution obtained at the output is more like a Gaussian distribution thermal light [6] and thus the $g^{(2)}(\tau = 0)$ I see is closer to 2 (thermal source) that is 2.22. However, the distribution starts to deviate from the ideal thermal light features for the lower noise amplitude (orange) and gives the $g^{(2)}(\tau = 0)$ value to be 2.83, which is much higher than 2. One prominent feature that can be hinted at directly from the intensity data is the intensity variance. As I know from Ch. 2 Sec. 2.3, if the variance is much larger than the mean square intensity, then $g^{(2)}(\tau = 0)$ will have a much higher value than 2. Consequently, if I observe the intensity time series of lower noise amplitude, only a few sharp peaks are observed compared to high noise amplitude. Hence, 0.4 V has a much larger variance compared to the mean square intensity, which results in the $g^{(2)}(\tau = 0)$ being much higher than the 2. This also validates that if the variance of the noise amplitude is wide and I am at the bottom of the fringe where the phase of the fringe is π, then the $g^{(2)}(\tau = 0)$ will be nearer to value 3.

Therefore, to investigate the dependency of the second-order temporal coherence function on the applied noise voltage at the bottom of the fringe, I measure the intensity time series at the output of the MZM. I then vary the peak-to-peak amplitude of the noise voltages, V_{pp} from 0.25 V to 5 V with a step size of 0.25 V for voltages 0.25 V to 1 V and step size of 0.5 V for voltages 1 V to 5 V. Eventually, I calculated the $g^{(2)}(\tau = 0)$ values of each respective noise amplitude intensity time series data.

Fig. 3.10c and 3.10d shows the measured $g^{(2)}(\tau = 0)$ values as a function of V_{pp} applied at the bottom fringe. The measured $g^{(2)}(\tau = 0)$ function in Fig. 3.10c is using the PSG FG noise generator and Fig. 3.10d is using the TEK FG noise generator.



S/N	Noise	$g^{(2)}(\tau=0)$
1	Gaussian	3.0
2	TEK FG	2.683
3	PSG FG	2.902

Figure 3.10: The experimental values of the $g^{(2)}(\tau = 0)$ at the bottom of the fringe of the MZM. (a) The table shows $g^{(2)}(\tau = 0)$ values for the standard Gaussian noise and the noise generated by the two noise generators TEK FG and PSG FG used for the measurement. (b) The theoretical calculation of $g^{(2)}(\tau = 0)$ for different values of noise amplitudes produced by a standard Gaussian noise generator. (c) and (d) Theoretical (orange) and Experimental (blue) $g^{(2)}(\tau = 0)$ values obtained for noise amplitude sent to the bottom of the modulator using PSG FG and TEK FG noise generator, respectively.

The blue solid diamonds are experimental data obtained using the intensities data from the photodiode, whereas the orange solid diamonds are the theoretical fittings of the function given in Eq. 3.6 calculated using the Gaussian noise distribution of each noise generator. A good match exists between the experimental and theory for higher values of noise amplitudes. Later, in Sec. 3.6, I explain the reason for the deflection in the measurement and prediction $g^{(2)}(\tau = 0)$ values at lower noise amplitudes. Moving forward, I observe that the plot shown in Fig. 3.10d for the TEK

FG noise generator when V_{pp} is decreased from 5 V to 1 V, the degree of second-order coherence of pseudothermal light goes from bunching to super-bunching. The measured values of $g^2(\tau = 0)$ equal to 1.99 $\pm$ 0.01 for 5 V and 2.62 $\pm$ 0.02 for 1.5 V. The uncertainty is the calculated mean error value. Similarly, for the PSG FG noise generator, the $g^2(\tau = 0)$ function is bunching and super-bunching for lower noise amplitudes. However, I was only able to achieve a maximum value of V_{pp} of 7 V for the PSG FG noise generator; as a result, I only have the super-bunching second-order coherence shown in Fig. 3.10d. The $g^2(\tau = 0)$ values for the PSG FG are 2.83 $\pm$ 0.01 and 2.86 $\pm$ 0.01 for 5 V and 1.5 V respectively.

Similarly, in Fig. 3.9d for the middle of the fringe, the $g^{(2)}(\tau = 0)$ equals 1, same as a coherent light source for both high 2 V and low 0.4 V noise amplitudes. As mentioned, this observation can be explained in terms of intensity variance. The variance of the intensity data series shown in Fig. 3.9c is extremely narrow and disappears for lower noise amplitudes. Moreover, from Eq. 3.8, I know that when φ_o equals to $\pi/2$ and has a zero variance, the $g^{(2)}(\tau = 0)$ is equal to 1 and for the wide variance the $g^{(2)}(\tau = 0)$ reaches a maximum value of 1.5.

Earlier, I demonstrated the relation between the second-order coherence function and noise amplitude at the bottom of the fringe by measuring the intensity time series at the output of the MZM. Similarly, I also study this relation for the noise amplitudes sent at the middle of the fringe. The relation is shown in the Fig. 3.11 where the $g^{(2)}(\tau = 0)$ is plotted as a function of varied noise amplitudes. The graph shows the second-order coherence at a delay time equal to zero for the experimental and theoretical values for different noise amplitude values. The blue solid diamonds are measured, and the orange is the predicted values of $g^{(2)}(\tau = 0)$ obtained using the

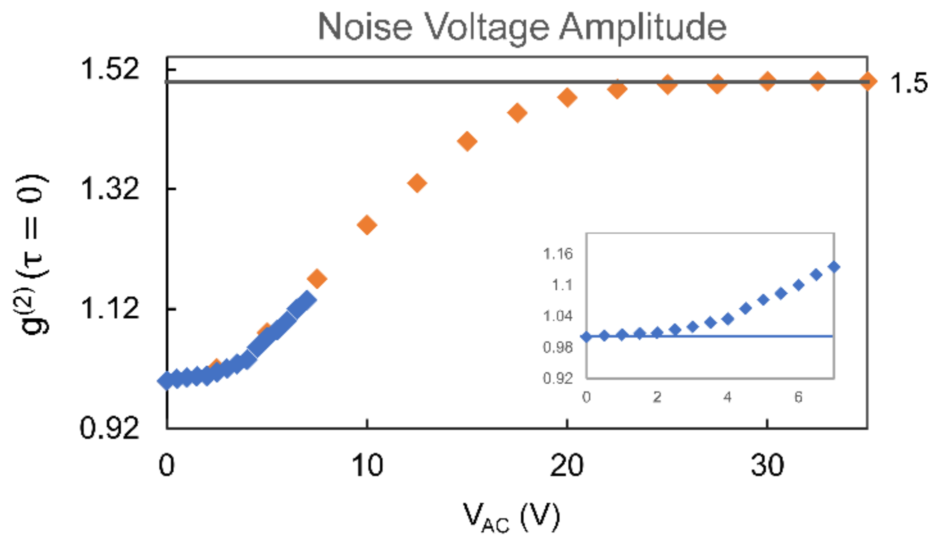


Figure 3.11: The second order coherence $g^{(2)}(\tau = 0)$ values when the Gaussian noise is applied at the middle of the fringe of the MZM. The theoretical (orange) and experimental (blue) $g^{(2)}(\tau = 0)$ plotted for different noise amplitude values.

expression in Eq. 3.8 for φ_o equals to $\pi/2$. The graph clearly shows a good agreement between the theoretical and experimental values for the lower V_{pp}, whereas, to observe the $g^{(2)}(\tau = 0)$ for higher V_{pp} I need a noise generator that can go up to 30 V or higher. However, as the noise generator I am using can do a maximum value of V_{pp} equal to only 7 V (PSG FG), I am not able to achieve the experimental value of $g^2(\tau = 0)$ greater than 1.

3.6 Discussion

In the above two sections, I theoretically and experimentally proved that a super-bunching pseudothermal light can be observed in my strategy. This realization is achieved using a bias voltage at one of the arms of the Mach Zehnder Modulator driven by a Gaussian noise on the other arm. The bias voltage helped me to change the phase of the fringe significantly. Thus, I was able to move from the top to the middle to the bottom of the fringe and go from a coherent $g^{(2)}(\tau = 0)$ equal to 1 to a super-bunching $g^{(2)}(\tau = 0)$ greater than 2 for the same setup, keeping all other parameters constant.

In Sec.3.5, I observed $g^2(\tau = 0)$ equal to 2.62 ± 0.02 and 2.86 ± 0.01 using the noise generators TEK FG and PSG FG respectively. These values are larger than the bunched thermal or pseudothermal light. Hence, I confirm that my experiment observed a super-bunching pseudothermal light. Also, from Eq. 3.11, it can be understood that the value of $g^2(\tau = 0)$ can be further increased and reached to 3 if a standard Gaussian noise generator is employed.

I also established the dependency of $g^2(\tau = 0)$ with the noise amplitude sent to the bottom or middle of the fringe. The plots shown in Fig. 3.10c and 3.10d for the bottom of the fringe explicitly displays the inverse relation where the $g^{(2)}(\tau = 0)$ values increases as the noise amplitude V_{pp} decreases. I observe a good match between the measured and predicted values within the error bars for the noise amplitudes higher than 1 V. There is a deflection of 1.75% for TEK FG and 3.73% for PSG FG at 1 V. This deflection increases as I further decrease the noise amplitude.

This behavior can be explained using the three graphs shown in Fig. 3.12. These graphs show how the scope noise can affect the intensity time series at the output of the MZM. The intensity time series corresponding to the low noise amplitudes sent at the bottom of the fringe is of comparable size to the scope noise. The scope noise is a Gaussian distribution with a zero mean. The $g^{(2)}(\tau = 0)$ of intensity with zero mean is ∞ from Eq. 1.2 in Ch. 1. Therefore, scope noise is problematic for the intensity of low noise amplitude as the $g^{(2)}(\tau = 0)$ equal to ∞ dominates, and thus the $g^{(2)}(\tau = 0)$

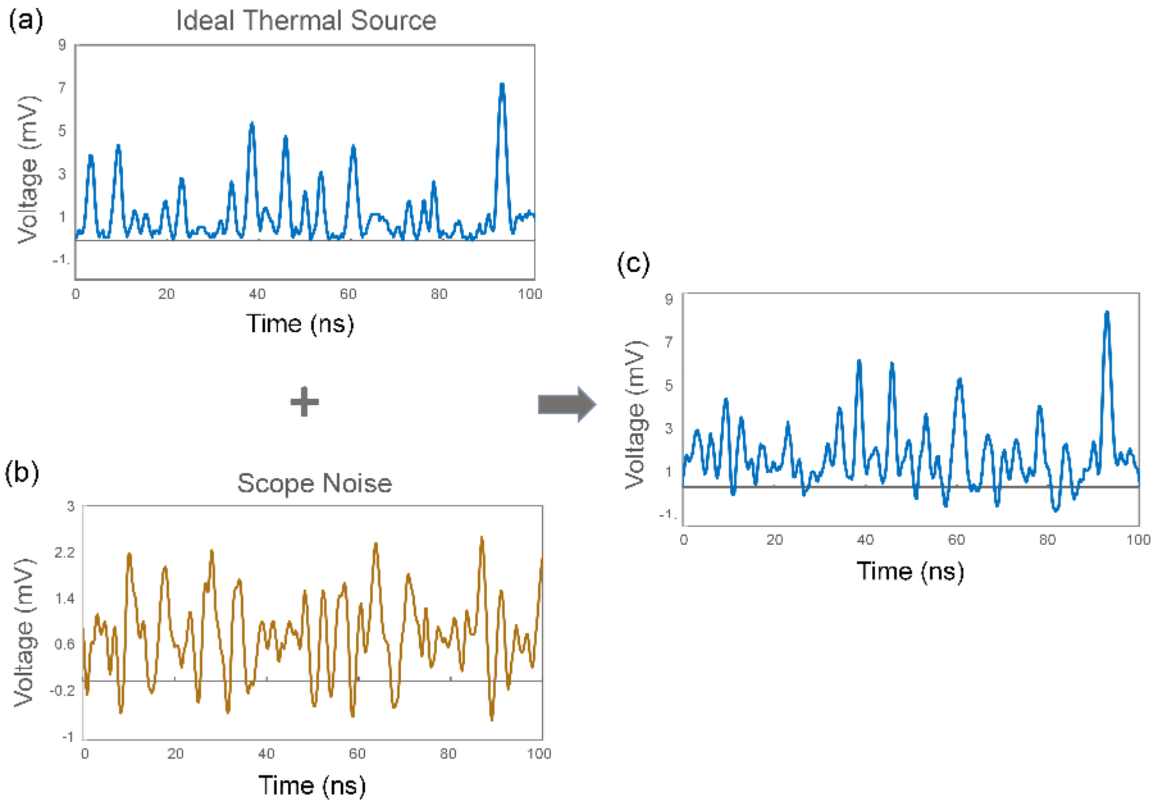


Figure 3.12: (a) Shows the ideal thermal source intensity data at low noise amplitude. (b) The scope noise data only. (c) The total intensity is when the ideal thermal source intensity and the scope noise are added together. This imitates the intensity signal at the output of the MZM when a lower noise amplitude is applied at the bottom of the fringe in our experiment.

for my measured intensity becomes non-reliable. Fig. 3.12a corresponds to an ideal thermal intensity, which ideally should correspond to my measured intensity time series for low noise. However, the intensity series I observe is shown in Fig. 3.12c, where the intensity values go below zero because of the scope noise, resulting in a different value of the $g^2(\tau = 0)$ than expected. Therefore, the calculated $g^{(2)}(\tau = 0)$ of such intensity data will deviate from the theoretical prediction as observed in Fig. 3.10c and 3.10d. At lower V_{pp}, 0.75 V to 0.25 V, the output voltage from the photodiode reads 20 mV to 5 mV, respectively, on the scope. These output voltage readings are close to scope noise (2 mV), and thus, I see the deviation in $g^{(2)}(\tau = 0)$ values. The deviation increases as the V_{pp} decreases.

For the first time, the second-order coherence function is calculated using the intensity-field correlation measured using a single-spatial mode. My experimental setup is completely single-mode fiber-based. Thus, the loss of light due to scattering in space is negligible compared to other research group experiments using a free-space setup. I also reported that the coherence time for the super bunching pseudothermal light created using my setup is around 10^{-8} to 10^{-6} s, which is very short compared to the 10^{-5} to 1 s that have been reported [16, 19, 27]. In addition, I also use a low amplitude Gaussian noise as the high voltage electro-optical modulator requires high power to operate. The MZM has a high modulation speed of around one to tens of GHz, which is very high compared to a few hundred MHz speed of an acoustic-optical modulator. In retrospect, MZM can be a better option than other optical modulators used to create super-bunching pseudothermal light.

3.7 Conclusion

In conclusion, I have successfully proved that by using a simple device like MZM, I am able to build a technology that creates random intensity fluctuations with high correlations. The second-order coherence $g^{(2)}(\tau = 0)$ of such highly correlated intensities is higher than 2, i.e., the created light source is a super bunching pseudothermal light source. The $g^{(2)}(\tau = 0)$ can be tuned from a coherent to bunching to super bunching by varying the noise amplitudes and bias voltage. The experimental setup is performed using single-mode fibers. It is a simple and economical setup, with no expensive equipment like high-voltage modulators. Such super bunching light sources can be used for applications such as studying higher-order coherence or increasing the visibility of ghost imaging.

Chapter 4

Intensity Correlation Function for Light Passing Through an Unequal-path Interferometer

4.1 Introduction

The Mach-Zehnder modulator (MZM), connected to a bias voltage and Gaussian noise, is an excellent technology for generating super bunching pseudothermal light. However, the small phase shift in MZM limits the system to only a single fringe for a swipe. Therefore, I introduce an unequal-path interferometer (UPI) system, a flexible and inexpensive device that allows access to more fringes when connected to a frequency-modulated diode laser. This system produces intensity fluctuations at the output of the UPI when a Gaussian noise generator modulates the laser coupled to the UPI. Thus, I aim to study the relation between the second-order coherence function of the intensity fluctuation at the output for varying noise amplitude applied directly to the laser using a noise generator. Some groups successfully have used such current-to-frequency converter laser systems to study chaos-based systems [41] and slow-light devices [42]. Later in this chapter, I will also explain how this device can be implemented in experimental setups that can generate a pseudothermal source. In conclusion, such fascinating characteristics of the UPI make it an interesting experimental system for studying and understanding chaotic or thermal sources.

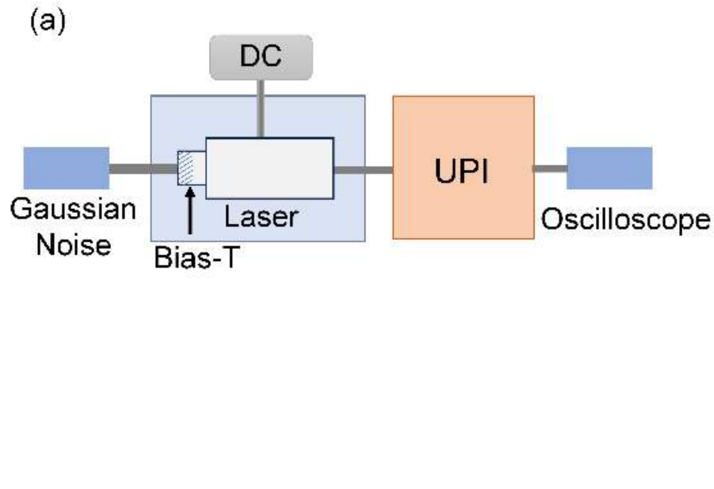


Figure 4.1: DC: Direct current Voltage and UPI: unequal-path interferometer. (a). The schematic diagram representing the Gaussian noise connected to the laser using a bias-T. The modulated laser is connected to the UPI system, which outputs fluctuating intensity. (b). The schematic diagram of bias -T is a 50 Ohm resistance connected to a grounded diode.

The setup is easy to construct using readily accessible pieces of equipment. It is a simple device built using a diode laser with a Gaussian noise generator applied to it through a device called bias-T, as shown in Fig. 4.1(a). Bias -T is a resistor with a diode that goes to the ground. The 50 Ohm is the resistance of the diode. The injected alternate current (I_{AC}) of the Gaussian noise voltage must be subtracted from the direct current (I_{DC}) supplied to the laser. The Gaussian noise connected to bias-T is used to modulate the injected current of the laser, which causes a change in laser frequency, a device that acts as a current-to-frequency converter. The most

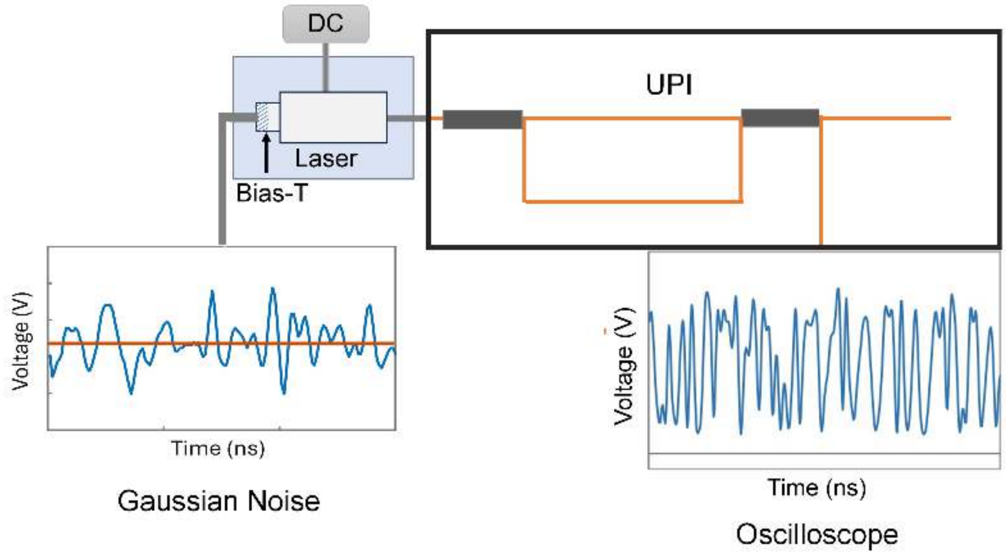


Figure 4.2: DC: Direct current Voltage and UPI: unequal-path interferometer. The schematic diagram represents the UPI setup. A Gaussian noise generator is connected to a laser using a bias-T. The modulated laser light propagates the UPI system, and as a result, a fluctuating intensity is observed at the oscilloscope.

important feature of this construction is that the laser frequency can be primarily modulated by varying its injecting current.

As shown in Fig. 4.2, I connect this current-to-frequency modulated laser to an unequal path interferometer to obtain an intensity fluctuation as a function of time. The MZM is an equal path interferometer; therefore, it is insensitive to any change in the laser frequency. As a result, I changed the internal phase of the MZM to observe a fringe pattern at the output. On the contrary, UPI is very sensitive to frequency changes; hence, I keep the laser outside the interferometer. I do not change any internal parameters like the phase of the interferometer to create fringes at the output of the UPI. Instead, the fringes are created by the interference between the frequency-modulated laser light propagating through an unequal path length interferometer.

However, the intensity pattern observed at the interferometer's output is similar to the MZM as shown in Fig. 4.4 but with more fringes.

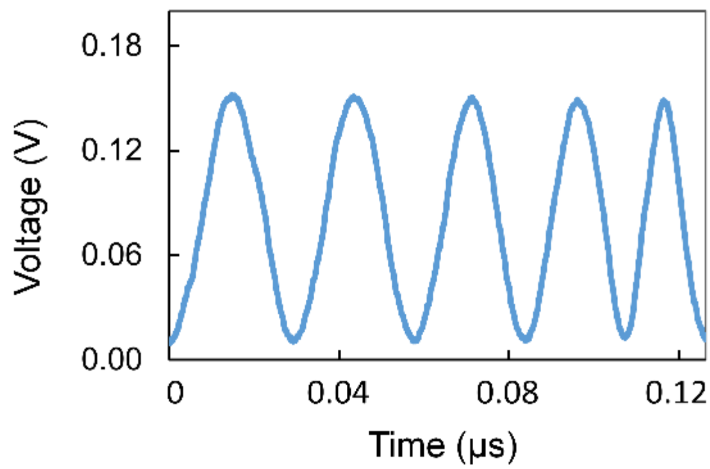


Figure 4.3: The fringes created by the delay interferometer setup. Here, I see more numbers of fringes compared to a single fringe in the Mach Zehnder modulator case.

4.2 Theory

In this section, I develop a mathematical model of the current-frequency-modulated laser and derive the second-order coherence function for the UPI connected to such a modulated laser system.

4.2.1 Current-Frequency Converter

The output power and frequency change as a function of the applied voltage at the AC port of the laser. If the laser is modulated at a timescale slower than a gigahertz, then the applied voltage has a linear relation with the power and frequency of the laser, and the relation is given as [41]:

$$P(t) = \kappa V(t) + P_{\mathrm{o}}. \tag{4.1}$$

$$\omega(t) = \eta V(t) + \omega_{\mathrm{o}} = \frac{\eta}{\kappa}(P(t) - P_{\mathrm{o}}) + \omega_{\mathrm{o}}. \tag{4.2}$$

where $P(t)$ and $\omega(t)$ are the total optical power and frequency of the laser, respectively, $V(t)$ is the noise voltage applied at the bias-T and κ, and η are empirically determined constants, i.e., the voltage-to-power and voltage-to-frequency conversion strength respectively. For a small change in the voltage, there is not much change in the power; thus, the dominant effect is observed only for the frequency of the laser.

The above Eq. 4.2 can also be written in terms of current [41, 42] as

$$\omega(t) = \alpha i(t) - i(t) \circledast h(t). \tag{4.3}$$

where α is current to frequency conversion strength and the $i(t)$ is the modulation current. Two mechanisms can describe the current and frequency change happening in the laser. The first one is the change in the carrier density denoted by the first term on the right side of Eq. 4.3, and the second one is the thermal effect, described by the second convolution term. Both these changes can shift the refractive index of the cavity material and can result in laser frequency modulation. The thermal effect is slower as it changes the frequency around or lower than 5 MHz [43, 44].

4.2.2 Second-order coherence in UPI

Now, to characterize the second-order correlation function $g^{(2)}$ for UPI, I first find the expression for the intensity at the output of the UPI to be

$$I_{\mathrm{UPI}}(t) = \frac{I_{\mathrm{UPI,o}}}{2}[1 + \cos(\omega(t)\Delta T)]. \tag{4.4}$$

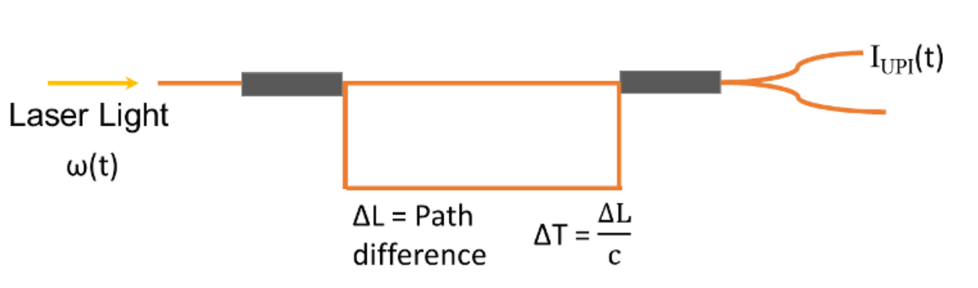


Figure 4.4: ΔT: Delay time, c: speed of light in vacuum. The schematic of an unequal-path interferometer.

where, $\mathrm{I_{UPI}(t)}$ and $\mathrm{I_{UPI,o}}$ are the output and input intensities of the UPI, respectively, and ΔT is the time difference. Using the frequency of the laser from Eq. 4.3 only for linear approximation we get

$$\omega(t) = \alpha i(t) + \omega_\mathrm{o} = \delta\omega(t) + \omega_\mathrm{o}. \tag{4.5}$$

It can be seen that $\delta\ \omega(t)$ equals $\alpha i(t)$. Also, $\delta\ \omega(t)$ is a Gaussian random variable as $i(t)$ is proportional to the noise voltage applied to the laser, which is a Gaussian random variable. Therefore, implying this, I get the modified intensity of UPI to be

$$I_\mathrm{UPI}(t) = \frac{I_\mathrm{UPI,o}}{2}[1 + \cos(\omega_\mathrm{o}\Delta T + \delta\omega(t)\Delta T)]. \tag{4.6}$$

Now if $\omega_o\Delta T$ equals to φ_o then

$$I_\mathrm{UPI}(t) = \frac{I_\mathrm{UPI,o}}{2}[1 + \cos(\varphi_\mathrm{o} + \delta\omega(t)\Delta T)]. \tag{4.7}$$

Here, φ_o is the initial phase. Now, using the second-order coherence $g^{(2)}(\tau = 0)$ equation from Ch. 1, Eq. 1.2 and above expressions for $\mathrm{I_{UPI,o}}$ and $\mathrm{I_{UPI}(t)}$ from Eq. 4.6 and Eq. 4.7, I get

$$g^{(2)}(\tau=0)=\frac{\langle(\cos^2(\frac{\delta\omega(t)\Delta T+\varphi_o}{2}))^2\rangle}{\langle\cos^2(\frac{\delta\omega(t)\Delta T+\varphi_o}{2})\rangle^2}. \tag{4.8}$$

The probability density, P($\delta\omega$) is Gaussian and is given as

$$P(\delta\omega)=\frac{1}{\sqrt{2\pi}\sigma}e^{\frac{-\delta\omega^2}{2\sigma^2}}. \tag{4.9}$$

Incorporating the probability density in Eq. 4.8, I finally, get the $g^{(2)}(\tau=0)$ to be equal to

$$g^{(2)}(\tau=0)=\frac{\frac{3}{2}+2\cos(\varphi_{\mathrm{o}})e^{\frac{-\sigma^2\pi^2}{2}}+\frac{1}{2}\cos(2\varphi_{\mathrm{o}})e^{-2\sigma^2\pi^2}}{[1+\cos(\varphi_{\mathrm{o}})e^{\frac{-\sigma^2\pi^2}{2}}]^2}. \tag{4.10}$$

Now, if the frequency deviation is large, i.e., $\delta\omega\ \Delta T >> \pi$ then variance $>> 1$ and thus $g^{(2)}(\tau=0)$ is equal to 1.5. This value agrees with Loudon's two emitters model in Ch. 2 Sec. 2.5.

4.3 Experiment

The experimental setup for the UPI is shown in Fig. 4.5, where I use the same Fitel laser used for the MZM experiment in Ch. 3. The laser is mounted on a Thorlabs LM14S2 (Butterfly Laser Diode Mount). The mount also has a bias-T equipped, which is used to connect the noise generator. The laser light is coupled into a polarization-maintained single-mode fiber (PM SMF), connected to an interferometer with a delay length of 2 m. The interferometer consists of two 50/50 couplers that are single-mode at 1550 nm, one variable attenuator (VA), and one fiber-based polarization controller (PC). Finally, the output of the interferometer goes into a photodiode. To collect data, I used two photodiodes, New Focus 1611 and Thorlabs DET08CFC. The photodiode is connected through a coaxial cable to an oscilloscope (Agilent Technologies) with a 20 GHz sampling frequency and 4 GHz bandwidth to collect the intensity time series data.

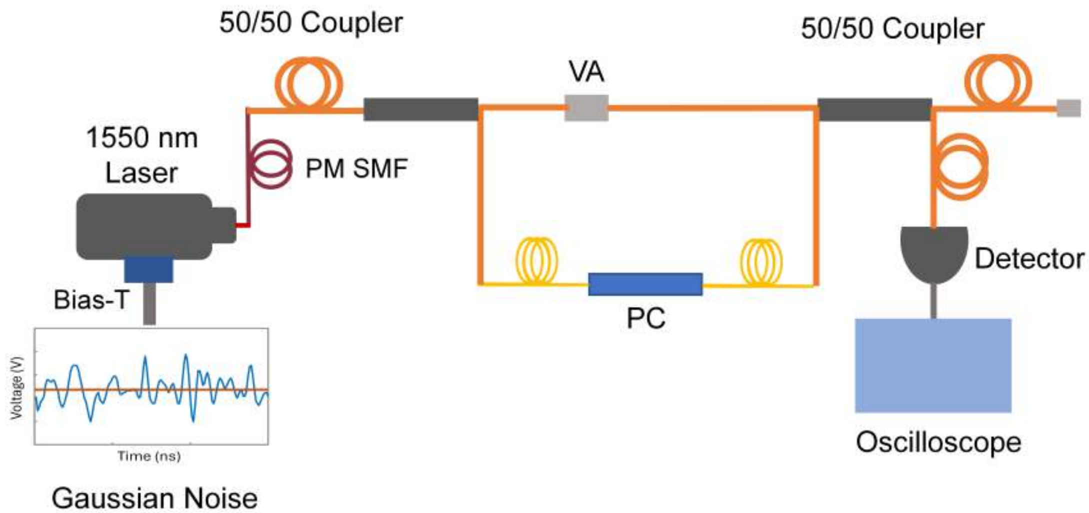


Figure 4.5: PM SMF: polarized single-mode fiber, VA: variable attenuator, PC: polarization controller. The schematic of the experimental apparatus of the unequal-path interferometer. The continuous single-mode 1550 nm laser is modulated using a Gaussian noise generator at bias-T. The laser is coupled through a polarization-maintained single-mode 50/50 coupler. One output of the coupler goes into a polarization controller (PC), whereas the other output is connected to a variable attenuator (VA). The PC and VA are coupled to each of the inputs of another 50/50 coupler. Finally, one end of the coupler is connected to a detector, i.e., a photodiode. The interference fringes are detected by a photodiode and recorded by a digital oscilloscope.

The injection current of the laser is modulated using a Tektronix Function generator-AFG3251 (TEK FG). The TEK FG has a waveform with a period of 5 ns; thus, I chose a path difference of 2 m for the interferometer. The time delay of the interferometer needs to be larger than the correlation length of the Gaussian voltage noise to acquire a Gaussian noise [42]. The noise voltage is sent from the TEK FG to the bias-T, where the bias-T converts the noise signal into a current, which gets added to the laser DC injection current. The modulation of the laser current helps to achieve oscillations in a frequency of around 800 MHz for 1 mA of change in the laser current. The setup is tightly mounted on an optic table to maintain mechanical stability, as the power at the interferometer's output is extremely sensitive to the path difference. A wavelength order of variation of the laser light (1550 nm) in the path difference can result in significant variations in the output power. Additionally, I wrapped the fibers with insulating foam boxes to reduce thermal expansion.

To investigate the $g^{(2)}(\tau = 0)$ function of the UPI, I varied the noise amplitude (V_{pp}) from 0.05 to 5 V with a step size of 0.1 V for the voltage 0.1 to 0.5 V and step size of 1 V for the voltage 1 to 5 V at bias -T. I collected the respective intensity long-time series at the output of the UPI using a photodiode. Using Ch. 1, Eq. 1.2, I calculated the $g^{(2)}(\tau = 0)$ for applied (V_{pp}) and plotted them in the graph shown in Fig. 4.6. I observe that $g^{(2)}(\tau = 0)$ to be 1.323 ± 0.01 for 5 V for both the NewFocus and Thorlabs photodiode. The value of $g^{(2)}(\tau = 0)$ increases and stabilizes to value 1.5 as the (V_{pp}) value decreases. Therefore, I see that experimental values of $g^{(2)}(\tau = 0)$ for the NewFocus detector are 1.496 ± 0.02, 1.514 ± 0.02 and 1.532 ± 0.02 for (V_{pp}) equal to 150, 100 and 50 mV respectively. Similarly, for the Thorlabs detector, I obtain 1.490 ± 0.02, 1.508 ± 0.02 and, 1.524 ± 0.02 for (V_{pp}) equal to 150, 100, and 50 mV respectively.

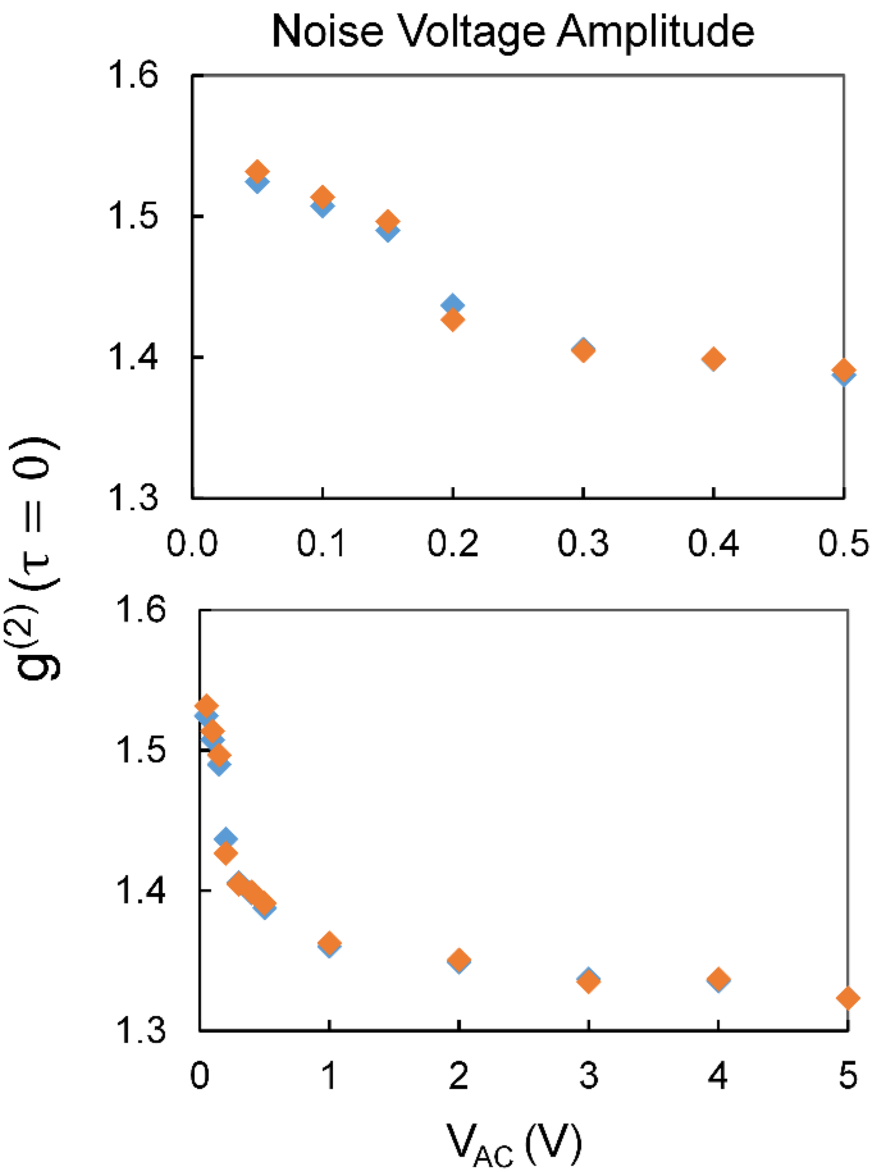


Figure 4.6: The measured $g^{(2)}(\tau = 0)$ values as a function of (V_{pp}) applied for the UPI experiment. The orange solid diamonds are $g^{(2)}(\tau = 0)$ values determined using the output intensity collected using the NewFocus detector, whereas the blue solid diamonds are using the Thorlabs detector. The bottom plot includes (V_{pp}) from 0 to 5 V, whereas the top plot is a magnified version that only shows the lower (V_{pp}) from 0 to 0.5 V.

4.4 Discussion

In the above section 4.3, I have shown that for both the detectors, for a low V_{pp} of 50 mV, I was able to obtain the $g^{(2)}(\tau = 0)$ values equal to 1.52 ± 0.02 for the Thorlabs and 1.53 ± 0.01 for the NewFocus. However, for the high V_{pp} of 5 V, I could only get $g^{(2)}(\tau = 0)$ values equal to 1.32 ± 0.06 for both the detectors. I found that the V_{pp} equal to or lower than 150 mV matches the theoretical prediction of $g^{(2)}(\tau = 0)$, i.e., equal to 1.5 whereas they start to deflect for the V_{pp} values above 150 mV. The reason for this deflection is the power of the laser. One thing to consider is that the laser does not only change its frequency as a function of the voltage applied at the AC port of the bias-T; the power also changes. The power change is negligible for a small change in the injected current, whereas a dominant effect is observed only for changes in the laser frequency. However, this is not true for large current injections. The larger values of noise amplitude affect both the power [42] and frequency, and thus $g^{(2)}(\tau = 0)$ at higher (V_{pp}) shows the deviation from 1.5.

4.5 Conclusion

In this experiment, I showed that the length-delayed interferometers' intensity fluctuations are chaotic when connected to a laser modulated by a Gaussian noise generator. I was able to achieve a second-order coherence function $g^{(2)}(\tau = 0)$ equal to 1.52 ± 0.02 and 1.53 ± 0.01 for the Thorlabs and NewFocus detectors respectively in a single-mode using the UPI setup. Therefore, it proves that this system is equivalent to the two emitters in Loudon's model, and thus the value of $g^{(2)}(\tau = 0)$ can be further tuned to create a bunching pseudothermal light source by adding a large number of delay interferometers.

Chapter 5

Coherent light propagating in a Multi-Mode Optical Fiber Manifest a Super Bunching Effect with Super Gaussian Intensity Distribution

5.1 Introduction

A typical multi-mode (MM)) optic fiber is a low-loss waveguide with a cylindrical core and a diameter much larger than the wavelength of light. It supports many transverse modes. They have a large acceptance angle, increasing light-gathering ability, and it is relatively easy to couple light into and out of the fiber. They find wide applications in building-scale communication systems, illumination systems, high-power lasers, and delivering light to optical devices. Also, MM fibers can be used as a device in their own right, such as temperature sensor [45, 46], for pseudothermal light source [28] and all-fiber optical spectrometer [47–52].

Because of the many applications in the last few decades, the study of light propagation in MM fibers has gained more attention. In this chapter, I demonstrate that when a coherent light propagates through a MM fiber, it develops into a speckle pattern that manifests a spatial super bunching effect without a particular input mode preparation. The factor that emerges as important is the mode-dependent loss (MDL) in the fibers, which is caused by material absorption and also leaky modes. Fig. 5.1 illustrates the ray diagram of light propagation in ideal and real multi-mode fibers and helps to understand the essence.

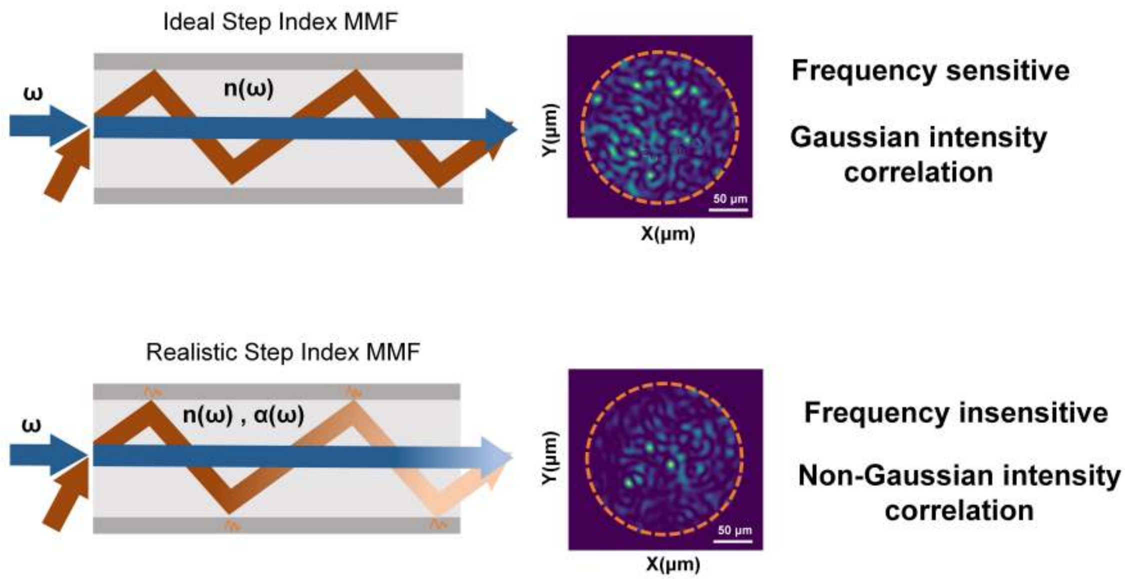


Figure 5.1: The ray diagram illustrating the ideal case and the case when absorption is introduced in a step-index multi-mode fiber. The top figure is the ideal mode representation. When light propagates through a fiber that does not have absorption, the speckles generated at the output are more sensitive to the frequency of the laser. There are more bright speckles. Whereas vice versa is observed if the absorption is present in the fiber, less sensitive to incident light frequency, and has fewer bright speckles.

The remaining part of the paper is categorized as follows: In section II, I describe the ideal mode theory in a step-index MM fiber followed by MDL and correlation functions used to model the speckles at the output of the MM fiber. In section III, I present the experimental setup; in section IV, I explain and discuss the results. Finally, in section V, I provide the conclusion.

5.2 Theory:

5.2.1 Ideal-Mode theory

I only consider a step-index (SI) MM fiber here, as all the experimental data is based on SI fiber. The ideal fiber is straight, with no variation along the length, cylindrically symmetric, and free from scattering (no mode-coupling). The refractive index profile $n(r)$ is cylindrically symmetric and depends only on the radial coordinate r of the fiber. Let the radius of the fiber be ρ, then for the SI fiber has the refractive index inside the core ($r < \rho$) a constant value, i.e., n_{co} and has a constant value of n_{cl} in the cladding ($r > \rho$). The light propagates inside the fiber when the $n_{co} > n_{cl}$. The fiber modes are linearly polarized modes using the scalar wave (Eqn. [5.1]) and assuming the weakly guiding approximation.

$$\nabla^2 \mathbf{E} = \frac{n^2}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}. \tag{5.1}$$

Here, n is the refractive index, and c is the speed of light in a vacuum. Using the scalar equation and separating the time and z-coordinate of each of the modes of transverse electric field ($\mathbf{E}$) (Eq. 5.2), I get the Helmholtz equation for the transversal field distribution $\psi(r,\phi)$in the SI fiber shown in Eq. 5.3

$$\mathbf{E}(r,\phi,z,t) = \psi(r,\phi)e^{i(\omega t - \beta z)}. \tag{5.2}$$

$$\{\triangle_t + n^2 k_o^2 - \beta^2\}\psi(r,\phi) = 0. \tag{5.3}$$

where ω is the angular frequency, β is the propagation constant, n is the refractive index, k_o is the wave number, $\triangle_t$ is the transversal part of Laplacian and $\psi(r, \phi)$ is the transverse field distribution at t and z equal to zero. The transversal field distribution is Bessel functions of the first and second kind for the SI fiber

$$\psi(r,\phi) = \begin{cases} \frac{A}{J_l(U)} J_l(\frac{U_r}{\rho}) \begin{bmatrix} cos(l\phi) \\ sin(l\phi) \end{bmatrix}, & r < \rho \\ \frac{A}{K_l(W)} K_l(\frac{W_r}{\rho}) \begin{bmatrix} cos(l\phi) \\ sin(l\phi) \end{bmatrix}, & r > \rho \end{cases} \tag{5.4}$$

where J_l are first and K_l are second kind Bessel functions. V, U, and W are defined as

$$U = \rho\sqrt{{k_o}^2{n_{co}}^2 - \beta^2} \tag{5.5}$$

$$W = \rho\sqrt{\beta^2 - {k_o}^2{n_{cl}}^2} \tag{5.6}$$

$$V = \sqrt{U^2 + W^2} = \rho k_o\sqrt{{n_{co}}^2 - {n_{cl}}^2} \tag{5.7}$$

5.2.2 Mode-Dependent Loss

Any mode loss caused by any medium inside the fiber is considered a mode-dependent loss (MDL). This loss can be a consequence of various factors, but I only discuss two types: the attenuation of modes due to leaky modes and due to material absorption. MDL is present in both absorbing and nonabsorbing optical fibers. In nonabsorbent fibers, the MDL is contributed by the leaky modes, whereas in the absorbing fiber, it is due to both the leaky modes and material absorption.

All the optical fibers support both leaky [53] and bound modes [54, 55]. Leaky modes attenuate in the direction of modes propagation, and the attenuation is significant for the higher-order modes compared to lower-order modes. Like leaky modes, the MDL caused by material absorption is also high for high-order modes. The length of the fibers is directly proportional to the material absorption; longer fiber has more absorption. Also, a longer fiber carries a significantly larger number of modes than a short one, which implies more higher-order modes. Therefore, high MDL is observed in long fibers. In some cases, the higher-order modes even become extinct due to the MDL.

Now, finding the relation between the MDL and the bound modes is essential to explain the behavior of the speckle patterns observed at the fiber exit. Therefore, two methods exist to find the mode-dependent absorption coefficient: by finding the complex roots of the eigenvalue equation or by implementing the perturbation theory directly to the U parameter defined in Eq. 5.5. In this chapter, I use the latter method by writing the real and imaginary part of the refractive index (n), propagation constant (β), and parameter U as

$$n = n_{(r)} + in_{(i)} \ , \qquad n^2 \approx {n_{(r)}}^2 + 2in_{(r)}n_{(i)}. \tag{5.8}$$

$$\beta = \beta_{(r)} + i\beta_{(i)} \ , \qquad \beta^2 \approx {\beta_{(r)}}^2 + 2i\beta_{(r)}\beta_{(i)}. \tag{5.9}$$

$$U = U_{(r)} + iU_{(i)} \ , \qquad U^2 \approx {U_{(r)}}^2 + 2iU_{(r)}U_{(i)}. \tag{5.10}$$

Here, $n_{(r)}$, $\beta_{(r)}$ and $U_{(r)}$ are real parts of n, β and U and $n_{(i)}$, $\beta_{(i)}$ and $U_{(i)}$ are the imaginary parts. Now, applying the real and imaginary parts of U in $U^2 = \rho {k_o}^2 {n_{co}}^2 - \beta^2$ from Eq. 5.5 gives

$$ {U_{(r)}}^2 + 2iU_{(r)}U_{(i)} = \rho[{k_o}^2({n_{co,r}}^2 + 2in_{co,r}n_{co,i}) - ({\beta_{(r)}}^2 + 2i\beta_{(r)}\beta_{(i)})] \tag{5.11} $$

and further simplifying, I get

$$ U_{(r)}U_{(i)} = \rho {k_o}^2 n_{co,r}n_{co,i} - \beta_{(r)}\beta_{(i)}. \tag{5.12} $$

where I use ${U_{(r)}}^2 = \rho {k_o}^2 {n_{co,r}}^2 - {\beta_{(r)}}^2$. Also, $n_{co,r}$ and $n_{co,i}$ are real and imaginary parts of the refractive index of the core. Snyder and Love [53] have given propagation constant in terms of normalized frequency parameters like U and V as

$$ \beta = \frac{V}{\rho\sqrt{2\Delta}}\left(1 - \frac{2\Delta}{V}U^2\right) \tag{5.13} $$

where, $\Delta = \frac{n_{co} - n_{cl}}{n_{co}}$ for the weakly guiding approximation.

Now, substituting the real and imaginary parts of β and then equating the imaginary parts of both sides, I am able to obtain the imaginary part of beta as β

$$ \beta_{(r)} + i\beta_{(i)} = \frac{V}{\rho\sqrt{2\Delta}}\left(1 - \frac{2\Delta}{V}\left({U_{(r)}}^2 + 2iU_{(r)}U_{(i)}\right)\right) \tag{5.14} $$

$$ \beta_{(i)} = -\frac{V}{\rho\sqrt{2\Delta}}\,\frac{2\Delta}{V}\left(2U_{(r)}U_{(i)}\right) = -2\frac{\sqrt{2\Delta}}{\rho}\,\left(U_{(r)}U_{(i)}\right) \tag{5.15} $$

$$ \tag{5.16} $$

Finally substituting Eq. 5.11 in Eq. 5.15, I get the expression for $\beta_{(i)}$ as

$$ \beta_{(i)} = -2\frac{\sqrt{2\Delta}}{\rho}\,\rho^2\,[{k_o}^2 n_{co,r}n_{co,i} - \beta_{(r)}\beta_{(i)}] \tag{5.17} $$

$$\beta_{(i)} = -2\frac{\sqrt{2\Delta}\rho {k_o}^2 n_{co,r} n_{co,i}}{1 - 2\sqrt{2\Delta}\rho\beta_{(r)}} \tag{5.18}$$

For my case $\sqrt{2\Delta}\rho\beta_{(r)} \gg 1$. Thus,

$$\beta_{(i)} = \frac{{k_o}^2 n_{co,r} n_{co,i}}{\beta_{(r)}} \tag{5.19}$$

The mode-dependent absorption coefficient is given as $\alpha_{lm} = 2\beta_{(i)}$ and finally, plugging in the expression of imaginary propagation constant, I get

$$\alpha_{lm} = 2\frac{{k_o}^2 n_{co,r} n_{co,i}}{\beta_{(r)}} \tag{5.20}$$

5.2.3 Correlation Functions

Photon statistics, spectral correlation, and probability density functions are essential for characterizing light properties. The sections below discuss these functions in detail.

I. Second Correlation Function

As discussed in previous chapters, the second-order coherence-intensity correlation is essential to classify a light source [6]. In this chapter, I incorporate the spatial second-order coherence function to characterize the speckle patterns at the output of SI MM fiber.

If the intensity $I(\vec{r})$ of the light source is measured at a point such that the statistical properties are invariant in space, then the normalized form of the spatial second-order coherence in terms of intensities is given as

$$g^{(2)}(\Delta r) = \frac{\left\langle I(\vec{r}_1)I(\vec{r}_1 + \Delta r)\right\rangle}{\left\langle I(\vec{r})\right\rangle^2}. \tag{5.21}$$

where $\Delta r = |\vec{r}_2 - \vec{r}_1|$

$$g^{(2)}(\Delta r = 0) = \frac{\langle I(\vec{r})^2\rangle}{\langle I(\vec{r})\rangle^2}. \tag{5.22}$$

Similar to temporal in Sec. 2.3 in Ch. 2, spatial second-order coherence can also be estimated in terms of variance. If $I(\vec{r}) = I + \delta\ I(\vec{r})$ and $\langle \delta I(\vec{r})\rangle = 0$ then correlation at $\Delta r = 0$ will be

$$\begin{aligned} g^{(2)}(0) &= \frac{\langle (I + \delta I(\vec{r}))^2\rangle}{\langle I + \delta I(\vec{r})\rangle^2} && (5.23)\\ &= \frac{\langle I\rangle^2 + \langle \delta I(\vec{r})^2\rangle}{\langle I\rangle^2} && (5.24)\\ &= 1 + \frac{\langle \delta I(\vec{r})^2\rangle}{\langle I\rangle^2}. && (5.25) \end{aligned}$$

Here at $\Delta r = 0$, the term $\langle \delta I(\vec{r})^2\rangle$ signifies the variance of the intensity fluctuation. Its value is always a positive number that results the $g^{(2)}$ to be in the region $1 \leq g^2(\Delta r = 0)$ and $g^{(2)}(\Delta r) \leq g^{(2)}(\Delta r = 0)$ for a classical source. Furthermore, the variances of the thermal light from bunching to super-bunching become larger than the mean value, making the $g^{(2)}$ value go from 2 to higher than 2, respectively.

II. Probability Density Function (PDF)

As discussed in Ch. 2, when light is produced from many point-like sub-sources, each sub-source emits light with the same frequency but a randomly changing phase;

such sources are standard descriptions of a thermal light source. The intensity of such sources has a Gaussian distribution, and the probability density function of such distribution is a negative exponential, as given in Eq. 5.26. The $g^{(2)}$ of such Gaussian intensity shows a bunching effect and is equal to 2. A similar intensity profile is observed for the speckles formed by the interference of different modes supported by a MM fiber and superposed at the output of the fiber. The PDF is a negative exponential for such speckle patterns. This is because when light propagates through the fiber, it obtains a random and different phase from the modes it interacts with, resembling the behavior of point-like sub-sources of a thermal light source.

However, speckles can also have a non-Gaussian intensity distribution, which may no longer fit negative exponential statistics. I have generated speckles that manifest as super-Gaussian intensity distribution as the PDF they fit to is K-Distribution given by Eq. 5.28. The light source is super thermal and the value of $g^{(2)}(\Delta r = 0) > 2$. The intensity PDF for Gaussian distribution is given as

$$P_x(x) = \frac{1}{\overline{I}} exp\big[- \frac{x}{\overline{I}} \big] \tag{5.26}$$

where $\overline{I}$ is average intensity. The intensity distribution that fits the super-Gaussian intensity distribution is given as

$$\begin{aligned} P_I(I) &= \tfrac{1}{\overline{I}} \int_0^{\infty} \frac{1}{x} exp\left[- \left(\frac{1}{x} + \frac{x}{\overline{I}} \right) \right] dx && (5.27) \\ &= \tfrac{2}{\overline{I}} K_0 \left(2 \sqrt{\tfrac{I}{\overline{I}}} \right) && (5.28) \end{aligned}$$

where $\overline{I}$ is average intensity, and K_0 is the modified Bessel function of the second kind, order zero. This type of PDF is also called K-Distribution.

III. Spectral-Correlation Function

The interesting statistics of the speckles formed at the end of the MM fiber signify the spatial correlation; however, more information is related to these grainy patterns by exploring its spectral correlation. Researchers have previously estimated the MM Fiber's bandwidth using the spectral correlation function (SCF) technique. In the above sections, I used the monochromatic input light to model the light propagation inside the MM Fiber. However, in reality, the angular frequency of the source can have a finite width; thus, the speckles at the fiber's output can have well-defined spatial and spectral features. SCF in the normalized form can be defined as the correlation function of two speckle patterns at source light frequencies ν and $\nu + \Delta\nu$. The cross-correlation between the speckle patterns decreases when the frequency difference $\Delta\nu$ increases. Therefore, it can be implied that the amount of decorrelation of the speckle patterns is based on the modes dispersion of the fiber for a given $\Delta\nu$. Thus, SCF can be used to study the spectral characteristics of the output modes as the bandwidth of the SCF can provide a good measure of their types. The SCF can be written in the normalized form as [56]

$$SCF(x, \Delta\nu) = \frac{C(x, \Delta\nu)}{C(x, \Delta\nu = 0)} \tag{5.29}$$

$$= \frac{\langle I(x,\nu) \cdot I(x,\nu+\Delta\nu)\rangle - \langle I(x,\nu)\rangle\langle I(x,\nu+\Delta\nu)\rangle}{\langle I(x,\nu) \cdot I(x,\nu)\rangle - \langle I(x,\nu)\rangle\langle I(x,\nu)\rangle}. \tag{5.30}$$

where $I(x, \nu)$ is the speckle intensity distribution at $\nu = 0$, and $I(x, \nu + \Delta\nu)$ is the intensity distribution after the source frequency is changed by $\Delta\nu$. The angle bracket notation is for statistical averages.

5.3 Experiment

A coherent light source is propagated through the MM fiber to generate the speckles at the exit of SI MM fiber shown in Fig. 5.2. The light source is a monochromatic continuous wave (CW) stabilized laser (Fitel, F0L 15DCWC-A82-19340-B) centered at wavelength 1550 nm with an output power of 37.5 mWatt. The laser beam outputs through a polarization-maintaining single-mode fiber (PM-SMF) and is coupled to another PM-SMF to get an angled physical connect (APC) connector at the other end. The APC connector is coupled to a physical connect (PC) connector of SI MM fiber. The NA of the fiber is 0.22, which is a 12-degree angle; therefore, I connect APC to PC to excite as many modes as possible inside the MM fiber as the APC has a 8-degree angle. The slight angle of APC makes the incidence angle larger, which excites more modes inside the fiber due to internal reflection.

The SI MM fiber has a core radius of 200 μm and a numerical aperture (NA) of 0.22. The fiber supports 3784 modes. The MDL caused by material absorption is due to Hydroxyl ions (OH$-$) in the fiber. The OH$-$ ions form a bond with the glass structure in the fiber core, and this bond's molecular vibration causes the absorption. The fibers used to collect the experimental data have high and low quantities of OH$-$ ions present in them. The low OH fibers (FG200LEA) attenuate about 0.75 dB/km at 1550 nm, whereas the high OH fibers (FG200UEA) have 100 dB/km at 1550 nm. I use four different lengths of fibers in both categories, i.e., 5 m, 25 m, 50 m, and 75 m, respectively to collect speckle data. When coherent light is coupled into the MM fiber, interferences occur among the many modes carried by the fiber, and the superposition of these modes is observed as a grainy speckle pattern in the far field behind the fiber exit. The speckles are recorded using a camera (Hamamatsu, C144041-10U, InGaAs

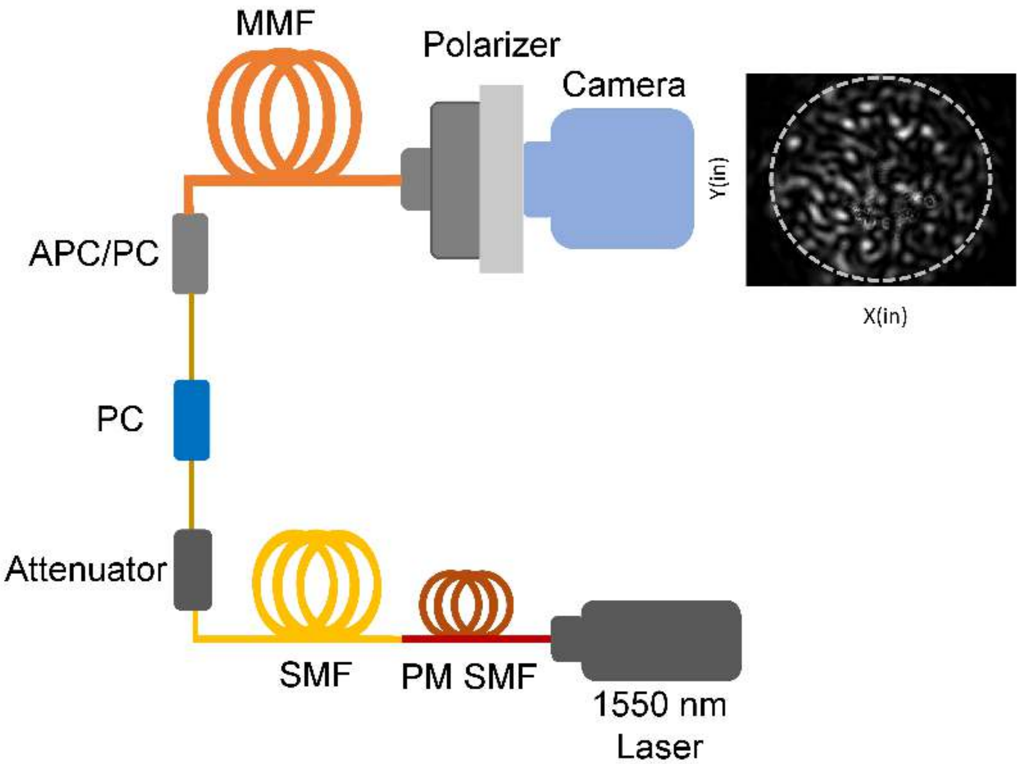


Figure 5.2: The schematic of the experimental set-up used to generate speckles at the output of a multi-mode fiber. PM SMF: Polarization maintained single-mode fiber, SMF: Single-mode fiber, PC: Physical connect, APC: Angled Physical Connect, MMF: Multi-mode fiber. A continuous single-mode 1550 nm laser is coupled through PM SMF into an MMF using an APC to PC connector. The speckles are created at the output of the MM fiber. The intensity of the speckles is recorded using a CCD camera.

sensor) with cell size 20 μm x 20 μm and exposure 100 μs. A polarizer (LPIREA100-C, N-BK7, 1050-1700 nm) was placed before the camera as the polarization of the laser light is not maintained during its propagation inside the MM fiber. The intensity distribution of the speckles depends on the degree of the polarization; a measurement without a polarizer has a different distribution than with a polarizer [28]. Hence, it is essential to employ a polarizer while collecting speckle images.

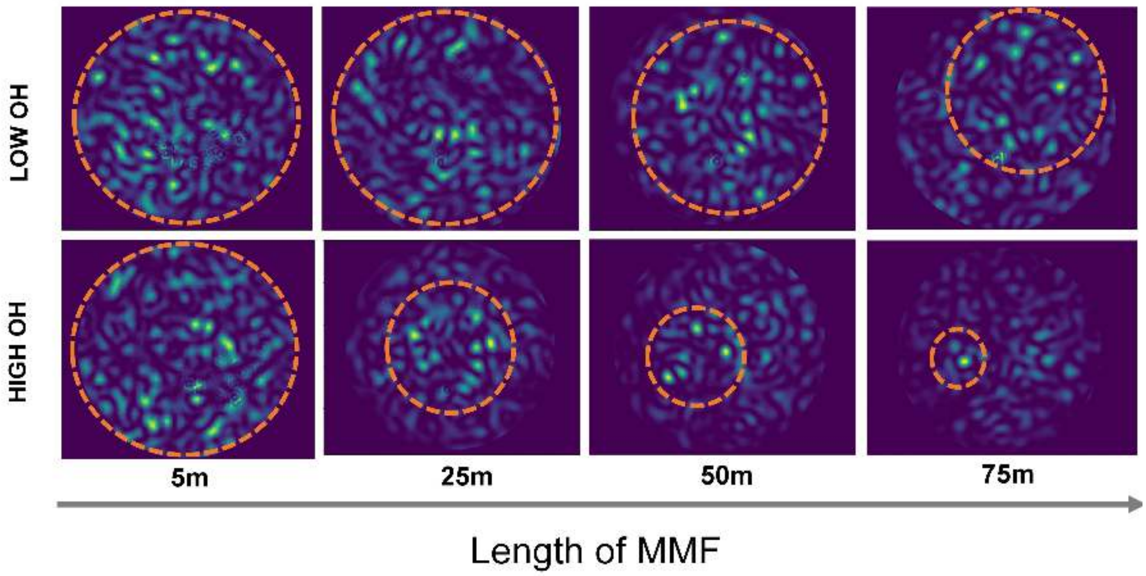


Figure 5.3: The intensity profiles of the speckles recorded at the end of MMF for the Low OH (top row) and High OH (bottom row) fibers with different lengths: 5m, 25m, 50m, and 75m, respectively.

Firstly, I collect the speckle images in the far field (as I do not use any lens or microscope objective for imaging) using this experimental setup for different types of fibers. The images I collect can be categorized into two kinds. The first is based on the fibers' material absorption or amount of OH, and the second is based on the length of the fibers. The fibers I use have high and low OH, and each of these high and low OH fibers is available at four different lengths, as mentioned above. The images shown in Fig. 5.3, absorption vs. length shows the speckles images obtained for different lengths for both high and low absorption MM fibers. The images show different speckle patterns for both the categories: absorption and length. When the absorption of the fibers is changed from low to high, there are fewer bright speckles, and similarly, when the length is varied, keeping the same OH, a low density of bright speckles is observed for longer fibers.

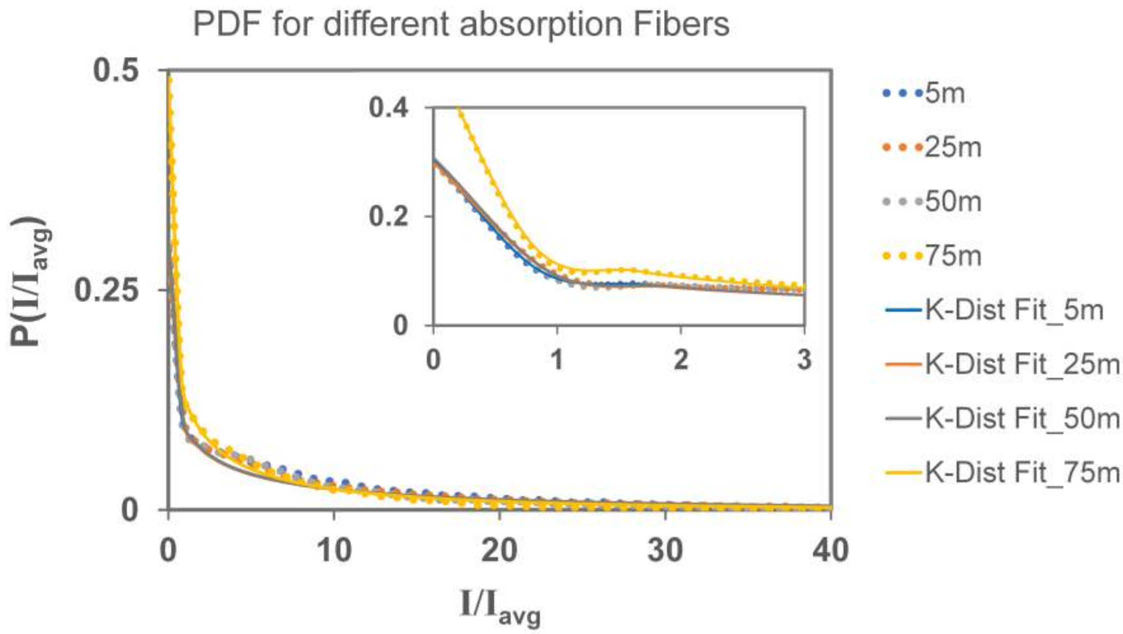


Figure 5.4: The plot shows the K-distribution fit for High OH fibers with different lengths.

As the images of the speckle patterns show variation in space, it is interesting to characterize their statistics. I first subtract the background from the speckle image before using the images for analysis. Then, to study the statistics of these speckle patterns, I measure their intensity distribution $P_I(I)$ and then find the distribution's probability density function (PDF). To do that, I construct a normalized histogram of the measured pixel intensities and then apply a fitting curve. The speckle generated by a MM fiber has a Gaussian intensity distribution, which fits a negative exponential PDF given by Eq. 5.26 [28]; however, the speckles I observed at the fiber

Length (m)	Reduced Chi-Square values
5	1.00
25	1.05
50	1.20
75	1.01

Table 5.1: The summary of reduced chi-square values for different lengths of high absorption fibers

exit gives a poor fit to such PDF. The curve that fits the intensity distribution is a K-distribution discussed in Sec. 5.2.3. The K-distribution PDF fit well, and the result for high OH fibers is shown in Fig. 5.4, and the reduced chi-square values are shown in the table. 5.1. The Low OH fibers also fit the K-distribution and thus show analogous results. Hence, it signifies the speckle patterns have a non-Gaussian intensity distribution.

To further confirm my analysis, I calculated the spatial second-order coherence $g^{(2)}(\Delta r)$ function of these speckles. The $g^{(2)}(\Delta r)$ of speckle images collected with and without a polarizer differ; thus, I calculate the $g^{(2)}(\Delta r)$ for both scenarios using the Eq. 5.21. The $g^{(2)}(\Delta r)$ correlation plots of each of these cases are shown in Fig. 5.5 and Fig. 5.6 as a function of distance from the core of the fiber. I observe $g^{(2)}(\Delta r = 0) > 2$ for all the fibers; however, the fibers with high absorption have higher values compared to low absorption. The $g^{(2)}(\Delta r = 0)$ equal to 2 is observed for the thermal bunching source and higher than 2 for super bunching. Therefore, it signifies that the speckles I observe have a super bunching effect. It also agrees with the non-Gaussian intensity distribution.

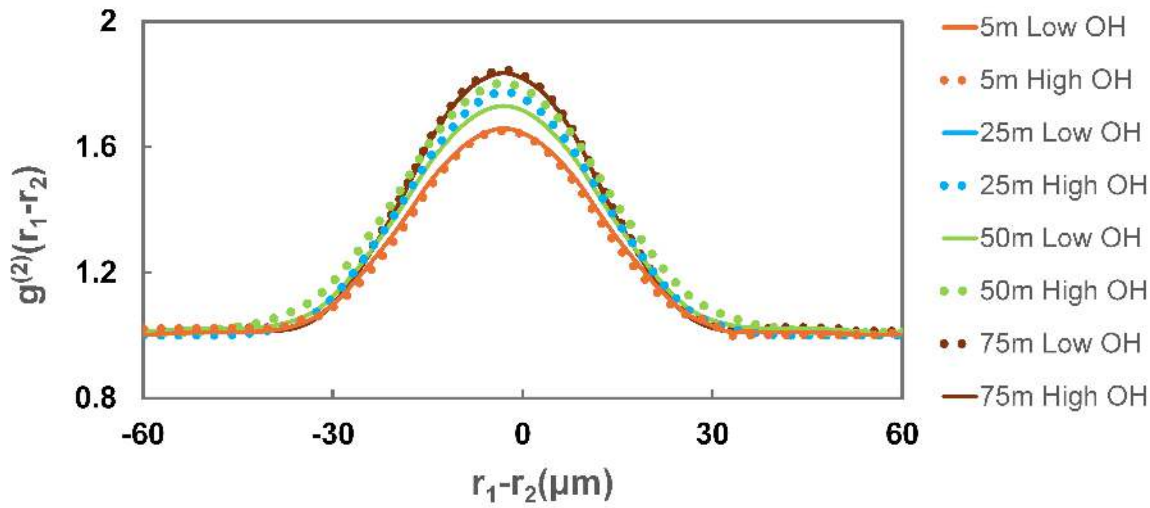


Figure 5.5: $g^{(2)}(\Delta r)$ plot for Low and High OH MM fibers with different lengths. The images of speckles used for the analysis are taken without a polarizer placed before the CCD camera.

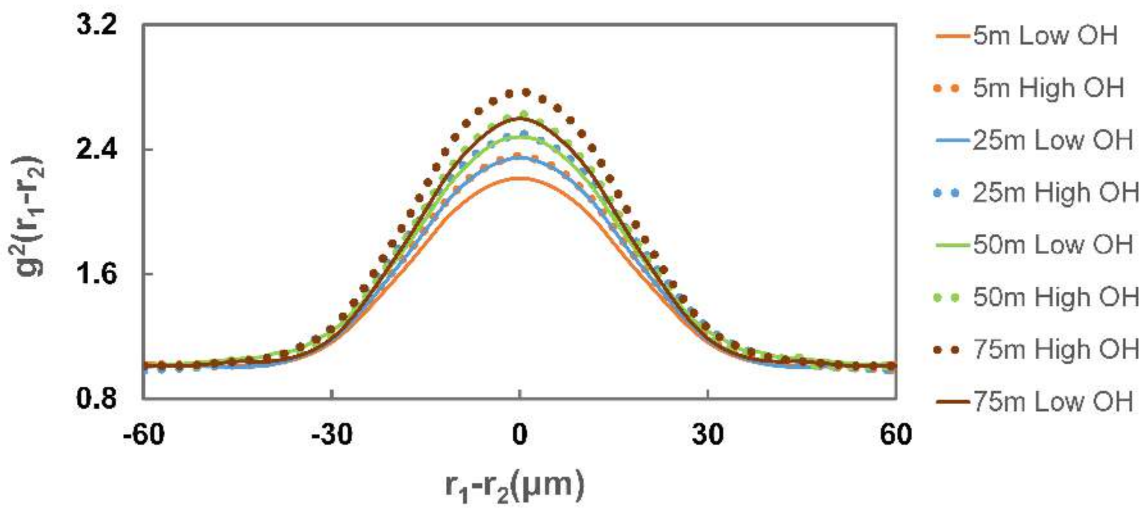


Figure 5.6: $g^{(2)}(\Delta r)$ plot for Low and High OH MM fibers with different lengths. The images of speckles used for the analysis are taken using a polarizer before the CCD camera.

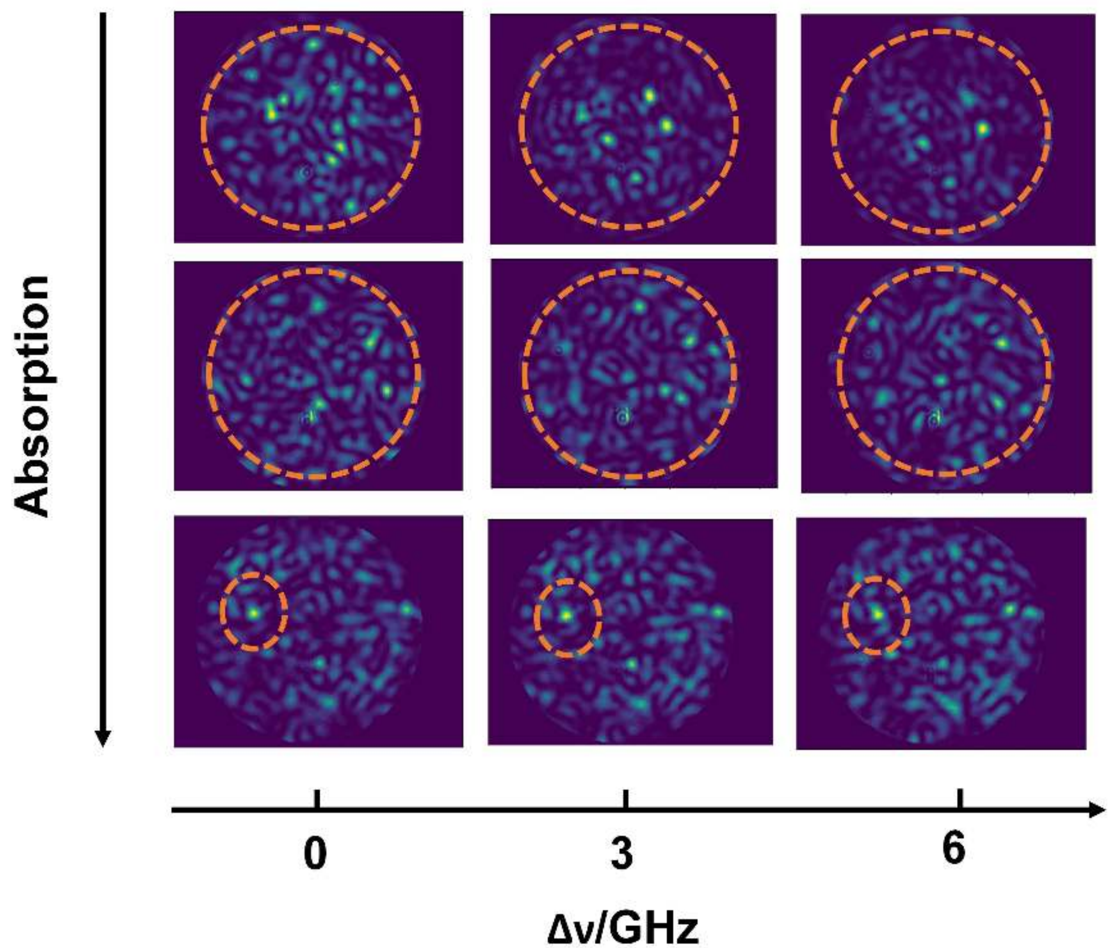


Figure 5.7: The intensity profiles of speckles were generated using 75m MM fiber. The top row refers to Low OH bare fiber, the middle row refers to Low OH jacked fiber, and the bottom row refers to High OH jacked fiber.

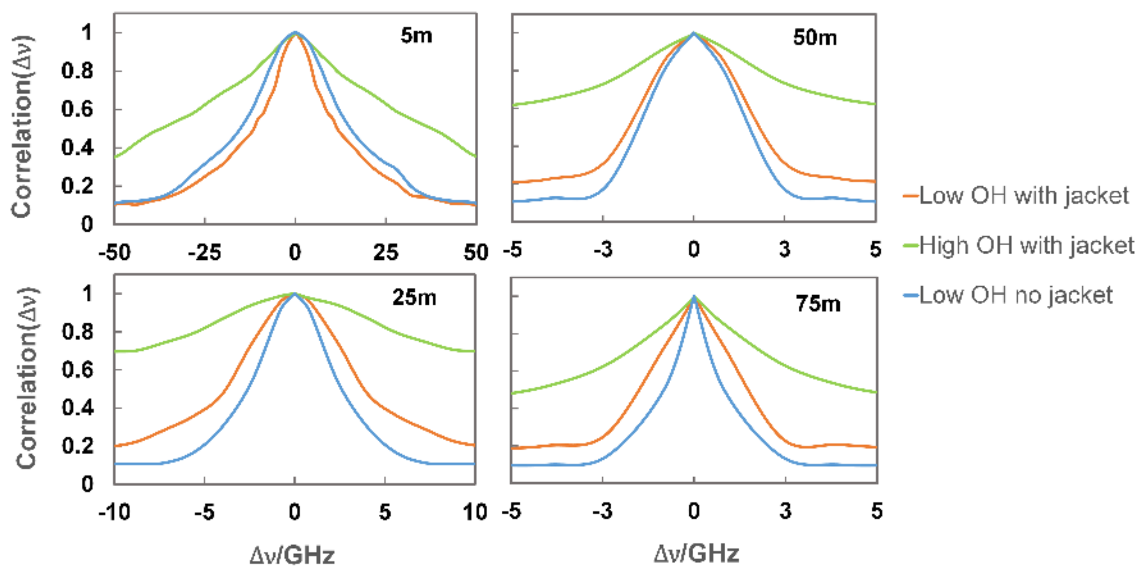


Figure 5.8: The frequency correlation function of bare and jacketed low OH fiber is plotted with jacked high OH fiber of different lengths. SCF is normalized and thus goes to unity at $\Delta\nu = 0$

Finally, I study the spectral correlation functions of speckles to understand them further to find out whether they are only spatially varying or if it has some spectral correlation, too. Studying these properties of speckles using experimental results helps to characterize the speckles more profoundly. To analyze the spectral correlation, I tuned the frequency of the incident laser light and then collected the far-field images of the speckles. The images shown in Fig. 5.7 are of the speckle patterns observed at three different frequencies: 0, 3, and 6 GHz for three different fiber types (bare Low OH, jacketed Low OH, and jacketed High OH). The images at these three frequencies are almost identical for high-absorption fiber compared to low-absorption fiber. To calculate the spectral correlation function, SCF($\Delta\nu$), I first recorded the speckle intensity distribution at $\Delta\nu$ equals zero. Then, I changed the frequency of the laser light by $\Delta\nu$ and recorded the new intensity distribution. I calculated spectral correlation SCF($\Delta\nu$) using Eq. 5.30 and these two data. I repeated this process for other non-zero values of $\Delta\nu$, and finally, obtained spectral correlation plots for different length fibers as shown in Fig. 5.8 for different length fibers. I observe that the spectral decorrelation of each type of fiber is different. The fibers with high absorption are decorrelating slowly compared to those with low absorption.

5.4 Discussion

In 2017, Mehringer [28] proposed a Gaussian distribution for speckles generated by a SI MM fiber in far-field imaging. However, as shown in the above Sec. 5.3, I discovered a non-Gaussian intensity distribution for the speckle patterns created by SI MM fiber. Moreover, I observed a relation between the spatial distribution of the speckles and the material absorption. As the absorption in the fiber was increased, the number of bright speckle grains decreased. The manifestation of fewer bright spots signifies that the absorption changes the correlation of speckle patterns; it generates a super-Gaussian intensity-intensity correlation instead of a Gaussian. Furthermore, the $g^2(\Delta\text{r}=0)$ of such intensity correlation is higher than 2, inferring a super bunching spatial second-order coherence. This can also be understood from the variance of the intensity of the speckles. The less bright spots are fewer sharp peaks in the intensity profile, meaning the variance is higher than the mean square of the intensity from Eq. 5.25 and thus $g^2(\Delta\text{r}=0) > 2$. Therefore, I deduce from the experimental results that I have created speckles with a super-Gaussian intensity distribution showing a super-bunching second-order correlation.

To generate the Gaussian intensity distribution speckles with $g^{(2)}(\Delta \mathrm{r}=0)$ equal to 2, Mehringer [28] applied vibration of 1 Hz to a 1 m SI MM fiber. However, I only propagated a coherent light into a SI MM fiber and observed $g^{(2)}(\Delta \mathrm{r}=0) > 2$ for the speckles created at the exit of the MM fiber. The value of $g^{(2)}(\Delta \mathrm{r}=0)$ was higher for the fibers with high material absorption. Therefore, I obtained a minimum value of $g^2(\Delta \mathrm{r}=0)$ equal to 2.17 ± 0.04 for the 5 m low OH jacketed fiber, and the maximum value of 2.70 ± 0.03 for the 75 m high OH jacked fiber. To explain this phenomenon, I propose that it is caused by the interferences of the fiber modes under a mode-dependent loss (MDL) caused by leaky modes in low OH fibers and leaky modes and material absorption in high OH fibers [57–59].

In addition, I also observed different values of $g^2(\Delta \mathrm{r}=0)$ as shown in Fig 5.6 and Fig 5.10 for the fibers with the same material absorption. For instance, a 25 long low OH bare fiber has $g^2(\Delta \mathrm{r}=0)$ equal to 2.19 ± 0.03 whereas a 25 long low OH jacketed fiber $g^2(\Delta \mathrm{r}=0)$ is equal to 2.33 ± 0.03. This fact can be explained using the stain-induced effect [60, 61]. The jacketed fiber has a strain induced by the jacket, and therefore, the leaky modes MDL is higher in the jacketed fibers compared to bare fibers. The high MDL increases the $g^2(\Delta \mathrm{r}=0)$ value, and thus different values are observed for fibers with the same material absorption. Therefore, putting it all together, I propose that both leaky modes and material absorption play a role in increasing the MDL and thus resulting in different values of $g^{(2)}(\Delta \mathrm{r}=0)$. In addition, the jacketed fiber MDL is also increased by the strain induced, which is absent in the bare MM fibers.

To understand the spatial correlations of the speckles in detail, I also plotted $g^{(2)}(\Delta \mathrm{r}=0)$ for the different lengths jacketed low and high OH fibers as shown in

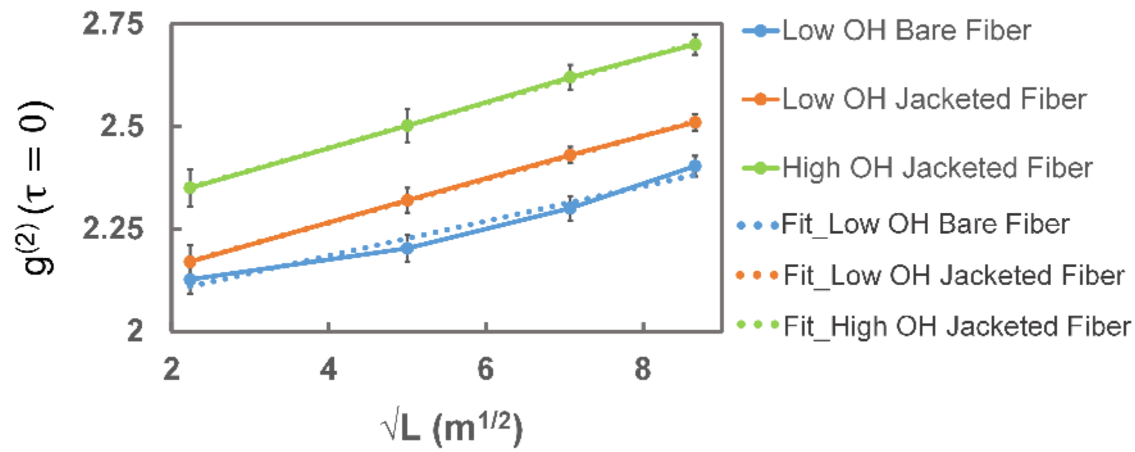


Figure 5.9: The second-order correlation at (Δr=0), as a function of the square root of fiber length L for low and high OH jacketed MM fibers. The linear fits are the dashed lines and solid lines with dots are the experimental data points. The plot directly relates $g^{(2)}(\Delta\text{r}=0)$ with $\sqrt{L}$.

Fiber	Reduced Chi-Square values
Bare Low OH	2.70
Jacketed Low OH	1.08
Jacketed High OH	1.05

Table 5.2: The values of reduced chi-square for bare low OH, jacketed low OH, and jacketed high OH MM fibers for $g^{(2)}(\Delta\text{r}=0)$ vs $\sqrt{L}$ plots shown in Fig. 5.9.

Fig. 5.9. I observed that $g^{(2)}(\Delta\text{r}=0)$ is directly proportional to the square root of the length of the jacketed fibers. Whereas $g^{(2)}(\Delta\text{r}=0)$ of the bare low OH fibers is directly proportional to the length of the fibers. The relation indicates that the fiber's physical properties are related to the spatial second-order correlation. It would be interesting to study this behavior in detail and characterize it further to find other correlations.

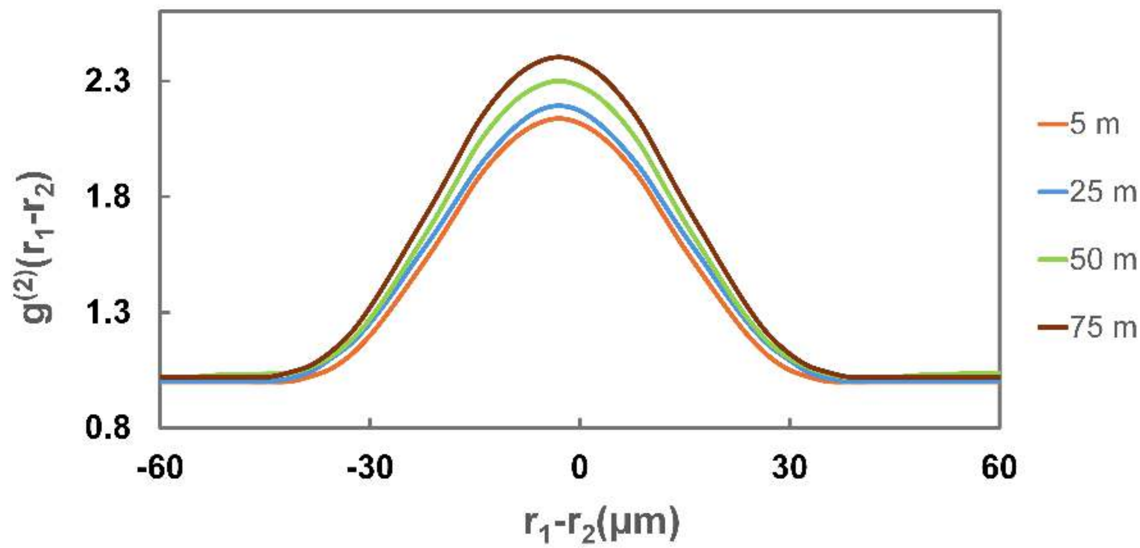


Figure 5.10: $g^{(2)}(\Delta r)$ plot for Low OH bare fibers with different lengths. The images of speckles used for the analysis are taken using a polarizer before the CCD camera.

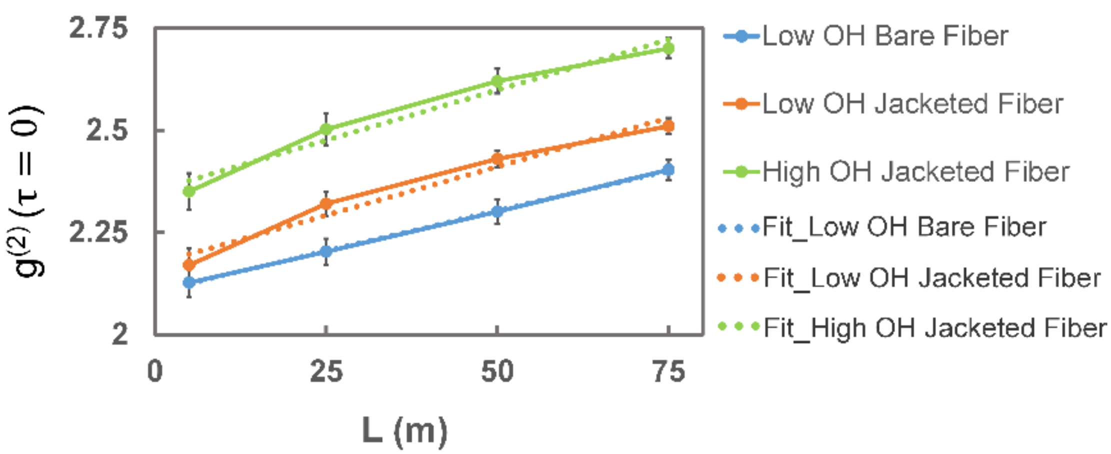


Figure 5.11: The second-order correlation at (Δr=0), as a function of fiber length L for Low OH bare MM fibers. The linear fits are the dashed lines and solid lines with dots are the experimental data points. The plot directly relates $g^{(2)}(\Delta\mathrm{r}=0)$ with L.

Fiber	Reduced Chi-Square values
Bare Low OH	1.06
Jacketed Low OH	2.80
Jacketed High OH	3.10

Table 5.3: The values of reduced chi-square for bare low OH, jacketed low OH and jacketed high OH MM fibers for $g^{(2)}(\Delta\mathrm{r}{=}0)$ vs L plots shown in Fig. 5.11.

Another interesting phenomenon observed in these SI MM fibers is the spectral correlations. The high OH fibers are less sensitive to the incident laser light frequency than the low OH jacketed and bare fibers. The fiber with high OH has a spectral width ($\Delta\nu$) broader than low OH fibers. From the experimental measurement and result of 75m fiber, I have the value of $\Delta\nu$ to be 21.40 GHz for Low OH jacketed fiber and 65.00 GHz for high OH jacked fiber, three times broader than that of Low OH fiber. This establishes the high OH fibers are less sensitive to incident light frequency. This could be an interesting phenomenon to study as there are modes that are less sensitive to changes in the frequency of light propagating through the fiber.

5.5 Conclusion

To summarize, I presented experimental results and numerical calculations on generating frequency-invariant modes with a super-Gaussian intensity correlation. The proposed method includes a simple setup with only a SI MM fiber and a camera. It is a compact, direct, inexpensive method to create super bunching speckles, i.e., the second-order correlation $g^{(2)}(\Delta\mathrm{r}{=}0) > 2$. I conclude that to observe this effect, the fibers should exhibit mode-dependent loss. The super-bunching impact increases with the increase of MDL in the fiber. This simple device can be applied to increase the visibility of ghost imaging and study and characterize the higher-order coherence theory.

Chapter 6

Summary and Future Work

In this thesis, I experimentally and theoretically demonstrated that super-bunching pseudothermal light can be created in a single-mode fiber with a short coherence time. I also described the super-Gaussian intensity-intensity correlation that implemented a super-bunching second-order coherence in a MM fiber.

In this chapter, I summarize all the essential accomplishments realized in this thesis and outline future investigations that can be conducted using the technology developed here.

6.1 Summary

In Ch. 1, I discuss the need for an alternate method to create a pseudothermal light source (PTLS). I highlight the present techniques, which are widely used for application purposes. However, the coherence time of these PTLS is longer than that of a true thermal source, and thus, it is crucial to develop a short-coherence PTLS to utilize the source potential fully.

In Ch. 2, I outline the importance of spatial and temporal coherence theory and the characterization of lights in a classical regime using intensity correlation, as these are the tools to investigate the PTLS. I also discussed in detail the background of PTLS by describing the different methods of creating these light sources.

In Ch. 3, I present an alternate technology to generate a PTLS with a tunable short coherence time of 10^{-8} to 10^{-6} s compared to 10^{-5} to 1 s achieved by other research groups. I also demonstrated that this technology gives second-order coherence higher than 2; thus, it is a super bunched PTLS. I conclude that this device is simple and efficient to use and also resolves issues like loss of light due to scattering in space and uncertainty in the calculation of the second-order coherence function.

In Ch. 4, I demonstrate the current-to-frequency converter device I built and describe an efficient technique for modulating the frequency of a diode laser by changing its current. I also discuss how this device can be implemented in an experimental setup to create a PTLS.

In Ch. 5, I describe a crucial technology based on multi-mode fiber. I demonstrate that spatially correlated speckles are created if a coherent light is propagated through a step-index multi-mode fiber. These spatially correlated speckles are characterized by a super-Gaussian intensity-intensity correlation function, which manifests as a speckle pattern of the light illuminating from the fiber that contains fewer bright speckles. I also discuss that these speckles are invariant to the change in frequency of the incident light. Therefore, these speckles are evolving both in space and in the frequency domain.

6.2 Future Work

The technologies developed in this thesis can be used to study interesting physics problems. The single-mode PTLS can be implemented in an experimental setup with a rotating ground glass disc to study high-order coherence theories. This technology can also increase imaging visibility in ghost imaging systems.

Furthermore, speckles' spatial and spectral evolution at the exit of multi-mode fiber can be investigated to determine whether these evolutions correlate with any physical properties of the fiber.

Bibliography

[1] D. R. Paschotta, "Encyclopedia of laser physics and technology-coherence, coherent, light, spatial and temporal coherence, monochromaticity," Retrieved 2018-10-29, from https://www. rp-photonics. com/coherence. html, Tech. Rep.

[2] R. Hanbury Brown and R. Q. Twiss, "A test of a new type of stellar interferometer on sirius," in *A Source Book in Astronomy and Astrophysics, 1900–1975.* Harvard University Press, 1979, pp. 8–12.

[3] M. Born *et al.*, "E. wolf principles of optics," *Pergamon Press*, vol. 6, no. 188, pp. 36–111, 1980.

[4] R. H. Brown and R. Q. Twiss, "Correlation between photons in two coherent beams of light," *Nature*, vol. 177, no. 4497, pp. 27–29, 1956.

[5] R. J. Glauber, "Nobel lecture: One hundred years of light quanta," *Reviews of modern physics*, vol. 78, no. 4, p. 1267, 2006.

[6] R. Loudon, *The quantum theory of light.* OUP Oxford, 2000.

[7] L. Mandel and E. Wolf, *Optical coherence and quantum optics.* Cambridge university press, 1995.

[8] ——, "Coherence properties of optical fields," *Reviews of modern physics*, vol. 37, no. 2, p. 231, 1965.

[9] W. Martienssen and E. Spiller, “Coherence and fluctuations in light beams,” *American Journal of Physics*, vol. 32, no. 12, pp. 919–926, 1964.

[10] T. Asakura, “Spatial coherence of laser light passed through rotating ground glass,” *Opto-electronics*, vol. 2, pp. 115–123, 1970.

[11] B. Crosignani, B. Daino, and P. Di Porto, “Light scattering by a rotating disk,” *Journal of Applied Physics*, vol. 42, no. 1, pp. 399–403, 1971.

[12] J. Churnside, “Speckle from a rotating diffuse object,” *JOSA*, vol. 72, no. 11, pp. 1464–1469, 1982.

[13] T. Gonsiorowski and J. C. Dainty, “Correlation properties of light produced by quasi-thermal sources,” *JOSA*, vol. 73, no. 2, pp. 234–237, 1983.

[14] T. Yoshimura, “Statistical properties of dynamic speckles,” *JOSA A*, vol. 3, no. 7, pp. 1032–1054, 1986.

[15] J. C. Marron and K. S. Schroeder, “Speckle from rough rotating objects,” *Applied optics*, vol. 27, no. 20, pp. 4279–4287, 1988.

[16] Y. Zhou, F.-l. Li, B. Bai, H. Chen, J. Liu, Z. Xu, and H. Zheng, “Superbunching pseudothermal light,” *Physical Review A*, vol. 95, no. 5, p. 053809, 2017.

[17] B. Bai, J. Liu, Y. Zhou, H. Zheng, H. Chen, S. Zhang, Y. He, F. Li, and Z. Xu, “Photon superbunching of classical light in the hanbury brown–twiss interferometer,” *JOSA B*, vol. 34, no. 10, pp. 2081–2088, 2017.

[18] Y. Zhou, X. Zhang, Z. Wang, F. Zhang, H. Chen, H. Zheng, J. Liu, F.-l. Li, and Z. Xu, “Superbunching pseudothermal light with intensity modulated laser light and rotating groundglass,” *Optics Communications*, vol. 437, pp. 330–336, 2019.

[19] J. Liu, R. Zhuang, X. Zhang, C. Wei, H. Zheng, Y. Zhou, H. Chen, Y. He, and Z. Xu, “Simple and efficient way to generate superbunching pseudothermal light,” *Optics Communications*, vol. 498, p. 127264, 2021.

[20] A. Gatti, E. Brambilla, M. Bache, and L. A. Lugiato, “Ghost imaging with thermal light: comparing entanglement and classicalcorrelation,” *Physical review letters*, vol. 93, no. 9, p. 093602, 2004.

[21] F. Ferri, D. Magatti, A. Gatti, M. Bache, E. Brambilla, and L. A. Lugiato, “High-resolution ghost image and ghost diffraction experiments with thermal light,” *Physical review letters*, vol. 94, no. 18, p. 183602, 2005.

[22] S. Crosby, S. Castelletto, C. Aruldoss, R. Scholten, and A. Roberts, “Modelling of classical ghost images obtained using scattered light,” *New Journal of Physics*, vol. 9, no. 8, p. 285, 2007.

[23] K. W. C. Chan, M. N. O’Sullivan, and R. W. Boyd, “High-order thermal ghost imaging,” *Optics letters*, vol. 34, no. 21, pp. 3343–3345, 2009.

[24] W. Gong and S. Han, “High-resolution far-field ghost imaging via sparsity constraint,” *Scientific reports*, vol. 5, no. 1, p. 9280, 2015.

[25] Y. Bromberg, Y. Lahini, E. Small, and Y. Silberberg, “Hanbury brown and twiss interferometry with interacting photons,” *Nature Photonics*, vol. 4, no. 10, pp. 721–726, 2010.

[26] Z. Ficek and S. Swain, *Quantum interference and coherence: theory and experiments*. Springer Science & Business Media, 2005, vol. 100.

[27] I. Straka, J. Mika, and M. Ježek, "Generator of arbitrary classical photon statistics," *Optics Express*, vol. 26, no. 7, pp. 8998–9010, 2018.

[28] T. Mehringer, S. Oppel, and J. von Zanthier, "An optical multimode fiber as pseudothermal light source," *Applied Physics B*, vol. 123, pp. 1–7, 2017.

[29] M. Imai, "Statistical properties of optical fiber speckles," vol. 130, pp. 89–104, 1986.

[30] D. Gloge, "Weakly guiding fibers," *Applied optics*, vol. 10, no. 10, pp. 2252–2258, 1971.

[31] J. Liu, J. Wang, H. Chen, H. Zheng, Y. Liu, Y. Zhou, F.-l. Li, and Z. Xu, "High visibility temporal ghost imaging with classical light," *Optics Communications*, vol. 410, pp. 824–829, 2018.

[32] M. Hong, A. Miller, R. d. J. León-Montiel, C. You, and O. S. Magaña-Loaiza, "Engineering super-poissonian photon statistics of spatial light modes," *Laser & Photonics Reviews*, vol. 17, no. 10, p. 2300117, 2023.

[33] T. B. Pittman, Y. Shih, D. Strekalov, and A. V. Sergienko, "Optical imaging by means of two-photon quantum entanglement," *Physical Review A*, vol. 52, no. 5, p. R3429, 1995.

[34] A. F. Abouraddy, B. E. Saleh, A. V. Sergienko, and M. C. Teich, "Role of entanglement in two-photon imaging," *Physical review letters*, vol. 87, no. 12, p. 123602, 2001.

[35] R. S. Bennink, S. J. Bentley, and R. W. Boyd, ““two-photon” coincidence imaging with a classical source,” *Physical review letters*, vol. 89, no. 11, p. 113601, 2002.

[36] A. Gatti, E. Brambilla, M. Bache, and L. A. Lugiato, “Ghost imaging with thermal light: comparing entanglement and classical correlation,” *Physical review letters*, vol. 93, no. 9, p. 093602, 2004.

[37] A. Valencia, G. Scarcelli, M. D’Angelo, and Y. Shih, “Two-photon imaging with thermal light,” *Physical review letters*, vol. 94, no. 6, p. 063601, 2005.

[38] G. Scarcelli, V. Berardi, and Y. Shih, “Can two-photon correlation of chaotic light be considered¡? format?¿ as correlation of intensity fluctuations?” *Physical review letters*, vol. 96, no. 6, p. 063602, 2006.

[39] P. Ryczkowski, M. Barbier, A. T. Friberg, J. M. Dudley, and G. Genty, “Ghost imaging in the time domain,” *Nature Photonics*, vol. 10, no. 3, pp. 167–170, 2016.

[40] ——, “Ghost imaging in the time domain,” *Nature Photonics*, vol. 10, no. 3, pp. 167–170, 2016.

[41] J. Blakely, L. Illing, and D. J. Gauthier, “High-speed chaos in an optical feedback system with flexible timescales,” *IEEE journal of quantum electronics*, vol. 40, no. 3, pp. 299–305, 2004.

[42] Y. Zhu, M. Lee, M. A. Neifeld, and D. J. Gauthier, “High-fidelity, broadband stimulated-brillouin-scattering-based slow light using fast noise modulation,” *Optics Express*, vol. 19, no. 2, pp. 687–697, 2011.

[43] S. Kobayashi, Y. Yamamoto, M. Ito, and T. Kimura, "Direct frequency modulation in algaas semiconductor lasers," *IEEE Transactions on Microwave Theory and Techniques*, vol. 30, no. 4, pp. 428–441, 1982.

[44] C.-Y. Tsai, C.-H. Chen, T.-L. Sung, C.-Y. Tsai, and J. M. Rorison, "Theoretical modeling of carrier and lattice heating effects for frequency chirping in semiconductor lasers," *Applied physics letters*, vol. 74, no. 7, pp. 917–919, 1999.

[45] H. Gao, H. Hu, Y. Zhao, J. Li, M. Lei, and Y. Zhang, "Highly-sensitive optical fiber temperature sensors based on pdms/silica hybrid fiber structures," *Sensors and Actuators A: Physical*, vol. 284, pp. 22–27, 2018.

[46] H. Gao, Y.-X. Zhang, W.-G. Zhang, and T.-Y. Yan, "High sensitivity optical fiber temperature sensor based on pdms-filled with extended measuring range," *Optik*, vol. 248, p. 168181, 2021.

[47] B. Redding and H. Cao, "Using a multimode fiber as a high-resolution, low-loss spectrometer," *Optics letters*, vol. 37, no. 16, pp. 3384–3386, 2012.

[48] B. Redding, S. M. Popoff, and H. Cao, "All-fiber spectrometer based on speckle pattern reconstruction," *Optics express*, vol. 21, no. 5, pp. 6584–6600, 2013.

[49] B. Redding, S. M. Popoff, Y. Bromberg, M. A. Choma, and H. Cao, "Noise analysis of spectrometers based on speckle pattern reconstruction," *Applied optics*, vol. 53, no. 3, pp. 410–417, 2014.

[50] B. Redding, M. Alam, M. Seifert, and H. Cao, "High-resolution and broadband all-fiber spectrometers," *Optica*, vol. 1, no. 3, pp. 175–180, 2014.

[51] ——, "High-resolution and broadband all-fiber spectrometers," *Optica*, vol. 1, no. 3, pp. 175–180, 2014.

[52] F. Arecchi, E. Gatti, and A. Sona, "Time distribution of photons from coherent and gaussian sources," *Physics Letters*, vol. 20, no. 1, pp. 27–29, 1966.

[53] A. W. Snyder, J. D. Love *et al.*, *Optical waveguide theory.* Chapman and hall London, 1983, vol. 175.

[54] N. Kapany, *Optical waveguides*, 1972.

[55] D. Marcuse, "Light transmission optics; van nostrand reinhold," *New York*, vol. 19822, p. 466, 1972.

[56] B. Moslehi, J. W. Goodman, and E. G. Rawson, "Bandwidth estimation for multimode optical fibers using the frequency correlation function of speckle patterns," *Applied Optics*, vol. 22, no. 7, pp. 995–999, 1983.

[57] A. W. Snyder, D. Mitchell, and C. Pask, "Failure of geometric optics for analysis of circular optical fibers," *JOSA*, vol. 64, no. 5, pp. 608–614, 1974.

[58] R. Olshansky and D. Nolan, "Mode-dependent attenuation of optical fibers: excess loss," *Applied optics*, vol. 15, no. 4, pp. 1045–1047, 1976.

[59] A. R. Mickelson and M. Eriksrud, "Mode-dependent attenuation in optical fibers," *JOSA*, vol. 73, no. 10, pp. 1282–1290, 1983.

[60] H. Grebel and G. Herskowitz, "Effect of strain in periodically deformed single-mode optical fibers," *Applied optics*, vol. 26, no. 11, pp. 2155–2158, 1987.

[61] H. Grebel and M. Mezhoudi, "Effect of periodic strain on the leaky modes in single-mode fiber," *Applied optics*, vol. 27, no. 23, pp. 4819–4821, 1988.